Thermal Science

Engineering thermodynamics | Thermal Engineering | Heat & Mass transfer

Thermodynamics

It is the branch of science which deals with the energy transfer. According to Stephen W. Hawkin's definition thermodynamics is the macroscopic science which deals with the energy transfer notably heat and mass interaction and its affect on the physical properties of subtances which are perceptible of measurable.

Application of thermodynamics

1. **Power Producing device**
   Eg. Steam engine, internal combustion (9c) engine, Nuclear reactor.
2. **Power consuming device**
   Eg. Refrigerator, heat pump.
3. **Thermodynamic system**
   System: Any fixed region in a space upon which our concentration is to analyze the problem is known as thermodynamic system
   - There should be the limited region in order to establish any thermodynamic system.
   - The schematic representation of a thermodynamic system is given as

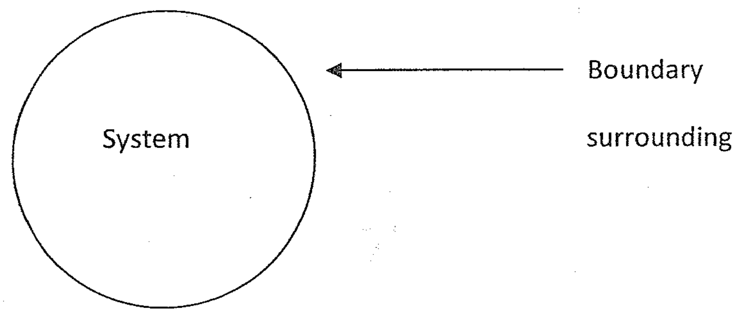


Fig. thermodynamic system

Boundary:

The separation line which separates system from the surrounding is known as boundary.

- There are two types of boundary
  i) dithermal boundary
  ii) adiabatic boundary

i) Dithermal boundary:
The boundary which allow the heat transfer to take place in between the system and surrounding.
- All the conductors are the examples of dithermal boundary

ii) Adiabatic boundary:
The boundary which doesn't allow the heat transfer to take in between the system and surrounding.
- All the insulators are the examples of adiabatic boundary.

There are three different types of thermodynamic system.

1. Open system
2. Closed system
3. Isolated System

1. Open system

- The system which interacts both mass and energy with the surrounding is known as open system.
- It is also known as control volume system.
- The control volume boundary is the main responsible factor to operate any open system devices.
  Eg. Air compressor.
  Most of the engineering devices are the examples of open system.
- The schematic representation of an open system is

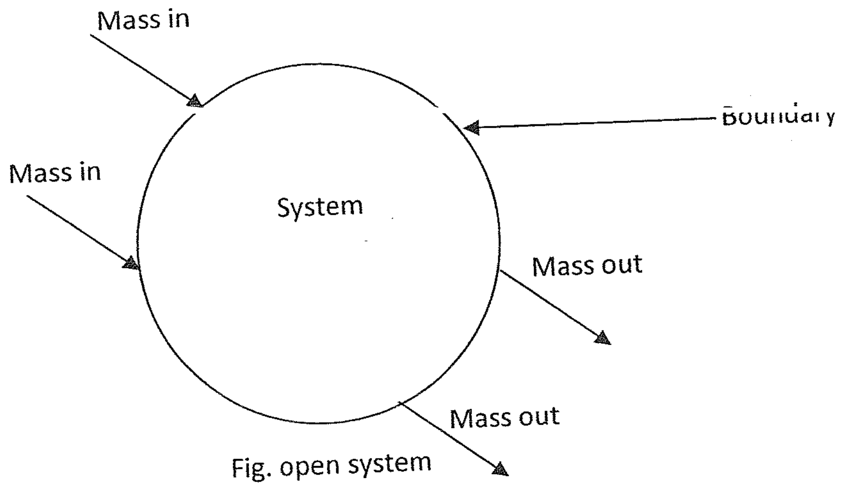
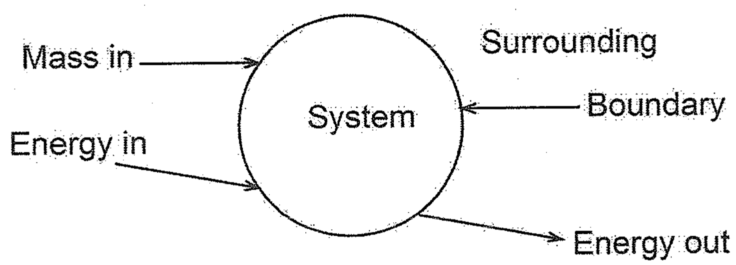
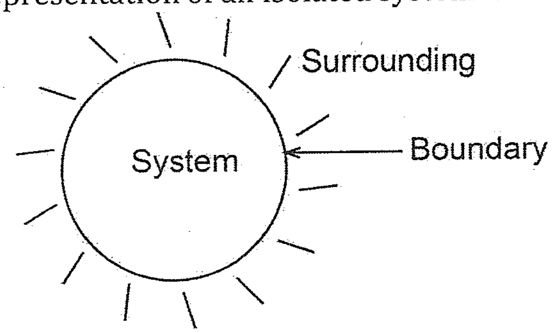


Fig. open system

2. **Close system**
    - The system which interacts energy in between the system and surrounding but doesn't allow mass to interact is known as closes system.
    - It is also known as closed mass system.
    - Hydraulic brake, cylinder fitted with piston.
    - The schematic representation of closed system is



3. **Isolated system**
    - The system which doesn't allow both mass and energy with the system and surrounding is known as isolated system.
    - There should be the complete restriction to interact anything in between the system & surrounding.
    - Eg. Thermas flask
    - The schematic representation of an isolated system is



Thermodynamic properties:

- Any parameter or variable which is used to describe the condition or a state of a system is known as thermodynamic properties.
- If the value of all the thermodynamic properties is fixed for a certain time then the system is said to be in a particular system.
- The most commonly used thermodynamic properties are pressure, temperature and volume.
- If the value of any of the thermodynamic property is changed then, the system changes its state from one to two, which is particularly known as change of state.

System 1

State one		State two
$P_1 = 1\ Pa = \lvert N \rvert m^2$		$P_1 = 0.8Pa = 0.8\ N/m^2$
$V_1 = 0.5m^3$	$\xrightarrow{\text{Heat supply}}$	$V_1 = 0.7m^3$
$T_1 = 15^0c$		$T_1 = 20^0c$

In thermodynamics there are three commonly used various properties. They are
1. Intensive Property
2. Extensive property
3. Specific property

1. **Intensive Property**
   The property which is expressed in order to describe the parameter of system which are independent to the mass of a system is known as intensive property. Eg. Pressure, temperature.

2. **Extensive property**
   The property which is expressed in order to describe the parameter of system which are dependent to the mass of a system is known as extensive property. Eg. Volume, enthalpy (H), entropy (S), internal energy (U)

3. **Specific property**
   The property which is expressed in order to describe the parameter of system which are expressed per unit mass of the system is known as specific property.
   - When the extensive property is expressed per unit mss of a system then the specific property is formed.
   - The specific property is always expressed in small letters.
     For eg.
     Extensive property H $= \frac{Extensive}{mass} = \frac{H}{m} = \frac{KJ}{kg} = KJ/kg = h$
   - The three requirements or characteristics of a thermodynamic property is given as
     a) It must depend upon the states only.
     b) For a given particular state the value of all the thermodynamic properties remains constant.
     c) Their differential should be exact.

Thermodynamic equilibrium
For a given system is said to be in a thermodynamic equilibrium if
a) Thermal equilibrium, mechanical equilibrium, chemical equilibrium is satisfied.
- Thermal equilibrium
  For a given system to be in a thermal equilibrium there must be the equality of temperature is maintain.
- Mechanical equilibrium
  For a given system to be in a mechanical equilibrium there must be the absence of unbalanced force.
- Chemical equilibrium
  For a given system to be in a chemical equilibrium there must be the absence of unbalanced reaction.

State, path, process & cycle

The locus of various point in which a particular system changes its state then the particular locus is known as path.

- It is the way which represent how the state changes.

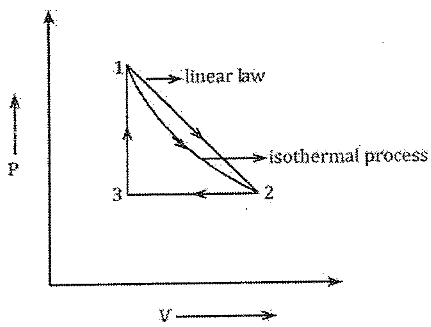
Process:

- If the path is fully identified which represent the changes of state from one to another then the path is known as process.
- The most commonly used thermodynamic properties are
  a) Isothermal process (T is constant)
  b) Isobaric process (P is constant)
  c) Isochoric process (V is constant)

Cycle:
A cycle consists of more than one process among which all must be the indentified process combinely form a cycle in which initial state is identical to the final state. For a given cycle the change in the value of any properties remains constant.

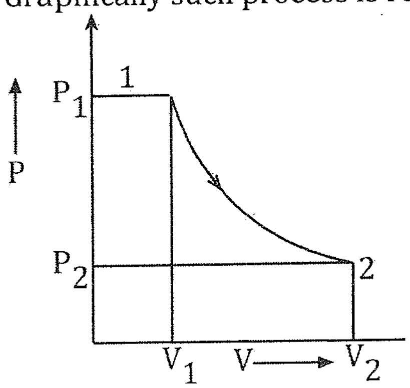
Draw a cycle by using the tree different process that you know.

1 – 2: linear law
2 – 3: Isobaric
3 – 1: Isochoric



V.V.I. Quassi – equilibrium process OR quassi static process

1. The process in which every state pass through by the system is an equilibrium process is known as quassi – equilibrium process.
2. Infinite slowness is the characteristic feature of a quassi- equilibrium process.
3. Graphically such process is represented by a continuous curve in a pv diagram.



Explanation of a quassi – equilibrium process.

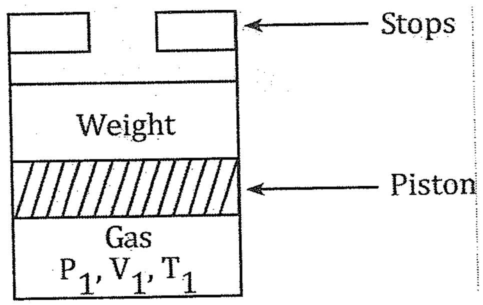
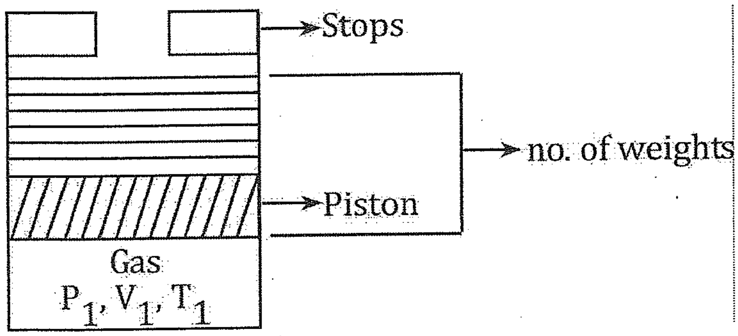


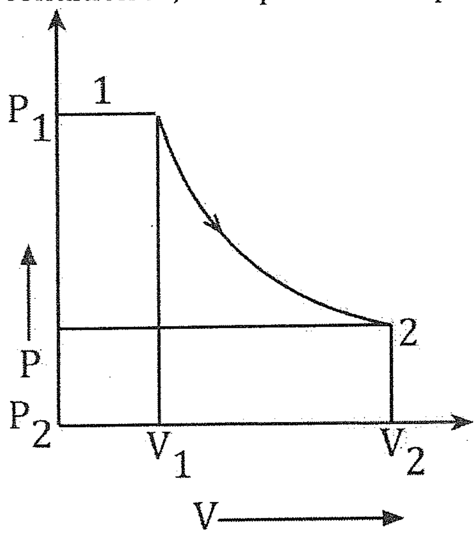
Fig. 1

- Consider a piston cylinder assembly containing a gas which is having P_1, V_1, T_1 as a initial properties. At this condition the system is having only one state at an equilibrium condition. As shown in figure no.1 the upward pressure exerted by the gas is balanced by the weight which is placed above the piston. Mathematically
  $\therefore$ Mass of weight = pressure of gas.
  Or, mg = P gas
- If we remove the weight from the piston, there is the presence of ublanced force between the system and surrounding which results the movement of piston in upward direction tiu it hits the stops.
- At this condition, the gas comes in another equilibrium condition which is represented by the properties P2, V2, T2. The number of state pass through by the piston non-equilibrium state because we cannot locate each & every state pass though by the system shown in fig(1)

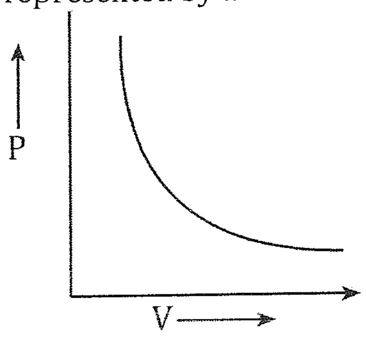
- In order to locate each and every state passed through by the system we divided the single weight into many parts and kept above the piston as shown fig (2)



- If we remove the first weight above the piston, then there will be a small unbalanced force which result the small displacement of piston & we can locate that particular state.
- If we continue the same process again, we can locate each and every point passed through by the system ($P_1, V_1, T_1, P_2, V_2, T_2, \ldots\ldots\ldots P_n, V_n, T_n$)
- So, in this process, every state passed through by the piston is an equilibrium condition so, such process is represented by a continuous curve in a PV diagram.

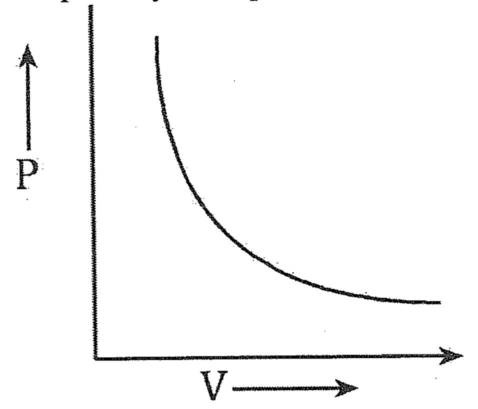


- If we represent the process which is shown in fig (1) in a PV diagram. Then it is represented by a dash line.



Reversible process:
The process in which every state pass through by the system is an equilibrium state is known as reversible process

- In reversible process the initial state must be restored again and again.
- In reversible process, the system must pass though the forward path when it is reversed.
- Graphically such process is represented by a continuous curve in a P-V diagram

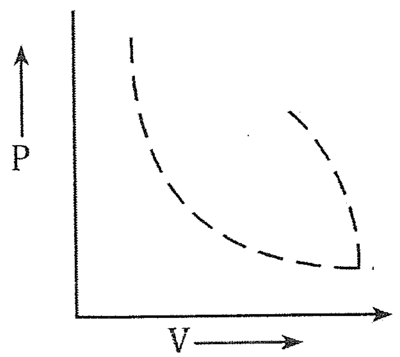


Characteristics of reversible process

1. There is no reversible process exist in the real nature
2. All the spontaneous process are not considered a a reversible process.
3. The efficiency of a reversible process is always 100%, so it is considered a quality standard to measure other practical devices.
4. If we apply the reversible process in a power producing device then, it will produce maximum work by consuming less amount of heat, again, if we apply the reversible process on power consuming devices then such devices will consume minimum amount of work to produce maximum effect.
   For eg: frictionless motion, magnetic effect, expansion & compression of spring.

Irreversible process:
The process in which every state passed through by the system, when it underfoing a certain process is not in an equilibrium condition is known as irreversible process.

- In this process the forward path & backward path are non-identical with each other.
- Graphically, such process is represented by a discontinuous curve in a P-V diagram.



Characteristics of an irreversible process.

1. All the real process exist in the practical nature is considered as an irreversible process.
2. If we develop an analyze the graph in between any thermodynamic property, the forward path and the backward path may vary due to change in thermodynamic properties.
3. In every irreversible process, when the system actuate at first, the movement of various mechanism creates the friction which finally result the low performance measure of an irreversible process.
4. The standard of any mechanism in the irreversible process is maintained & analyzed with reference to the performance measure of a reversible process. For eg: Mixing process, combustion process piston movement during an internal combustion engine.

Some common thermodynamic properties.

Pressure (P): It is defined as force acting per unit area Mathematically,

$$P = \frac{F}{A} = \frac{N}{m^2} = N/m^2$$

$1N/m^2 = 1Pa$

$1KPa = 1 \times 10^3 N/m^2$

$1bar = 1 \times 10^5 N/m^2$

$1MPa = 1 \times 10^6 N/m^2$

$1atm = 1.015 \times 10^5 N/m^2$

$= 101.15Kpa$

$= 101.15 \times 10^3 N/m^2$

There are 3 types of pressure

1. Atmospheric pressure (Patm)
2. Gauge pressure (P gauge)
3. Absolute pressure (P abs)

1. Atmospheric pressure:

The pressure excreted by the atmospheric air in a certain area of any object is known as atmospheric pressure.

- It is measured with the help of barometer
- The displacement of height of a mercury inside the barometer gives the reading of atmospheric pressure
  Mathematically,
  (P) = Sgz
  Where,
  S= density of mercury (13600 kg/m^3)

g = acceleration due to gravity (9.8 m/s^2)

z = altitude (760mm) = $\frac{760}{1000}m$

Patm = Sgz baro

$= 13,600 \times 9.8 \times \frac{760}{1000}m$

Patm = 101.29 x 10^3 N/m^2

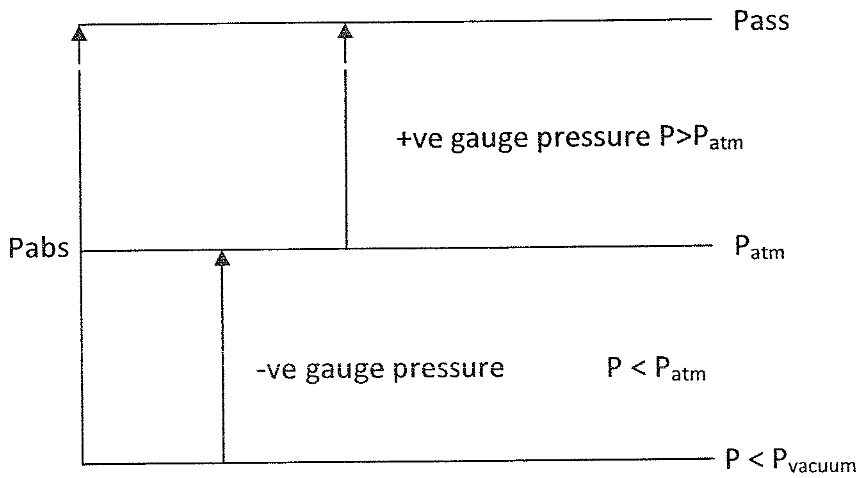
Patm = 101.29 KPa

2. **Gauge pressure (Pgauge):**
   - The pressure which is measured with reference to the atmospheric pressure is known as gauge pressure.
   - It is measured with the help of U- to be manometer
   - If the pressure is more than the atmospheric pressure then it is known as positive gauge pressure.
   - If the pressure is less than the atmospheric pressure then it is known as negative gauge pressure

     Mathematically, the gauge pressure is given by,

     P_{gauge} = sgz mano

3. Absolute pressure (P_{abs})
   - The pressure which is measured with reference to the vacuum pressure is known as absolute pressure.
   - In vacuum pressure, the resulting pressure is perfectly zero.



- In case of +ve gauge pressure, the absolute pressure is given as,

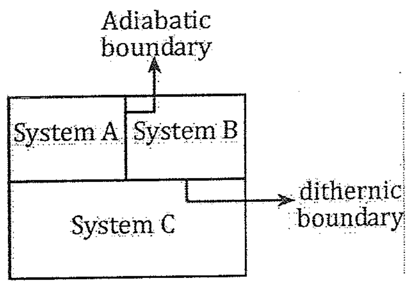
  $(P_{abs}) = P_{atm} + P_{gauge}$
- If the gauge pressure is –ve then, the absolute pressure is given as,

  $(P_{abs}) = P_{atm} - P_{gauge}$
- Negative gauge pressure is also expressed in ferms of vacuum pressure

Zeroth law of thermodynamics

When the two system is separately thermal equilibrium with the third system then all of the systems are in the condition of thermal equilibrium with each other.

Verification of zeroth law



- Let us consider a rigid compartment which consists of three system separated by their own boundary.
- For this instant the system is isolated from the surrounding. In the figure, system A & system B is separated by the adiabatic boundary where as system a & C, B & C individually connected with the dithermic boundary
- System A & B is not having the interaction of heat due to the presence of adiabatic boundary where as system A & B separately can perform the interaction of heat with the system C due to the presence of dithermic boundary.
- If we replace the adiabatic boundary with the dithermic boundary in between system A & B then no further change will occur in terms of temperature which ultimately verify zeroth low thermodynamic
- Finally, we can write
  $T_A = T_B = T_C$
  Specific volume (v)
  $\therefore \frac{V}{m} = \frac{m^3}{kg} = m^3/kg = v$

Microscopic approach and Macroscopic approach

- In thermodynamics all of the system and their properties can be studied or analyzed with the help of a certain approach
- The commonly used approach are:
  1. Microscopic approach
  2. Macroscopic approach
- At first these two approach is similar for their output but the presence of unique technique which is possessed by these two approach made them different.
- In this section we are going to find out the difference in between them

Microscopic Approach/statistical approach	Macrosopic Approach/Calssical Approach
1. In this approach the structure of matter is considered.	1. In this approach the structure of matter is not considered.
2. If we have to apply this approach to measure certain physical	2. If we have to apply this approach to measure certain physical quantities it

quantities it involves large number of variable.	involves less number of variable.
3. By using this approach, to measure a pressure of a gas which is kept inside the cylinder then it must consider the every molecules and their behavior. And it also include the pressure exerted by every gas molecule to the cross sectional area of cylinder wall	3. By using this approach, to measure the pressure of gas inside the cylinder then it directly measure the reading of standard gauge which is mounted above the cylinder.
4. It is more accurate process.	4. It is less accurate process
5. It is time consuming process which makes difficulty in calculation.	5. If is time saving process which brings simplicity in calculation.
6. It is used by scientist.	6. If is used by an engineers.

Formula numerical.

1. $v = \frac{v}{m}$

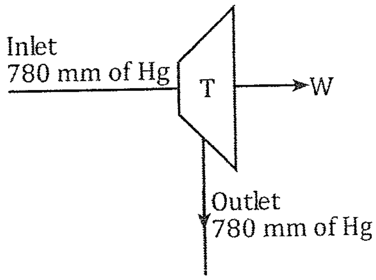
2. P = Sgz

3. If gauge pressure +ve $P_{abs} = P_{atm} + P_{gauge}$ If gauge pressure –ve $P_{abs} = P_{atm} - P_{gauge}$

Numerical

1. A steam turbine is designed to give the **power output of 10 kw**. In the inlet section the steam is maintained at the gauge reading of 780 mm of Hg, where the outlet is maintained at a vacuum which reads 720 mm of Hg. Determine the inlet and the outlet pressure in absolute form with the mechanism diagram.

Given data:
Inlet gauge reading (P_{gauge}) = 780 mm of Hg



Outlet vaccum reading
Pvacuum = 720 mm of Hg
i) (P_{abs}) inlet
ii) (P_{abs}) outlet
* P_{gauge} = 780 mm of Hg
= Sgz
$= 13,600 \times 9.8 \times \frac{780}{1000}$

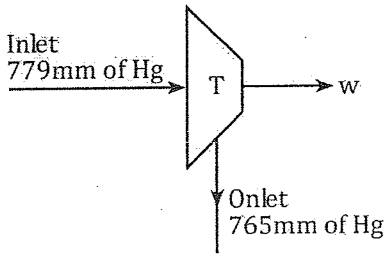
$\therefore P_{gauge} = 103958.4$ Pa
$\therefore (P_{abs})$ inlet $= P_{atm} + P_{gauge}$
$= 101.29 \times 10^3$ Pa + 103958.4 Pa
= 205248.4 Pa

* Pvaccum = Sgz

$$=13,600 \times 9.8 * \frac{780}{1000}$$

$P_{vaccum} = 95961.6$ Pa
* (P_{abs})out $= P_{atm} - P_{vaccum}$
$=101.29 \times 10^3$ Pa − 95961.6
= 5328.4 Pa

2. A steam turbine is developed in which the inlet of steam is maintained at 770 mm of Hg. After the production of work the steam is supplied to the condenser which is maintained at the vaccum which reads 720 mm of Hg. The barometer reading is 765 mm of Hg. Express the inlet & the outlet pressure in absolute form & in kilo pascal.
Given data:
Inlet gauge reading
$P_{gauge} = 770$ mm of Hg



Outlet gauge reading
$P_{gauge} = 720$mm of Hg

To find:
i) (P_{abs}) inlet
P_{gauge} = Sgz

$$= 13,600 \times 9.8 \times \frac{770}{1000}$$

= 102625.6 Pa
Patm = Sgz

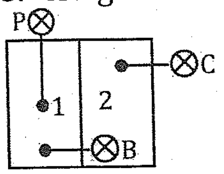
$$= 13,600 \times 9.8 \times \frac{765}{1000}$$

= 101959.2 Pa
Pvaccum = Sgz

$$=13,600 \times 9.8 \times \frac{720}{1000}$$

=95961.6 Pa
(P_{abs}) inlet $= P_{atm} + P_{gauge}$
=(101959.2 + 102625.6) Pa
= 204584.8 Pa
= 204.5848 kpa
(P_{abs})outlet $= P_{atm} - P_{vaccum}$
= (101959.2 - 9(P_{abs})outlet $= P_{atm} - P_{vaccum}$
= (101959.2 − 95961.6)Pa
=5998.2 pa = 5.9982 kpa

3. A rigid container have two compartments 1,2 as shown in figure.



The pressure gauge A reads (two bar) pressure gauge c reads 4 bar. Determine

i) The absolute pressure of compartment 1 } 2.

ii) Determine the gauge reading of B. (density of mercury S_{Hg} = 13600 kg/m^3, g = 9.8m/s^2 { Zbarrmeter = 760mm of Hg)

Given data:

$P_A = 2$ bar $= 2 \times 10^5$ Pa

$P_C = 4$ bar $= 4 \times 10^5$ Pa

$S_{Hg} = 13600$kg/m^3

9 = 9.8 m/s^2

Zbarometer = 760 mm of Hg

To find,

i) $(P_{abs})_1$, $(P_{abs})_2$

ii) $P_B = ?$

$(P_{abs})_1 = P_A + P_{atm}$

OR

$(P_{abs}) = P_B + (P_{abs})_2$

$$= 2 \times 10^5 + \frac{760}{1000} \times 13600 \times 9.8$$

$= 2 \times 10^5 + 101.13 \times 103$

$= 301130$ Pa

$= 301.13$ KPa

$(Pabs)2 = PC + Patm$

$= 4 \times 10^5 + 1010.13 \times 10^3$

$= 501130$ Pa

$= 501.13$ KPa

Now,

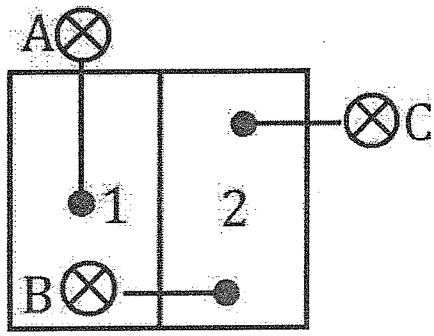
$(P_{abs})_1 = P_B + (P_{abs})_2$

$\therefore PB = (P_{abs})_1 - (P_{abs})_2$

$= (301.13 - 501.13)$ KPa

$= - 200$KPa

4. A rigid container have two compartments 1 & 2 as shown in figure
The pressure gauge A reads 780 mm of Hg and the pressure gauge B reads 778 mm of Hg. Determine the absolute pressure of compartment no. 1 & 2 in kpa & determine the gauge reading of C. (take S_{Hg} = 13600kg/m³ g = 9.8m/s², Zbarometer = 760 mm of Hg)



Given

$P_A = 780 \text{ mm of Hg} = Sgz = 13600 \times 9.8 \times \frac{780}{1000}$ Pa

= 103958.4 Pa

$P_B = 778 \text{ mm of Hg} = Sgz = 13600 \times 9.8 \times \frac{778}{1000}$ Pa

= 103691.84 Pa

P_{atm} = 760 mm of Hg = 101.13 x 103 Pa

S_{Hg} = 13600

G = 9.8 m/s2

Z_{baro} = 760 mm of Hg

$(P_{abs})_2 = P_{atm} + P_C$

$(P_{abs})_1 = P_B + (P_{abs})_1$

$(P_{abs})_1 = P_A + P_{atm}$

= (103958.4 + 101.13 x 103)Pa

205251.2Pa

205.2512 kpa

$(P_{abs})_2 = P_B + (P_{abs})_1$

$(P_{abs})_2$ = 103691.84 Pa + 205251.2 Pa

= 308943.04Pa

=308.94304kpa

$(Pabs)_2$ = Patm + PC

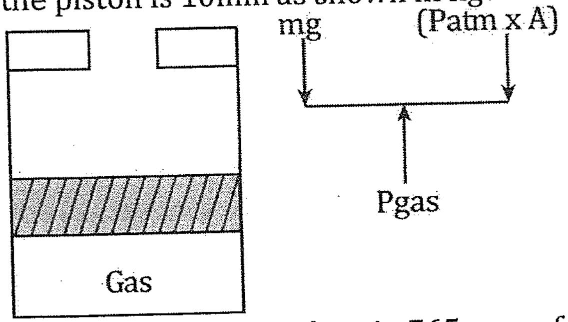
∴ PC = (Pabs)2 – Patm

= (308942.04 – 101.13 x 103)Pa

= 207813.04Pa

= 207.81304kpa

5. A piston cylinder assembly is having the mass of piston is 5kg and the diameter of the piston is 10mm as shown in figure.



If the barometer reading is 765 mm of Hg, determine the gas pressure to hold the pistion.
(Take S_{Hg} = 13600 kg/m^3, g = 9.8m/s^2)

Given data
Mass of piston (m) = 5kg
Diameter of the piston (d) = 10mm = 0.01m
Z_{baro} = 765 mm of Hg
$P_{atm} = S_{gzbaro}$

$$= 13600 \times 9.8 \times \frac{780}{1000} \text{ Pa}$$

$$= 101959.2 \text{ Pa}$$

Now
Area of piston (A) = $\frac{\pi}{4}\times(0.01)^2$
$= 7.857 \times 10^{-5}\ m^2$

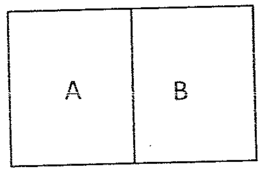
We have,
Pgas x A = mg + Patm X a
OR

$$\text{Pgas} = \frac{mg}{A} + \text{Patm}$$

$$= \frac{5\times9.8}{7.857\times10^{-5}} + 101959.2Pa$$

= 725606.90Pa
= 725.6069kpa

6. A rigid container have two compartments A & B container A has 2m^3 of air with the specific volume of 0.85m^3/kg and compartments B has 1.5m^3 of air with 0.5m^3/kg of specific volume. If the separation meter between A and B is broken then determine the final specific volume.



Solution:
Given data
V_A = 2m^3

Specific volume of A(ϑ_A) = $0.85 m^3/kg$
$V_B = 1.5 m^3$
Specific volume of B(ϑ_B) = $0.5 m^3/kg$
To find
Resultant specific volume (ϑ) = ?
OR

$$\vartheta = \frac{V}{m}$$
$$= \frac{V_A + V_B}{m_A + m_B}$$

To find mass:

OR, $\vartheta = \frac{V}{m}$

$$\vartheta A = \frac{V_A}{m_A}$$

$\therefore$ mA = $\frac{V_A}{\vartheta_A}$ = 2.3kg

$\therefore$ mB = $\frac{V_B}{\vartheta_B} = \frac{1.5}{0.5}$ = 3kg

OR. $\vartheta = \frac{V_A + V_B}{m_A + m_B}$

$$= \frac{2+1.5}{2.3+3}$$
$$= 0.66037 m^3/kg$$

7. A rigid container have two compartments A & B as shown in figure. Initially the volume of each compartments are equal with a total volume of $1m^3$. The specific volume of compartment a has 100 m^3/kg of specific volume and B has $50m^3/kg$. If the member is moved such that x is one fourth of the entire length. Determine the specific volume of compartment A & B in this condition.

Solution
Givne data:
$V = 1m^3$
$V_A = 0.5m^3$
$V_B = 0.5m^3$
$\vartheta_A = 100\ m^3/kg$
$\vartheta_B = 50m^3/kg$
Now,

$$m_A = \frac{V_A}{\vartheta_A} = \frac{0.5}{100} = 0.005kg$$

$$m_B = \frac{V_B}{\vartheta_B} = \frac{0.5}{100} = 0.01kg$$

From the question

x is $\frac{1}{4}$L

OR, V = A x L

From this condition

$V_A = \frac{1}{4} \times 1 = 0.25 m_3$

VB = 0.75m^3

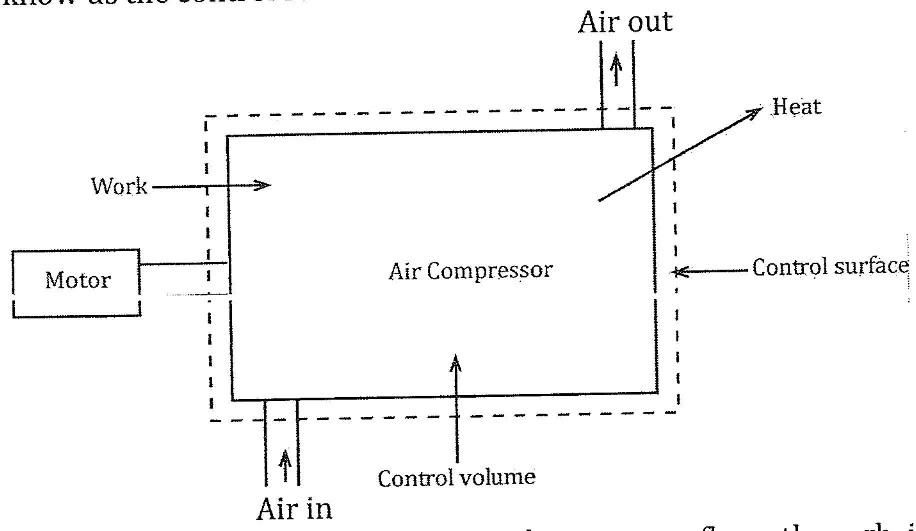
Again,

$\vartheta_A = \frac{V_A}{mA}$ $\qquad$ $\vartheta_B = \frac{V_B}{m_B}$

$= \frac{0.25}{0.005}$ $\qquad$ $= \frac{0.75}{0.01}$

$= 50 m^3 / kg$ $\qquad$ $= 75 m^3 / kg$

For thermodynamic analysis of an open system, such as an air compressor as shown in fig1, attention is for used on a certain volume in space surrounding the compressor, know as the control surface. Matter as well as energy closes the control surface.



A closed system is a system closed to matter flow, through its volume can change against a flexible boundary. When these is matter flow, then the system is considered to be a volume of fixed identify, the control volume. Thus these is no difference between an open system and a control volume.

Thermo statics

- The science of thermodynamics deals with systems existing in thermodynamic equilibrium states which are specified by properties.
- Infinitely slow quassi – static processes executed by systems are only, meaningful in thermodynamic plots. The name thermodynamics is thus said to be a misnomer, since it does not deal with the dynamic of heat, which is non quassi – static.

- The name 'thermo statics' then seems to be more appropriate. However, most of the real processes are dynamic and non quassi – static, although the initial and final states of the system might be in equilibrium.
- Such process can be successfully dealt with by the subject. Hence the term 'thermodynamics' is not appropriate.

Density (δ) is the mass per unit volume of a substance, which has been discussed curlier and is given in kg/m^3.

$$\int = \frac{m}{\vartheta}$$

In addition to m^3, another commonly used unit of volume is the litre (l). $1l = 1 \times 10^{-3} m^3$.

Macroscopic and Microscopic Viewpoint.

- These are two points of view from which the behavior of matter can be studied; the macroscopic and the microscopic.
- In the macroscopic approach, a certain quantity of matter is considered, without the events occurring at the molecular level being taken into account.
- From the microscopic point of view, matter is composed of myriads of molecules. If it is a gas, each molecule at a given instant has a certain position, velocity, and energy, and for each molecule at a given instant has a certain position, and these change very frequently as a result of collisions.

Macroscopic Approach	*Microscopic Approach*
i) In this approach certain quantity of matter is considered without considering the molecules.	i) In this approach, the events occurring at the molecular level is considered.
ii) If we have to measure the pressure of a gas, which is kept inside the cylinder, then from this approach, we directly take the value of standard pressure gauge.	ii) If we have to measure the pressure of a gas, which is kept inside the cylinder, then from this approach, every molecules and their properties such as position, velocity and energy is considered, which gives the final reading of pressure.
iii) This approach involves less number of variables.	iii) This approach involves large number of variables.
iv) Calculation is relatively faster and simpler in this approach.	iv) Calculation is time consuming and tedious in microscopic approach.
v) This approach is usually used in applied or classical thermodynamics.	v) This approach is usually used in statistical thermodynamics.
vi) It is used by an engineers.	vi) It is used by scientist.

Chapter 2

Heat & Work

Heat and work both are the forms of energy.

Energy:
- It is the capacity to do certain work
- It is always measured in joule
- In thermodynamics energy is classified into two types.
  a) Stored energy
  b) Transient energy

a) Stored energy:
- The form of energy which is always stored in certain object and cannot cross the boundary of that object is known as stored energy.
- Stored energy is always remain itself in that object, no any external factors can affect it.
- Stored energy is considered as a inferior form of energy.
  Eg. P.E, K.E, Internal energy

b) Transient Energy:
- The forms of energy which can cross the boundary of the system is known as transient energy.
- Transient energy is considered as a superior form of energy.
- It uses stored energy of certain object to form the transient forms of energy.
  Eg. Heat and work

Potential energy (P.E): Produced due to its position
$\therefore$ PE = mgz
m = mass
g = gravity
z = elevation or altitude

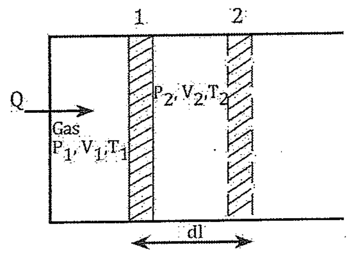
Kinetic energy (K.E): produced due to its motion
$\therefore \text{K.E} = \frac{1}{2}mv^2$
m = mass
v = velocity
internal energy (U) = K.E + P.E
Total energy (F) = K.E + P.E + U

$$F = \frac{1}{2}mv^2 + mgz + U$$

If the total energy $F = \frac{1}{2}mv^2 + mgz + U$, then the required equation for the specific energy(e) $= \frac{1}{2}v^2 + gz + U$

$$= \frac{1}{2}v^2 + gz + U$$

Work done in the Quassi-Static process
OR
Work done in the reversible process



- Let us consider a piston cylinder assembly which contains a gas.
- Initially, the gas is in an equilibrium condition having the thermodynamic properties(P1, V1, T1).
- When heat is supplied to the piston, due to the expansion process the piston will move a small distance and after sometime the system will attain another equilibrium condition having the thermodynamic properties (P2, V2, T2).
- When the piston will move a small distanced, the force acting on the piston is given as F = P.A - 1
  Where P = pressure acted by the gas to the piston.
  A = Area of cross – section of piston
  - Due to the movement of piston from 1 to 2, the work done by the gas is given as
    W = F.dl
    =P.A dl
    = P.dVw = P.dV - 2

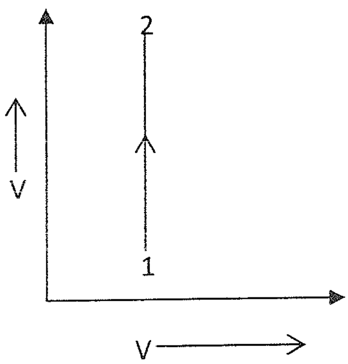
  Equation 2 represents that the work done by the gas in terms of change in volume.
  Equation 2 can be written as

  $$\int \omega = \int_{v_1}^{v_2} P.dV$$

  This is the required equation of work done during a reversible process.

Work done during a different types of reversible process

a) Insochoric process: The process is which the volume remains constant is known as isochoric process. The graphical representation of an isochoric process in P.V diagram is



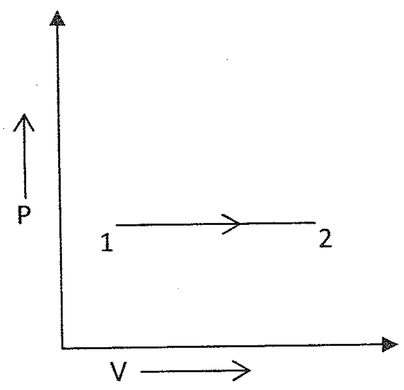
- The work done during an isochoric process is known as

$$\delta w = \int_{v_1}^{v_2} P.dV$$

Dv = 0

$$\therefore \delta w = 0$$

b) Isobaric process: The process in which pressure remains constant is known as isobaric process. The graphical representation of an isobaric process in P-V diagram is



- The work done during an isobaric process is

$$\delta w = \int_{v_1}^{v_2} P.dV$$

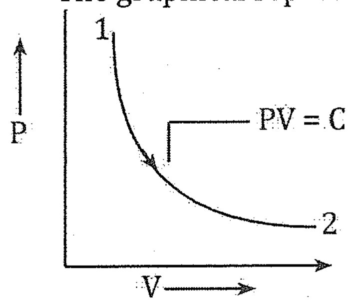
$$= P\,\delta \int_{v_1}^{v_2} dV$$

$$\delta w = \int_{v_1}^{v_2} P.dV$$

$$w = P\,[v_2 - v_1]$$

c) Isothermal process: The process in which the temperature remains constant is called isothermal process.

- The graphical representation of an isothermal process is



- The work done during an isothermal process is given as

$$\therefore \delta w = \int_{v_1}^{v_2} P.dV$$

As we know that,

PV = C

$P_1V_1 = P_2V_2 = PV$

$\therefore$ The intermediate pressure P is given as

$$P = \frac{P_1V_1}{V}$$

Now,

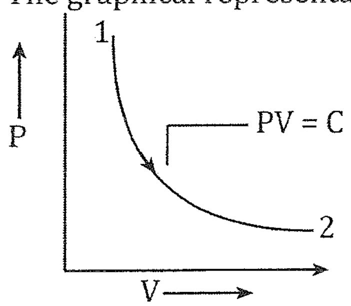
$$\delta w = \int_{v_1}^{v_2} \frac{P_1V_1}{V}.dv$$

$$= P_1V_1 \int_{v_1}^{v_2} \frac{dv}{v}$$

$$w = P_1V_1 \ln\left[\frac{v_2}{v_1}\right]$$

d) Polytropic process: The process which follow the gas equation PV^n = constant is known as polytropic process. Where n = polytropic index.

- This is the process which can represent all the reversible process.
- The graphical representation of a polytropic process is



- The work done during a polytropic process is given as

$$\therefore \delta w = \int_{v_1}^{v_2} P.dV$$

As we know that

$PV^n = C$

$P_1V_1^n = P_2V_2^n$

$$\therefore P = \frac{P_1V_1^n}{v_n}$$

Now,

$$\therefore \delta w = \int_{v_1}^{v_2} P.dV$$

$$= \int_{v_1}^{v_2} \frac{P_1V_1^n}{v^n}.dv$$

$$= P_1V_1^n \int_{v_1}^{v_2} \frac{1}{v^n}.dv$$

$$= P_1V_1^n \int_{v_1}^{v_2} v^{-n}.dv$$

$$= P_1V_1^n \left[\frac{V^{-n+1}}{-n+1}\right]_{v_1}^{v_2}$$

$$= P_1V_1^n \left[\frac{V^{1-n}}{n-1}\right]_{v_1}^{v_2}$$

$$= \frac{P_1V_1n}{1-n}[v_2^{1-n} - v_1^{1-n}]$$

$$= \frac{P_1V_1.v_2^1.v_2^{-n} - P_1V_1^n.v_1^1.v_1^{-n}}{1-n}$$

$$= \frac{P_1V_1.v_2^1.v_2^{-n} - P_1V_1^n.v_1^1.v_1^{-n}}{1-n}$$

$$= \frac{P_2V_2^n.v_2.v_2^{-n} - P_1V_1^n.v_1.v_1^{-n}}{1-n}$$

$$w = \frac{P_2V_2 - P_1V_1}{1-n}$$

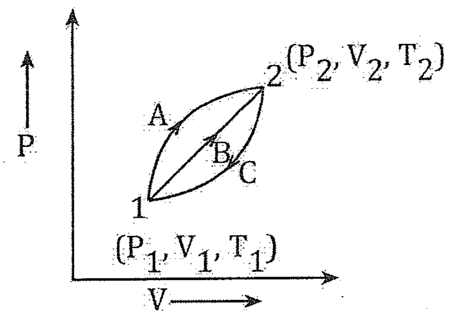
This is the required equation of work done during a polytropic process

Path & Point function:

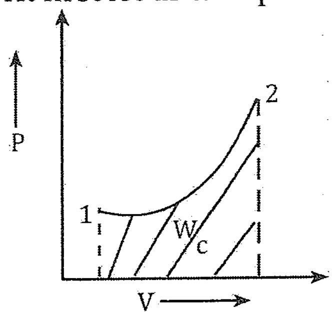
Point function: Any variable which depends upoun the state followed by a system is known as point function. All the thermodynamic property is considered as a point function. (Point function represented by d.)

Path function: Any variable which is dependent upon the path followed by them is known as a path function. For eg: Heat and work is considered as a path function. (All the path function is denoted by δ)

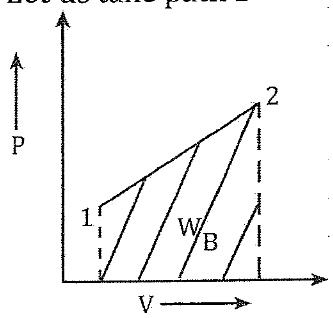
Why is the work done (w) is not considered as a thermodynamic property?



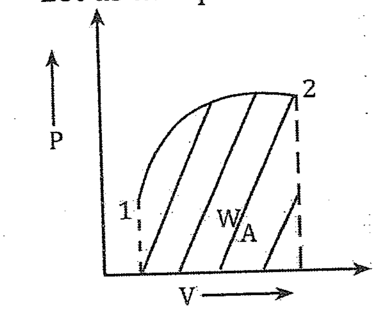
- Let us consider a cycle which consists of three different paths A, B & C.
- The system undergoing a cyclic process which changes its state from 1 to 2.
- The state 1 having the properties (P_1,V_1,T_1) & state 2 having the properties (P_2,V_2,T_2)
- As we know that the work done by any process is the area covered by that particular process under the P-V diagram
- If we analyze the path then we can evaluate the work done by them.
- At first let us take path C.



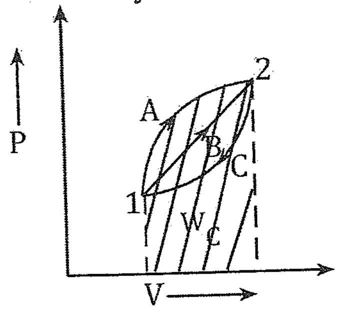
- Let us take path B



- Let us take path A



Combinely



From the figure above, we can calculate the work done by each process, we get

$W_A > W_B > W_C$

$\therefore$ So, work done (w) is a path function.

Formulas for numerical:

1. $\therefore w = \int_{v_1}^{v_2} P.dv$

2. Dependent upon the process.

   Use the formula

3. Language – cycle – process formula.

1. A piston cylinder assembly which contains a gas is in the expansion process where the volume changes from $0.5m^3$ to $0.8m^3$, in which the pressure change is given as P = $(v^2 + 6v)$ bar. Calculate the work done by the gas in kilo joule.

   Solution

   Given data

   $V_1 = 0.5m^3$

   $V_2 = 0.8m^3$

   $P = (v^2 + 6v)$ bar

   As we know that

   $$w = \int_{v_1}^{v_2} P.dv$$

$$= \int_{0.5}^{0.8} (v^2 + 6v).dv$$

$$= \left[\frac{v^3}{3}\right]_{0.5}^{0.8} + 6\left[\frac{v^2}{2}\right]_{0.5}^{0.8}$$

$$= \left[\frac{(0.8)^3 - (0.5)^3}{}3\right] + 6\left[\frac{(0.8)^2 - (0.5)^2}{2}\right]$$

= 0.1755 + 1.17

=1.3455 x 10^5 J

w = 134.55 x 10^3 J

w = 134.55 KJ

2. A piston cylinder assembly contains a gas in which volume changes from $0.8m^3$ to $1.2m^3$ in which the pressure change is $P = 1500\left(\frac{v}{100}+1\right)$ Pg where Pressure P is in pascal. Determine the initial pressure & the final pressure

Given data

$V_1 = 0.8m^3$

$V_2 = 1.2m^3$

The pressure change is given by

$$P = 1500\left(\frac{V}{100}+1\right)Pa$$

To find

The initial Y final pressure

Work done (w) = ?

To find the initial pressure P,

$$\therefore P = 1500\left(\frac{V}{100}+1\right)$$

$$P_1 = 1500\left(\frac{V}{100}+1\right)$$

$$= 1500\left(\frac{0.8}{100}+1\right)$$

= 1512 Pa

$$P_2 = 1500\left(\frac{V}{100}+1\right)$$

$$=1500\left(\frac{1.2}{100}+1\right)$$

= 1518 Pa.

To find work done

$$w = \int_{v_1}^{v_2} P.dv$$

$$= \int_{v_1}^{v_2} 1500\left(\frac{v}{100}+1\right)dv$$

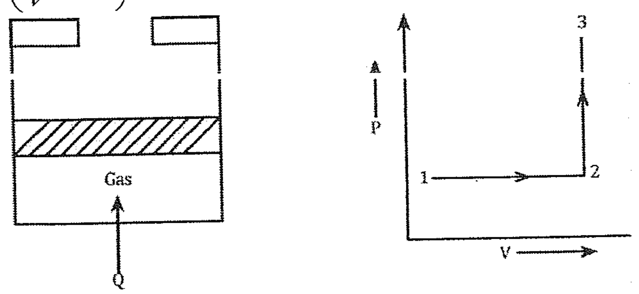
$$=1500\int_{v_1}^{v_2}\left(\frac{v+100}{100}\right)dv$$

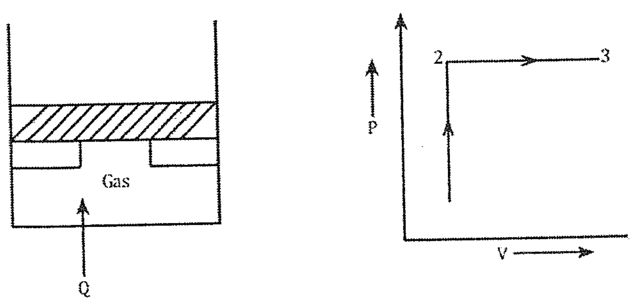
$$= \left\{\left[\frac{v_2}{2}\right]_{0.8}^{1.2} + 100\left[\frac{v}{1}\right]_{0.8}^{1.2}\right\}$$

$$= 15\left\{\left[\frac{(1.2)2-(0.8)^2}{2}\right]+100\left[\frac{1.2-0.8}{1}\right]\right\}$$

$= 15\ [0.4 + 100 \times 0.4$
$= 606J$
$= 0.606\ KJ.$

3. A gas is kept on the cylinder where the initial volume is $1.2m^3$ when heat is supplied to the gas the volume changes upto $106.m^3$, where the pressure changes is $P = \left(\frac{2}{v}+1.5\right)$ bar. Determine the initial & the final pressure & the work done.





4. Oxygen is kept in the cylinder in which the initial pressure is 0.1 mega pascal and 250C as shown in fig no. (1)

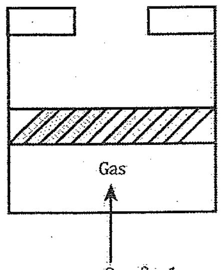
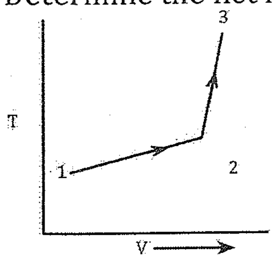


fig. 1

Heat is supplied to the gas upto when the piston hits the stops. There is an additional amount of heat transfer in which the final pressure is 8 mega pascal and temperature is 900^0c. Determine the work down by the gas in KJ. (take R= 0.287 KJ/Kg K, mass = 0.1kg)

Determine the net heat interaction TV diagram banaure



Given data

1^{st} data:

$P_1 = 0.1$ Mpa

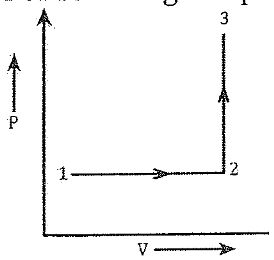
$= 0.1 \times 10^6$pa

$T_1 = 250$c

$= (25 + 273)$k

298k

Form ideal gas equation



$P_1V_1 = mRT$

$$V_1 = \frac{mRT_1}{P_1}$$

$$= \frac{0.1 \times 0.287 \times 10^3 \times 298}{0.1 \times 10^6}$$

$V_1 = 0.085m^3$

State 2:

When the piston hits the stops from the P – v diagram $P_1 = P_2 = 0.1 \times 10^6$ Pa
Initial pressure = 5 mega pascal
Final pressure = 8 mega pascal
Final temp. = 800^0c
V2 = V3 (to calculate)
State 3:
There is an additional amount of heat transfer, which results
$P_3 = 8$ mpa $= 8 \times 10^6$ Pa
$T_3 = (900 + 273) = 1173$ k
$P_3V_3 = mRT_3$

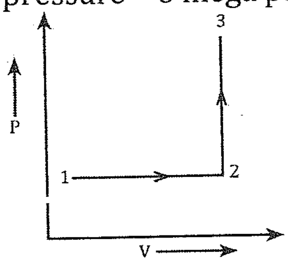
$$V_3 = \frac{mRT_3}{P_3}$$

$$= \frac{0.1 \times 0.287 \times 10^3 \times 1173}{8 \times 10^6}$$

$= 0.0042$ m^3

In this condition the final volume is smaller than the initial volume which cannot exist in real nature. This question is wrong

5. Oxygen is kept .... same data change; initial pressure = 5 mega pascal final pressure = 8 mega pascal and final temp. = 800^0c.



Given data

$P_1 = 5$ Mpa

$= 5 \times 10^6$Pa

$T_1 = 25^0c = 298k$

From ideal gas equation,

$P_1V_1 = mRT_1$

$$V_1 = \frac{mRT_1}{P_1}$$

$$V_1 = \frac{0.1 \times 0.287 \times 10^3 \times 298}{5 \times 10^6}$$

$= 0.00171$m^3

State 2:

When the piston hits the stops from P – V diagram
$P_1 = P_2 = 5 \times 10^6$ mpa.
$V_2 = V_3$ (to calculate)

State 3:

There is an additional amount of heat transfer, which results
$P_3 = 8Mpa = 8 \times 10^6$ pa
$T_3 = 800^0c = 800 + 273 = 1073k$

$P_3V_3 = mRT_3$

$$V_3 = \frac{mRT_3}{P_3}$$

$$V_3 = \frac{0.1 \times 0.287 \times 10^3 \times 1073}{8 \times 10^6}$$

$$V_3 = 0.0038m^3$$

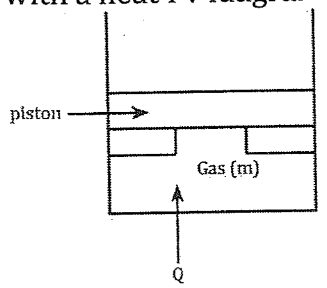
To find work done
$W = w_{1-2} + w_{2-3}$

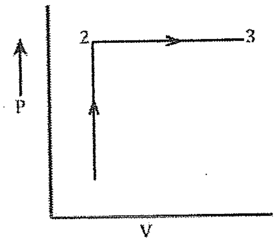
W_{1-2} = Isobaric process
$= P_1 (V_2 - V_1)$
$= 5 \times 10^6 (0.0038 - 0.0017)$
$= 10500J$

W_{2-3} = Isochoric process
$= 0$

$\therefore W = W_{1-2} + W_{2-3}$
$= (10500 + 0)J$
$= 10500J$
$10.5kJ$

6. A piston cylinder assembly containing a gas of 0.1kg. The initial temperature & pressure of the gas is 25^0c, 3 megapascal. The heat is supplied to the gas until the lift of piston from the stop. There is an additional heat transfer to the gas upto the system reaches 5.5 mega pascal and 800^0c. Caluculate the work done by the system with a neat Pv idagra.





Given data:

State 1:

m = 0.1kg

P_1 = 3Mpa = 3 x 10^6 pa

T_1 = 25 + 273 = 298k

$\therefore$ by using the gas equation

PV = mRT_1

OR

$P_1V_1 = mRT_1$

OR

$$V_1 = \frac{mRT_1}{P_1}$$

$$= \frac{0.1 \times 0.287 \times 10^3 \times 298}{3 \times 10^6}$$

$\therefore$ V_1 = 2.85 x 10^{-3}m^3

State 2:
$P_2 = P_3$ (from the diagram)
$V_2 = V_1$

State 3:
P_3 = 5.5Mpa = 5.5 x 106pa
T_3 = 800 + 273 x 1073k
By using ideal gas equation
$P_3V_3 = mRT_3$

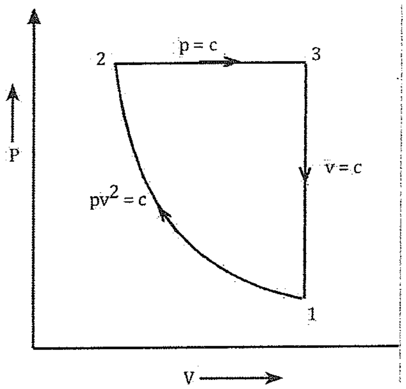
$$V_3 = \frac{mRT_3}{P_3}$$

$$= \frac{0.1 \times 0.287 \times 10^3 \times 1073}{5.5 \times 10^6}$$
$= 5.599 \times 10^{-3} m^3$

To find the work done

$\therefore$ Wnet = $W_{1-2} + W_{2-3}$
W_{1-2} = Isocheric Process; W = 0
W_{2-3} = isobaric process
$= P(V_3 - V_2)$
W_{2-3} = 5.5 x 106 (5.5 x 10^{-3} – 2.85 x 10^{-3})
= 15.1 x 103j
=15.1kj

7. Air undergoes the three following process in a piston cylinder assembly. Process 1-2: compression with PV^2 = constant with the initial pressure is 200kpa and volume changes from an initial volume of $0.6m^3$ to $0.2m^3$. Process 2-3: constant pressure process. Process 3-1: constant volume process.
Given data
Process 1 – 2: Polytropic process
$PV^2 = C$



Process 2-3: P = C Isobaric process
Process 3-1: V =C Isochoric process.

To find the work done
$$W_{1-2} = \frac{P_2V_2 - P_1V_1}{1-n}$$
When n = 2
$P_1 = 200 \times 10^3$pa
$V_1 = 0.6m^3$
$V_2 = 0.2m^3$
Now,
$P_2V_2n = P_1V_1n$
$P_2 (0.1)^2 = (200 \times 10^3) \times (0.6)^2$
$$P_2 = \frac{200 \times 10^3 \times (0.6)^2}{(0.2)^2}$$
$= 10 \times 10^5$pa
$$W_{1-2} = \frac{18 \times 10^5 \times 0.2 - 200 \times 10^3 \times 0.6}{1-2}$$
$\therefore W_{1-2}$ = -2.4 x 105J
=-240kJ

Process 2-3: isobaric process
$W_{2-3} = P(V_3 - V_2)$
$= 18 \times 10^5 (0.6 - 0.2)$
$W_{2-3} = 7.2 \times 10^5 J$
$= 720kJ$

W_{3-1}: Isochoric process
$\therefore W_{3-1} = 0$
$\therefore$ Net work done (Wnet) = $W_{1-2} + W_{2-3} + W_{3-1}$
$= -240 + 720 + 0$
Wnet = 480KJ

Note: sign convention for work
1. W is positive when work is done by the system.
2. W is negative when work is done on the system.

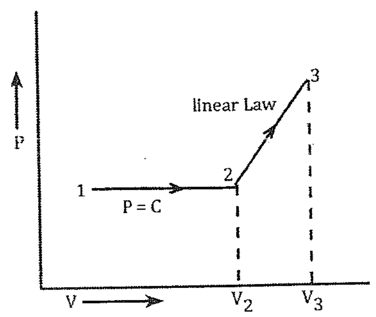
Sign convention for heat
1. Q is positive, when heat is gain by the system
2. Q is negative, when heat is lost by the system

Definition of work in thermodynamics.

In thermodynamics work is said to be done when there is the interaction between system & surrounding and the property changes other than the temperature difference in which energy travels by keeping the system mass constant.

8. A system which contains gas at the initial pressure of 150kpa, and volume of $2m^3$. The gas is allowed to an expansion process by keeping the pressure constant where the volume changesupto $4m^3$. Another process in which pressure and volume rising linearly upto 300kpa, $6m^3$. Determine the net work done by the combine process.

Solution
Given data
$P_1 = 150kpa$
$V_1 = 2m^3$
$= 150 \times 10^3 pa$
$V_2 = 4m^3$
$P = C$
$W = P(V_2 - V_1)$
$= 150 \times 10^3 (4 - 2)$
$= 300 \times 10^3 J$
$= 300KJ$



$\therefore W_{2-3}$ = linear law
$P_1 = P_2 = 150 \times 10^3 pa$
$V_3 = 6m^3$, $V_2 = 4m^3$
$P_3 = 300 \times 10^3$

$= 101559.36Pa + 105291.2Pa$
$= 206850.56Pa$
$= 206.85056kpa$

$\therefore (P_{abs})_{outlet} = P_{atm} - P_{vacuum}$
$= 101559.36Pa - 98627.2Pa$
$= 2932.16Pa$
$= 2.93216kpa$

$P_{atm} = 762mm$ of Hg
Sgz

$$= 13600 \times 9.8 \times \frac{762}{1000}$$
$= 101559.36Pa$

2. Given data:
$V_1 = 2m^3$
$V_2 = 4m^3$
The pressure change is given by

$$P = \left(3v^2 + \frac{1}{v}\right) kpa$$

To find;
* The work done by the gas
* Initial & the final pressure

To find the initial pressure P,

$$P = \left(3v^2 + \frac{1}{v}\right) kpa$$

$$P_1 = \left(3v_1^2 + \frac{1}{v_1}\right) kpa$$

$$= \left(3 \times (2)^2 + \frac{1}{2}\right) kpa$$

$$= \left(12 + \frac{1}{2}\right) kpa$$

$$= \left(\frac{25}{2} \times 10^3\right) pa$$

$= 12500pa$

$$P_2 = \left(3v_2^2 + \frac{1}{v_2}\right) kpa$$

$$= \left(3 \times (4)^2 + \frac{1}{4}\right) kpa$$

$\therefore W_{2-3} = \frac{1}{2}[P_1 + P_3] \times [V_3 - V_2]$

$= \frac{1}{2}[150 \times 10^3 + 300 \times 10^3] \times [6-4]$

$= \frac{1}{2} 450 \times 10^3 \times 2$

$= 450 \times 10^3 J$

$\therefore W_{2-3} = 450KJ$

$W = W_{1-2} + W_{2-3}$

$= 300 + 450$

$=750KJ$

1 A steam turbine is supplied with the steam at a gauge pressure of 790mm of Hg. After expansion in the turbine the steam is supplied to the condenser which is maintained at a vacuum which reads 740mm of Hg. Determine the absolute pressure at the inlet & out let of the turbine where the barometric reading is 762mm of Hg. ($S_{Hg} = 13600kg/m^3$, $g = 9.8m/s^2$)

2. A piston cylinder assembly containing a gas which undergoes an expansion process where the volume changes from $2m^3$ to $4m^3$ and the pressure is given by $P = \left(3v^2 + \frac{1}{v}\right)$kpa. Determine the work done by the gas in joule. Also find the initial & final pressure.

3. A system in which working medium is gas which undergoing an expansion process with $PV^{2.5} = C$, where the initial volume is $1.5m^3$ and final volume is $3m^3$. Another process which is combined with the first process is a constant pressure process and final is constant volume process. Determine the work done by the gas in given process. (initial pressure 150kpa)

1. Given data

Pgauge = 790mm of Hg

= sgZ

$= 13600 \times 9.8 \times \frac{790}{1000}$

$\therefore$ Pgauge = 105291.2Pa

Pvacuum = 740mm of Hg

= Sgz

$= 13600 \times 9.8 \times \frac{740}{1000}$

= 98627.2Pa

$\therefore$ (Pass)inlet = Patm + Pgauge

$= \left(3\times16+\frac{1}{4}\right)$kpa

$= \left(48+\frac{1}{4}\right)$kpa

$= 48.25$kpa

$= 48.25 \times 10^3$pa

$= 48250$pa

To find work done

$$w = \int_{v_1}^{v_2} P.dv$$

$$= \int_{v_1}^{v_2}\left(3v^2+\frac{1}{v}\right)dv$$

$$= \int_{v_1}^{v_2}3v^2dv + \int_{v_1}^{v_2}\frac{1}{v}dv$$

$$= 3\left[\frac{v^3}{3}\right]_2^4 + [\log v]_2^4$$

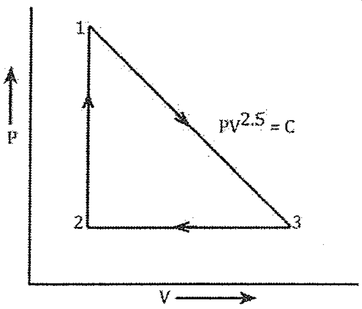
$=[4^3 - 2^3] + [un4 - un2]$

$= 56 + [1.386 - 0.693]$

$= 56 + 0.693$

$= 56.693$kJ

$= 56.693 \times 10^3$J



3. Given data

Process 1 – 2 polytropic process

$PV^{2.5} = C$

Process 2 – 3: P = C Isbaric process

Process 3-1: V = C Isochoric process

To find work done,

$$w_{1-2} = \frac{P_2V_2 - P_1V_1}{1-n}$$

Where, n = 2.5

$P_1 = 150$kpa $= 150 \times 10^3$

$V_1 = 1.5m^3$

$V_2 = 3m^3$

Now,

$P_2V_2^n = P_1V_1^n$

$P_2 x(3)^{2.5} = 150 \times 10^3 (1.5)^{2.5}$

$$P_2 = \frac{150\times10^3\times(1.5)^{2.5}}{(3)^{2.5}}$$

$= 2.65 \times 10^4$pa

Now,

$$W_{1-2} = \frac{2.65\times10^4\times3-150\times10^3\times1.5}{1-2.5}$$
$$= \frac{7.954\times10^4-2.25\times10^5}{1-2.5}$$
$$= 9.69 \times 10^4 J$$

Process 2-3: Isobaric process

$W_{2-3} = P(V_3 - V_2)$

$= 2.65 \times 10^4 [1.5 - 3]$

$= -3.97 \times 10^4 J$

-ve sign indicates that work is done on system

Now,

Process 3-1: Isochoric process

$\therefore w_{3-1}=0$

Net work done (wnet) = $w_{1-2} + w_{2-3} + w_{3-1}$

$=9.69 \times 10^4 + (-3.97 \times 10^4)$

$5.72 \times 10^4 J$

57.2kJ

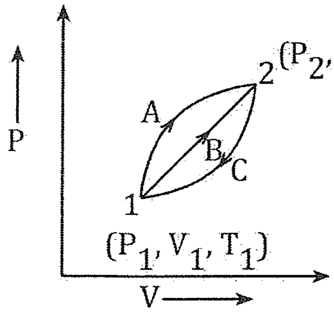
Chapter 3

First law of thermodynamics

* First law of thermodynamics deals with the conservation of energy principle.
* It states that energy cannot be created nor be destroyed but it can change its forms.
* We can increase the level of energy of any system by increasing the amount of mass, as we know that energy is an extensive property so it is dependent upon the mass.
* The mass of air entering in any system increases the level of energy and the amount of mass which will exit from the system decreases the energy level of any system.
* First law of thermodynamics applied to a process
    - First law of thermodynamics & its principle, when it is applied to a process it states that the total amount of heat which is supplied to any cycle is equal to the net work done by theat particular cycle.

$$\oint \delta Q = \oint \delta w$$

Derivation:



- Let us consider a thermodynamic cycle which consist of two different cycle i.e. 1-A-2-C-1 & 1-B-2-C-1 state 1 has the initial properties represented by (P_1,V_1,T_1) and state 2 has the final properties represented by (P_2,V_2,T_2)
- The cycle was characterized by three different paths usually A, B & C
- From the statement of first law of thermodynamics we have,

  $$\oint \delta Q = \oint \delta w$$

- At first let us consider the first, cycle i.e. 1-A-2-C-1

  $$\int \delta Q_A + \int_2^1 \delta Q_C = \int_1^2 \delta W_A + \int_2^1 \delta W_C$$

Now, let us consider second cycle, i.e. 1 – B – 2 – (-1
We have,

$$\int_2^1 \delta Q_B + \int_2^1 \delta Q_C = \int_1^2 \delta W_B + \int_2^1 \delta W_C$$

Now, subtracting equation 2 from 1 we get

OR

$$\int_2^1 \delta Q_A + \int_2^1 \delta Q_C - \int_1^2 \delta Q_B - \int_2^1 \delta Q_C = \int_1^2 \delta W_A + \int_2^1 \delta W_C - \int_1^2 \delta W_B - \int_2^1 \delta W_C$$

OR

$$\int_2^1 \delta Q_A - \int_1^2 \delta Q_B = \int_1^2 \delta W_A + \int_1^2 \delta W_B$$

OR

$$\int_2^1 \delta Q_A - \int_1^2 \delta Q_A = \int_1^2 \delta W_B + \int_1^2 \delta W_B$$

OR

$$\int_1^2 \delta(Q-W)_A = \int_1^2 \delta(Q-w)_B$$

From equation @ we can conclude the different points.

i) As we know that Q } W both are the path function but their difference is a point function so, it is a thermodynamic property.

ii) The equation Q – W his generated a new kind of property which is a stored energy which is represented by dE. So, we can write, δ Q - δ W = dE
Further, it can be written as
$\delta Q = dE + \delta W$
This is required equation when the first law of thermodynamics is applied to a certain process.
Equation b also written as
$\therefore \delta Q = dU + \delta W$

Specific heat capacity

- It is defined as the amount of heat required to raise the temperature of a unit mass by 1^0c.
- In gas, there are two types of specific heat capacity.
  Types: a) specific heat capacity at constant volume (Cv)
  b) specific heat capacity at constant pressure (Cp)
- Gas is having two different specific heat capacity because of its compressible nature.
  So, the specific heat capacity is different at constant volume & at constant pressure.
  a) Specific heat capacity at constant volume (Cv)

- It is defined as the amount of heat required to raise the temperature of unit mass by 1^0c by keeping the volume constant.
  Mathematically,
  $\delta Q = mc\,v\,dT$
  OR, $Cv = \frac{\delta Q}{mdT}$
- From the first law of thermodynamics for a given process, we have
  $\delta Q = dU + \delta W$
- As we know that at constant volume
  $\delta W = 0$
  So, the equation can be written as
  $\delta Q = dU = 0$
  $\delta Q = dU$
- Now, putting the value of δQ in equation 1 we get
  $C_V = \frac{dU}{m.dT}$
  $C_V = \frac{dU}{m.dT}$
  $C_V = \frac{du}{dT}$
- From this expression we can conclude that the specific heat capacity at constant volume is equation to the change in specific internal energy with temperature

Specific heat capacity at constant pressure:

It is defined as the amount of heat required to raise the temperature of unit mass by 1^0c by keeping the pressure constant
$\delta Q = MCpdT$
Or $C_P = \frac{\delta Q}{mdT}$
As we know that, when the 1st law of thermodynamics is applied to a process then,

$\delta Q = dU + \delta W$

At constant pressure,
$\delta W = P.dv$
$\delta W = dpV$

So, the 1st law of thermodynamics become
$\delta Q = dU + dpV$
Or $\delta Q = d(U+pV)$
Or, $\delta Q = dH$ ($H = U + pv$)

Now, the equation becomes
$C_P = \frac{dH}{m.dT}$

Or $C_P = \frac{dH}{m}\left(\frac{1}{dT}\right)$

$C_P = \frac{dh}{dT}$

From this expression, we can conclude that the specific heat capacity at constant pressure is equal to the change in specific enthalpy with temperature.

First law of thermodynamics for a control volume system

* conservation of energy for a control volume system:

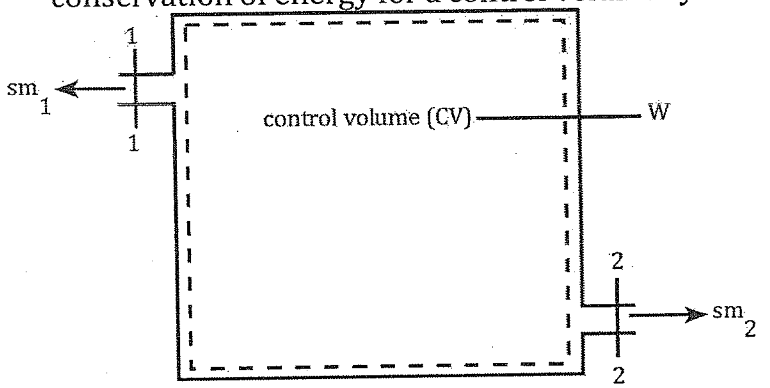


Fig. control volume system having one inlet & outlet parts.

- Let us consider a control volume system having the control volume boundary represented by CV. The control volume system has one number of inlet parts and one number of outlet part in which δm represent the total amount of mass entering into the system & δm2 represent the amount of mass which exit from the system.
- As we know that the amount of mass which is entering into the system increases the level of energy and the amount of mass which exit from the system decreases the level of energy of that particular system.
- Let Q is the amount of heat interaction with the system & W is the amount of work whichresults through the heat interaction.
- When the 1st law of thermodynamics is applied to a control volume system then it states that "The change of energy of a control volume system with the time is equal to the net energy transported by the system plus with interction with the system minus work interaction through out the system.

  Mathematically,

$$\frac{dEcv}{dt} = E_{net} + Q - W$$

- The net energy transported by the air is given as:

  $E_{net} = E_{in} - E_{out}$

  Where,

  $E_{in} = m(u + \frac{1}{2}v^2 + gz)$

OR, $E_1 = m(u + \frac{1}{2}v^2 + gz_1)$

OR, $E_2 = m(u_2 + \frac{1}{2}v^2 + gz_2)$

$E_{net} = E_{in} - E_{out}$

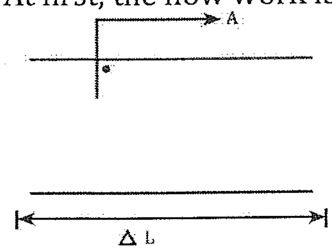
$= m(u_1 + \frac{1}{2}v_1^2 + gz_1) - m(u_2 + \frac{1}{2}v_2^2 + gz_2)$

Now, Q represents the amount of heat flow rate which is interacting with the system, and heat transfer occur when there is the temperature difference. So, the heat flow rate in this control volume system is given by,

Again, the work flow rate in an open system is given as,

$W = W_{flow} + W_{shaft} + W_{general}$

At first, the flow work is given as



$W_{flow} = F. \Delta L$

$= PA. \Delta L$

$= PV$

$W_{flow} = mp\upsilon$

At the inlet

$W_{flow} = -mP_1\vartheta_1$

At the outlet

$W_{flow} = mP_2\vartheta_2$

OR, $W_{flow} = -mP_1\vartheta_1 + mP_2\vartheta_2$

$W = W_{flow} + W_{shaft} + W_{general}$

$= -mP_1\vartheta_1 + mP_2\vartheta_2 + (W_{cv})$

Now, equation 1 becomes,

$$\frac{dEcv}{dt} = E_{net} + Q - W$$

$$= m(U_1 + \frac{1}{2}v_1^2 + gz_1) - m(U_2 + \frac{1}{2}v_2^2 + gz_2) + Qcv - (-mp_1\vartheta_1 + mp_2\vartheta_2 + Wcv)$$

$$= m(U_1 + \frac{1}{2}v_1^2 + gz_1) - m(U_2 + \frac{1}{2}v_2^2 + gz_2) + Qcv + mp_1\vartheta_1 - mp_2\vartheta_2 - Wcv$$

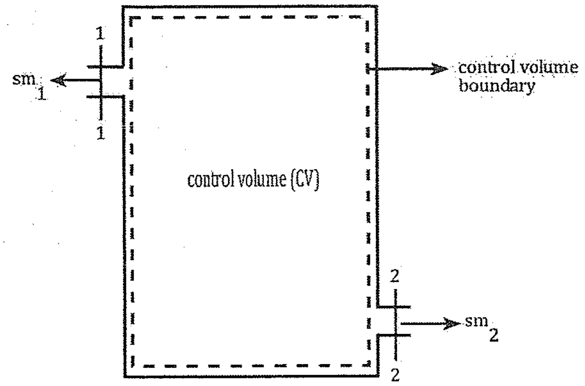
OR, $$\frac{dEcv}{dt} = \{m(U_1 + \frac{1}{2}v_1^2 + gz_1 + P_1\vartheta_1)\} - m\{(m(U_2 + \frac{1}{2}v_2^2 + gz_2 + P_2\vartheta_2) - Wcv$$

OR, $\frac{dEcv}{dt} = m\{(U1 + P1\vartheta1) + \frac{1}{2}v_1^2 + gz_1)\} - m\{U_2 - P_2\vartheta_2) + \frac{1}{2}v_2^2 + gz_2) - Wcv$

OR, $\frac{dEcv}{dt} = m\{(h_1 + \frac{1}{2}v_1^2 + gz_1)\} - m\{h_2 + \frac{1}{2}v_2^2 + gz_2) - Wcv$

Which represents that the basic equation that defines the first law of thermodynamics for an open system.

Conservation of mass for a control volume system



- As we know that a control volume system interacts both mass and energy with the surrounding.
- In this section we are going to find out the change of mass of any control volume system with the respect of time
- Let δm1 be the amount of mass which is intering into the system & δm2 be the amount of mass which exists from the system.
- As we know that, the amount of mass which is entering into the system increases the level of energy and the amount of mass which exit from the system decreases the level of energy of that particular system.
- The conservation of mass for a control volume system states that "The rate of change of mass in any control volume system is equal to the rate of mass flow at the inlet the rate of mass flow at the outlet

Mathematically,

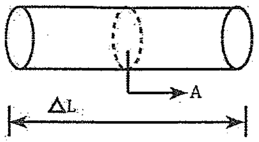
$\frac{dm_{cv}}{dt} = m_{in} - m_{out}$

OR, $\frac{dm_{cv}}{dt} = \sum_{in} m - \sum_{out} m$ (when the system consists of more than one no. of inlet & outlet parts.)

- The expression of mass flow rate (m) is calculated with the help of mass so,
  The mass of air is given as,

$\therefore$ m = volume x density
OR, m = V_{swept} x s

Again, the swept volume is also expressed as



Vswept = A×ΔL
OR, m = A×Δ×L
Now, dividig the equation by time 9t) we get,

OR, m = $A\times\frac{\Delta L}{\Delta t}\times\delta$

= $A\times\overline{V}\times\delta$ [where $\overline{V}$ is the velocity

OR, m = $A\times\overline{V}\times\frac{m}{v}$

OR, m = $A\times\overline{V}\times\frac{m}{\vartheta}$ {$\therefore \frac{m}{v}$ can be written as specific volume}

$\therefore m = \frac{A\times\overline{V}}{\vartheta}$

By putting the value of m in equation – (i)

$$\frac{dm_{cv}}{dt} = \sum_{in}\frac{A\times\overline{V}}{\vartheta} - \sum_{out}\frac{A\times\overline{V}}{\vartheta}$$

Which is the required equation that represent the conservation of mass for a control volume system.

Steady flow energy equation (SFEE)

Basic assumptions:

1. The change of any properties of the flow is considered as a zero.
   $\rightarrow \frac{dE_{cv}}{dt} = 0$
2. The control volume boundary doesn't move reletive to the coordinate frame of that particular system.
3. Because of the adiabatic boundary the heat flow rate & the work flow rate is considered as a zero.
   $\rightarrow$Q, W = 0
4. The change of P.E & K.E is considered as negligible
   As we have, the conservation of energy for a control volume system is

$$\frac{dE_{cv}}{dt} = \dot{m}\left(h_1 + \frac{v_1^{\,2}}{2} + gz_1\right) - \dot{m}\left(h_2 + \frac{v_2^{\,2}}{2} + gz_2\right) + Q - W$$

$$\text{Or, } o = \dot{m}\left(h_1 + \frac{v_1^{\,2}}{2} + gz_1\right) - \dot{m}\left(h_2 + \frac{v_2^{\,2}}{2} + gz_2\right) + Q - W$$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Which is the required equation that represents the steady flow energy equaiton.

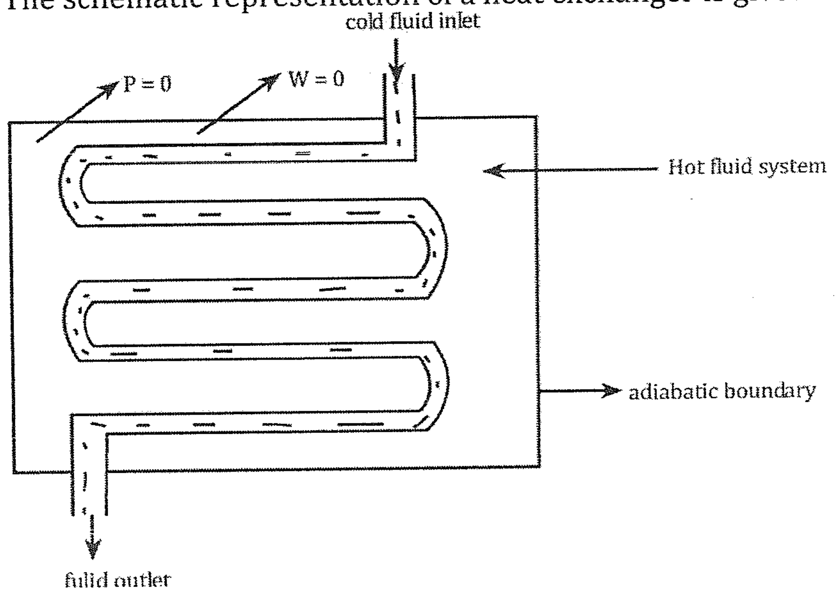
Steady flow process.
The process in which every properties remains constant during the fluid flow is known as steady flow process.

Applications:
1. Steady flow devices
2. Steady work devies.

1. Steady flow devices
   a) Heat exchanger
   b) Nozzle
   c) Diffuser
   d) Throttling device

2. Steady work devices
   a) yurbine
   b) compressor
   c) Boiler

1. Steady flow devices
   a) Heat exvhanger
      - Heat exchanger is a mechanical device which is operating under the steady flow condition & used to exchange the heat in between two different temperature bodies
      - Eg. Water colling system during an internal conbustion engine.
      - The schematic representation of a heat exchanger is given as:



Assumptions:

1. The mass flow rate at the inlet is equals to the mass flow rate at the outlet. (i.e. $m_{in} = m_{out}$)
2. Because of the adiabatic boundary, the heat flow rate & the work flow rate is considered as zero (i.e. Q = 0, W = 0)
3. The change is PF & KE is considered as negligible so PE = 0 & K.E = 0

From steady flow energy equation

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + 0$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right)$

Or, $m(h_1 + 0 + 0) = m(h_2 + 0 + 0)$

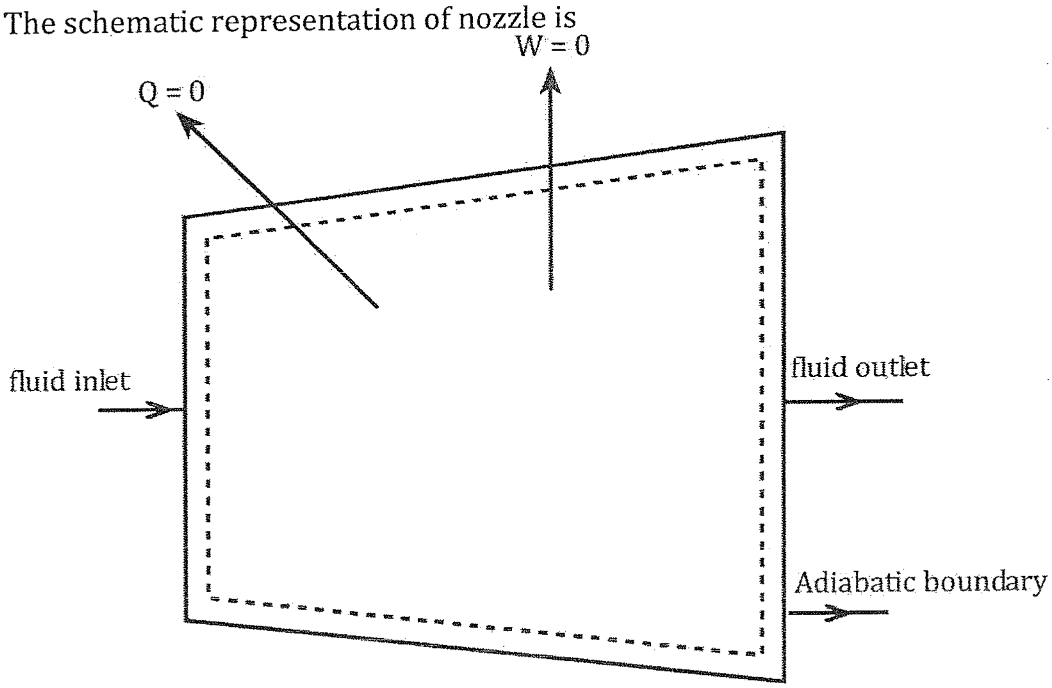
Or, $mh_1 = mh_2$

$\therefore h_1 = h_2$

Which is the required equation that represents the steady flow energy equation for heat exchanger.

2. Nozzle:

- Nozzle is a mechanical device which is operating under the steady flow condition, which is used to increase the velocity of the flow
- It increases the velocity with the expenses of pressure at the exit.
- The schematic representation of nozzle is



Assumptions:

- The mass flow rate at the inlet is equal to the mass flow rate at the outlet. ($m_{in} = m_{out}$)
- Because of the adiabatic boundary the heat flow rate & work flow rate is zero (Q = 0, W = 0)
- The change in PE is considered as negligible so it is zero (P.E = 0)
- In this device the exit velocity is higher than that of the inlet velocity so, v_1 is neglected

* From steady flow energy equation

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

$$\text{Or, } \dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + 0$$

$$\text{Or, } \dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right)$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} + gz_1 = h_2 + \frac{v_1^2}{2} + gz_2$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} + 0 = h_2 + \frac{v_1^2}{2}$$

$$\text{Or, } h_1 + 0 = h_2 + \frac{v_1^2}{2}$$

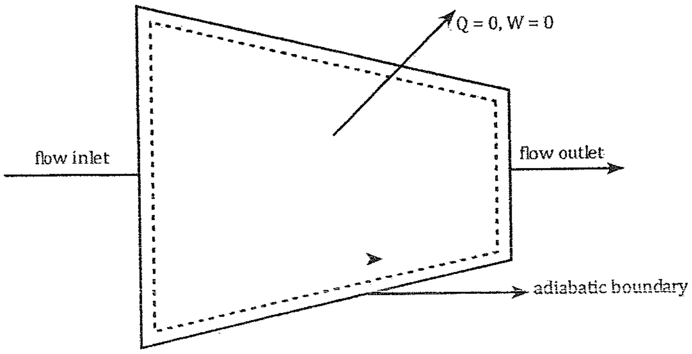
$$\text{Or, } h_1 = h_2 + \frac{v_1^2}{2}$$

$$\therefore v_1 = \sqrt{2(h_1 - h_2}$$

3. Diffuser:

It is a mechanical device which is operating under the steady flow condition, which is used to increase the pressure of the flow by the expenses of velocity at the outlet.

- The schematic representation of diffuser is



Assumption:

1. The mass flow rate at the inlet is equal to the mass flow rate at the outlet ($m_{in} = m_{out}$)
2. Because of the adiabatic boundary, the heat flow rate & the work flow rate is considered as zero (Q = 0, W = 0)
3. The change in PE is considered as negligible so, it is zero (i.e. PE = 0)
4. In this device the exit velocity is less than that of the inlet velocity so, v_2 is neglected.

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

$$\text{Or, } \dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + 0$$

$$\text{Or, } \dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right)$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} + gz_1 = h_2 + \frac{v_1^2}{2} + gz_2$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} + 0 = h_2 + \frac{v_1^2}{2}$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} = h_2 + 0$$

$$\text{Or, } h_1 + \frac{v_1^2}{2} = h_2$$

$$\text{Or, } v_1 = \sqrt{2(h_2 - h_1}$$

$$\therefore v_1 = \sqrt{2(h_2 - h_1}$$

4. Throttling device:

Throttling device is a mechanical device which is operating under the steady flow condition which restrict the fluid flow by using the variable area duct passage.

Assumption

1. The mass flow rate at the inlet is equals to the mass flow at the outlet. (min = mout)
2. Because of the adiabatic boundary, the heat flow rate & the work flow rate is considered as zero (i.e. Q = 0, W = 0)
3. The changes in PE & KE is considered as a negligible so PE = 0 & KE = 0

From the steady flow energy equation

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

$$\text{Or, } \dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + 0$$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right)$

Or, $\dot{m}$ $(h_1 + 0 + 0) = \dot{m}$ $(h_2 + 0 + 0)$

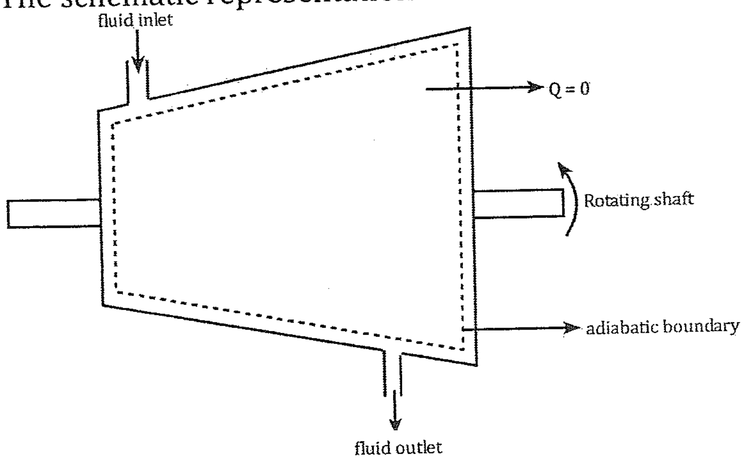
Or, $\dot{m}h_1 = \dot{m}h_2$

$\therefore h_1 = h_2$

Which is the required equation that represents the steady flow energy equation throttling device

2. Steady work device

- Turbine is a mechanical device which is operating under the steady work condition, which is used to convert the fluid energy to the mechanical energy & further converted into the electric form of energy.
- It is a work producing device
- The schematic representation of turbine is



Assumption:

1. The mass flow rate at the inlet is equal to the mass flow rate at the outlet (min = mout)
2. Because of the adiabatic boundary the heat flow rate is negligible so Q = 0
3. The changes in PE & KE is negligible so P.E = 0, & K.E = 0
4. In this device work is done by the system so work done is positive.

From steady flow energy equation

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $m(h_1 + 0 + 0) = m(h_2 + 0 + 0) + W$

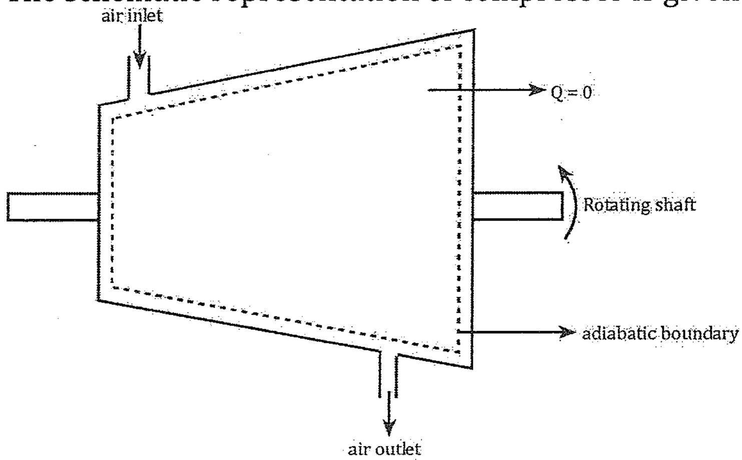
Or, $m(h_1) = m(h_2) + W$

Or, $W = m(h_1 - h_2)$

Which is the required equation that represents the steady work energy equation for turbine.

2. Compressor

- It is a mechanical device operating under the steady work condition, which is used to compress the flow generally, air.
- It is used to increase the energy level of any gas medium.
- The schematic representation of compressor is given as:



Assumptions:

1. The mass flow rate at the inlet is equal to the mass flow rate at the outlet. ($m_{in} = m_{out}$)
2. Because of the adiabatic boundary, the heat flow rate is negligible so Q = 0.
3. The changes in KE & PE is negligible so PE = 0 & KE = 0.
4. Because of the power consuming device the sign convention for work is negative.

From steady flow energy equation

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + 0 = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $m(h_1 + 0 + 0) = m(h_2 + 0 + 0) + W$

Or, $mh_1 = mh_2 - W$

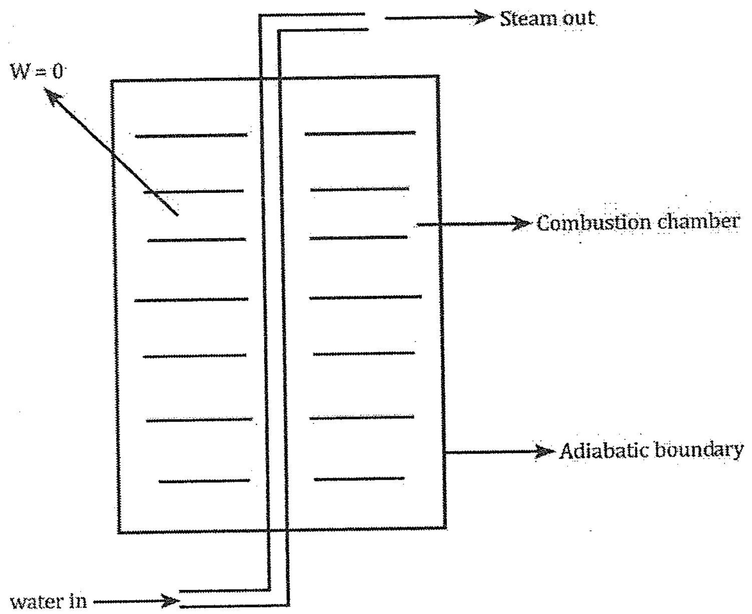
Or, $W = mh_2 - mh_1$

Or, $\therefore$ $W = m(h_2 - h_1)$

This is the required expression that represents the steady work device equation for compressor

3. Boiler

- It is a mechanical device working under the steady work device, which is used to produce the steam with the help of water.
- In this device, the chemical energy which is the resultant of the combustion process and finally converted into the heat energy
- The schematic representation of a boiler is



Assumptions:

1. The mass flow rate at the inlet is equal to the mass flow rate at the outlet. (min = mout)
2. Because of the adiabatic boundary the work flow rate is considerably negligible (w = 0)
3. The change in KE & is neglected so PE = 0 & KE = 0

From steady flow energy equation

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + 0$

Or, $\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right)$

Or, $m(h_1 + 0 + 0) = m(h_2 + 0 + 0) + W$

Or, $m(h_1)+Q = mh_2$

Or, $\therefore$ $W = m(h_2-h_1)$

This is the required equation that represent the steady work device for boiler.

Numerical
First law of thermodynamics
Or $\delta Q = dU + \delta w$
$dU = m.c_v.dT$
$= m.c_v(Tf-Ti)$
$\therefore \delta Q =$

Steady flow energy

Or, $m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

W?
h_1 = given
$h_2 = u_1 + P_1V_1$
$h_1 - h_2 = m.c_p(T_1-T_2)$
$\therefore C_V = \frac{du}{dt}$
$\therefore C_P = \frac{dh}{dt}$
dn = cp.dt

To find the value of h1, refer steam table.

$m = \frac{A.V}{\vartheta}$

$m_1 = \frac{A_1V_1}{\vartheta_1}$

$m_2 = \frac{A_2V_2}{\vartheta_2}$

2. A turbine is operating under the steady flow condition which receives 5000kg of steam per hour. The steam enters the turbine at a velocity of 300m/min, At an elevation of 5m and the specific enthalpy of 278KJ/Kg. It leaves the turbine at a velocity of $6000mmin^{-1}$, at an elevation of 1m and a specific enthalpy of 2259KJ/Kg. Heat loss from the turbine to the surrounding is 16736KJ/Kg. Determine the power output of the turbine.
Given data:

Inlet	outlet
m = 5000kg/hr	m = 500kg/hr
$\frac{5000}{3600} = 1.38 kg/\sec$	$\frac{5000}{3600} = 1.38 kg/\sec$
$V_1 = 3000m/min$	$V_2 = 6000m/min$

$\frac{3000}{60} = 50m/\sec$ $\qquad$ $\frac{6000}{60} = 100m/\sec$

$Z_1 = 5m$ $\qquad$ $Z_2 = 1m$

$h_1 = 278 \times 10^3 J/kg$ $\qquad$ $h_2 = 2259 \times 10^3 J/kg$

$Q = -16736 \times 10^3 J/kg$

From steady flow energy equation

$$m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

$$\text{Or, } 1.38(278\times10^3 + \frac{(50)^2}{2} + (-16736\times10^3) = 1.38\left(2259\times10^3 + \frac{(100)^2}{2} + 9.8\times1\right) + W$$

Or, -16350567.38 = 3124333.524 + W

Or, -16350567.38 – 3124333.524 = W

$\therefore$ W = -1.95 x 10^4KJ/Sec.

-ve sign indicates that the work is done on the system.

3. A steam enters a horizontally aligned well insulated nozzle of inlet area $0.15m^2$ & the velocity is 65m/s. At the inlet the specific enthalpy of steam is 3000KJ/kg. Specific volume of 0.187m3/kg & the outlet specific enthalpy is 2762 KJ/kg, specific volume is $0.498m^3/kg$. Determine (t) velocity of the steam at the outlet.
   i) mass flow rate of steam.
   ii) exit area of nozzle

Given data

Inlet	outlet
$A_1 = 0.15m2$	$h_2 = 2762kJ/kg$
$V_1 = 65m/sec$	$=2762 \times 10^3 J/kg$
$h_1 = 3000KJ/kg$	$\vartheta_2 = 0.498m3/kg$
$=3000 \times 10^3 J/kg$	
$\vartheta_1 = 0.187m^3/kg$	

To find

1) V2 = ?

2) m = ?

3) A2 = ?

As we know that
The exit velocity of a nozzle is

$\therefore V_2 = \sqrt{(h_1 - h_2)}$

$$V_2 = \sqrt{2(3000\times10^3 - 2762\times10^3)}$$

$\therefore\ V_2 = 689.927 m/\sec$

The mass flow rate of a steam is given as

$$m = \frac{A_1 V_1}{\vartheta_1}$$

$$= \frac{0.15\times65}{0.185}$$

$= 52.70$kg/sec

$$m = \frac{A_2 . V_2}{\vartheta_2}$$

$$52.70 = \frac{A_2 \times 689.927}{0.498}$$

$$A_2 = \frac{52.70\times0.498}{689.927}$$

$\therefore A_2 = 0.038m^2$

4. In an air compressor the air flow at the rate of 0.5kg/sec. In enters the compressor at 6m/sec with a pressure of 1 bar and specific volume of $0.85m^3/kg$ and leaves at 5m/sec with a pressure of 7 bar and a specific volume of $0.16m^3/kg$. The internal energy of the air leaving is 90kJ/kg greater than that of the air entering. Cooling water in the jacket surrounding alrorbs the heat from the air is at the rate of 60kJ/sec. Calculate the power required to drive the compressor and cross sectional area of inlet & outlet pipe.

Given data:

Inlet	outlet
m = 0.5kg/sec	m = 0.5kg/sec
V_1 = 6m/sec	V_2 = 5m/sec
P_1 = 1 bar	P_2 = 7 bar
$=1 \times 105 N/m^2$	$= 7 \times 105 N/m^2$
$\vartheta_1 = 0.85m^3/kg$	$\vartheta_2 = 0.16m^3/kg$

$\therefore U_2 - U_1 = 90kJ/kg$

$= 90 \times 10^3 J/kg$

$Q = -60 \times 10^{-3} J/sec$

From steady flow energy equation

$$m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

Or, $m\left[(U_1 + P_1\vartheta_1) + \frac{V_1^2}{2} + gz_1\right] + Q = m\left[(U_2 + P_2\vartheta_2) + \frac{V_2^2}{2} + gz_1\right] + W$

Or, $m\left[(U_1 - U_2) + (P_1\vartheta_1 - P_2\vartheta_2) + \frac{V_1^2}{2} - \frac{V_2^2}{2}\right] + Q = W$

Or, $0.5\left[-90\times10^3 + (1\times10^5\times0.85 - 7\times10^5\times0.15) + \frac{6^2}{2} - \frac{5^2}{2}\right] - 60\times10^3 = W$

$\therefore$ W = -147496.5J/sec

$$m = \frac{A_1V_1}{\vartheta_1}$$

$$A_1 = -\frac{m\vartheta_1}{V_1} = \frac{0.5\times0.85}{6} = 0.07m^2$$

$$A_1 = \frac{\pi}{4}d_1^2$$

$$d_1 = \sqrt{\frac{A_1\times4}{\pi}} = \sqrt{\frac{0.07\times4}{3.14}} = 0.298m$$

$$m = \frac{A_2V_2}{\vartheta_2}$$

$$A_2 = \frac{A_2\vartheta_2}{V_2} = \frac{0.5\times0.16}{5} = 0.016m^2$$

$$d_2 = \sqrt{\frac{A_2\times4}{\pi}} = \sqrt{\frac{0.016\times4}{3.14}} = 0.142m$$

Fixed values

C_P = 1.005kJ/kgk

C_V= 0.718kJ/kgk

R = 0.287 kJ/kgk = 0.287 x 10^3J/kgk

ψ = 1.4

5. In a gas turbine the gas enters at the rate of 5kg/sec with a velocity of 50ms^{-1} & enthalpy of 900kJ/kg and leaves the turbine with the velocity of 1m/s 150m/s & enthalpy of 400kJ/kg. The loss of heat from the gas to the surrounding is 25kJ/kg. (Assume R = 0.287kJ/kgk & C_P = 1.005kJ/kgk). Determine the power output of the turbine. If the inlet condition are 100Kpa, 27^{0}C. then determine the diameter of the inlet pipe.

Given data:

Inlet	outlet
m = 5kg/sec	m = 5kg/sec

$V_1 = 50 m/s$ $\qquad$ $V_2 = 150 m/s$

$h_1 = 900 kJ/kg$ $\qquad$ $h_2 = 400 kJ/kg$

$= 900 \times 10^3 J/kg$ $\qquad$ $= 400 \times 10^3 J/kg$

$Q = -25 kJ/kg$

$= -25 \times 5 kJ/sec$

$125 \times 10^3 kg/sec$

Or, $m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$

Or, $5\left\{900 \times 10^3 + \frac{(50)^2}{2} + 0\right\} - 125 = 5\left\{400 \times 10^3 + \frac{(150)^2}{2} + 0\right\} + W$

Or, $\{900 \times 10^3 + 1250\} - 125 \times 10^3 = \{400 \times 10^3 + 11250\} + W$

Or, $5.901250 - 125 \times 10^3 = 5.411250 + W$

Or, $4506250 - 125000 = 2056250 + W$

Or, $4381250 - 2056250 = W$

$\therefore W = 2325000 J/sec$

$= 2325 kJ/sec$

$m = \frac{A_1 V_1}{\vartheta_1}$

$\therefore P_1 V_1 = mRT_1$

$P_1 . \frac{V_1}{m} = RT_1$

$P_1 . \vartheta_1 = RT_1$

$\vartheta_1 . \frac{RT_1}{P_1} = \frac{0.287 \times 10^3 (27 + 273)}{100 \times 10^3} = 0.861 m^3/kg$

$A_1 . \frac{m\vartheta_1}{V_1} = \frac{5 \times 0.861}{50} = 0.861 m^2$

$A_1 = \frac{\pi}{4} d_1^2$

$d_1 = \sqrt{\frac{A_1 \times 4}{\pi}} = 0.331 m$

6. Air enters a compressor operating at steady state at 100kpa, 300k and leaves at 1000kpa, 400k with a volumetric flow rate of 1.5m^3/min. The work consumed by the compressor is 250kJ/kg of air. Neglecting the effects of PE & KE. Determine the heat transfer rate in KW.

Given data:

Inlet	outlet
$P_1 = 100kpa$	$P_2 = 1000kpa$
$= 100 \times 103pa$	$= 1000 \times 103pa$
$T_1 = 300k$	$T_2 = 400k$

$V = 1.5m^3/min$

$\frac{1.5m^3}{60\sec}$

$=0.025m3/sec$

$W = -250kJ/kg$

$= -25 \times 10^3 J/kg$

At exit

$P_2V_2 = mRT_2$

Or, $\frac{1000\times10^3\times0.025}{0.287\times10^3\times400} = m$

$\therefore$ m = 0.218kg/sec

Now,

$$m\left(h_1+\frac{v_1^2}{2}+gz_1\right)+Q=m\left(h_2+\frac{v_2^2}{2}+gz_2\right)+W$$

Where, $W = -250kJ/kg$

$= 250 \times 10^3 \times 0.21$

$= 55000J/sec$

$= -250 \times (0.218)kJ/sec$

$= 54.5kJ/sec$

$= - 54.5 \times 10^3 J/sec$

Or, $mh_1 + Q = mh_2 + W$

$h_1 - h_2 = P(T_1 - T_2)$

$m [C_P (T_1 - T_2)] + Q = W$

or, $0.218[1.005 \times 10^3 (300 - 400)] + Q = -54.5 \times 10^3$

or, $0.218 - 100500 + Q = -54\text{-}5 \times 10^3$

or, $-21909 + Q = 54.5 \times 10^3$

or, $Q = -54.5 \times 10^3 + 21909$

$\therefore Q = -32591W$

$= -32.59KW.$

7. Air flows at the rate of 1.5kg/sec through a turbine entering at 500kpa, 150^0C and with the velocity of 120m/sec. The exit condition of the air is 100kpa, 25^0C and with the velocity of 60m/sec. The power produced by the turbine is 180 megawatt. Determine

a) Heat loss from the turbine
b) diameter of the inlet and the outlet pipe. (take R = 287J/kgk)

Given data:

Inlet	Outlet
m = 1.5kg/sec	m = 1.5kg/sec
P_1 = 500kpa	P_2 = 100kpa
= 500×10^3pa	= 100×10^3pa
$T_1 = 15^0C = 423k$	$T_2 = 25^0C = 298k$
V_1 = 120m/s	V_2 = 60m/s

W = 180mw

$= 180 \times 10^6W$

$= 180 \times 10^6 J/sec$

Now,

$$m\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

$$m\left[cp(T_1 - T_2) + \frac{v_1^2}{2} - \frac{v_2^2}{2}\right] + Q = W$$

$$\text{Or, } 1.5\left[1.005\times10^3(423-298) + \frac{(120)^2}{2} - \frac{(60)^2}{2}\right] + Q = 180\times10^6$$

Or, $1.5[1.005 \times 10^3 \times 125 + 5400] + Q = 180 \times 10^6$

Or, $1.5[125625 + 5400] + Q = 180 \times 10^6$

Or, $1.5 \times 131025 + Q = 180 \times 10^6$

Or, $196538.5 + Q = 180 \times 10^6$

Or, $Q = 180 \times 10^6 - 196537.5$

$\therefore Q = 179803462.5 J/sec$

$\therefore Q = 179803.4625kJ/sec$

$= 179803\ 4625kw$

$= 179.8mw.$

$$m = \frac{A_1V_1}{\vartheta_1}$$

So, $P_1V_1 = mRT_1$

$$P_1.\frac{V_1}{m} = RT_1$$

$$\vartheta_1.\frac{RT_1}{P_1} = \frac{287\times423}{500\times10^3} = 0.2428$$

$$A_1.\frac{m\vartheta_1}{V_1} = \frac{1.5\times0.2428}{120} = 3.035\times10^{-3}m^2$$

$$\frac{\pi}{4}d_1^2 = 3.035\times10^{-3}$$

$$d_1 = \sqrt{\frac{4\times3.035\times10^{-3}}{\pi}}$$

$= 0.062m$

$$m = \frac{A_2V_2}{\vartheta_2}$$

So, $P_2V_2 = mRT_2$

$$P_2.\frac{V_2}{m} = RT_2$$

$$\vartheta_2 = \frac{RT_2}{P_2} = \frac{287\times298}{100\times10^3} = 0.85526$$

$$A_2 = \frac{m\vartheta_2}{V_2} = \frac{1.5\times0.85526}{60} = 0.021m^2$$

$$\frac{\pi}{4}d_2^2 = 0.021$$

$$d_2 = \sqrt{\frac{0.021\times4}{\pi}} = 0.16m$$

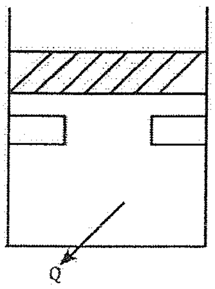
1. Air m = 0.5kg/sec flow in a compressor with steady state condition. The inlet condition is 5m/s with a specific volume of 0.83m³/kg, 150kpa and the exit condition is 7m/s velocity, 0.48m³/kg and 100kpa pressure. The internal energy of the air leaving is 85kJ/kg greater than that of the air entering. The water jacket absorbs the heat from the air is 120kJ/kg. Determine the power required to drive the compressor. Also determine the inlet & outlet diameter.

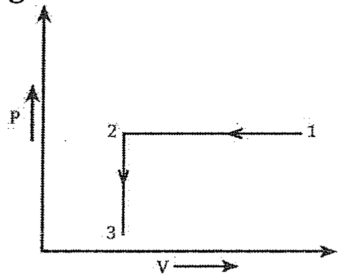
2. A piston cylinder assembly contains a fluid system which passes through a complete cycle of four different process. During a cycle the sum of all the heat transfer is –170kJ. The system completes 100 cycles per minute. Now determine the unknown value from the table.

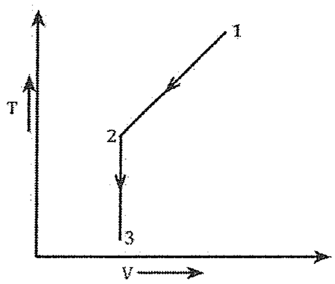
Process	Q(kJ/min)	W(kJ/min)	ΔE(kJ/min)
a – b	0	2,170	-
b – c	21,000	0	-
c – d	-2,100	-	-36,000
d – a	-	-	-

Determine whether it is a power cycle or refrigeration cycle.

3. A piston cylinder assembly which contains air (0.5kg) as shown in fig. The heat is lost by the system from an initial condition of 800k, 1500kpa. There is a heat lost from the system upto piston reaches the stops. There is a further heat loss so that the final condition is 0.045m³, 300k. Determine the net work done & the heat transfer with the PV & TV diagram.







1. **Given data:**

Inlet	Outlet
$V_1 = 5m/s$	$V_2 = 7m/s$
$\vartheta_1 = 0.83m3/kg$	$\vartheta_2 = 0.48m3/kg$
$P_1 = 150kpa$	$P_2 = 100kpa$
$=150 \times 10^3 pa$	$= 100 \times 10^3 pa$

$U_2 - U_1 = 85kJ/kg = 85 \times 10^3 J/kg$

$Q = -120kJ/kg = -120 \times 0.5kg/sec = -60kg/sec = -60 \times 10^3 J/sec$

$m = 0.5kg/sec$

To find;
Power = ?
d_1 = ?
d_2 = ?
From steady flow energy equation

$$m\left(h_1+\frac{v_1^2}{2}+gz_1\right)+Q=m\left(h_2+\frac{v_2^2}{2}+gz_2\right)+W$$

$$\text{Or, } m(U_1+P_1\vartheta_1)+\frac{V_1^2}{2}+gz_1)+Q=m(U_2+P_2\vartheta_2)+\frac{V_2^2}{2}+gz_2)$$

$$\text{Or, m}\left[(U_1-U_2)+(P_1\vartheta_1-P_2\vartheta_2)+\frac{V_1^2}{2}-\frac{V_2^2}{2}\right]+Q=W$$

$$\text{Or, } 0.5\left[-85\times10^3(150\times10^3-100\times10^3\times0.48)+\frac{5^2}{2}-\frac{7^2}{2}\right]+(-60\times10^3)=W$$

$$\text{Or, } 0.5[-85\times10^3(124500-48000)+12.5-24.5]-60\times10^3=W$$

$$\text{Or, } 0.5[-85\times10^3+76500+12.5-24.5]-60\times10^3=W$$

Or, 0.5 – 8512 – 60 x 10^3 = W
Or, -4256 – 60 10^3 = W
$\therefore$ W = -64256J/sec
= - 64.256kJ/sec
= - 64.256kw

$$m=\frac{A_1V_1}{\vartheta_1}$$

$$A_1=\frac{m\vartheta_1}{V_1}=\frac{0.5\times0.83}{5}=0.083m^2$$

$$A_1=\frac{\pi}{4}d_1^2$$

$$\text{Or, } d_1=\sqrt{\frac{A_1\times4}{\pi}}=\sqrt{\frac{0.083\times4}{3.14}}=0.325m$$

$$d_2=\sqrt{\frac{A_2\times4}{\pi}}=\sqrt{\frac{0.034\times4}{3.14}}=0.208m$$

$$A_2=\frac{m\vartheta_2}{V_2}=\frac{0.5\times0.48}{7}=0.034m^2$$

2. δQ = QE + δW

Process a – b
0 = dE + 2,170
$\therefore$ dE = -2,170kJ/min

Process b – c
21,000 = dE + 0
$\therefore$ dE = 21,000kJ/min

Process c – d
-21000 = -36000 +δW
Or, -21000 + 36000 = δW
$\therefore \delta$W = 33900

Process d – a
$\oint \Delta E = 0$
ΔEa-b + Δb-c + ΔEc-d + ΔEd-a = 0
-2170 + 2100 + (- 36000) + ΔE d – a = 0
Or, ΔE d-a = 2170 – 21000 + 36000
$\therefore \Delta$Ed-a = 17170kJ/min

Now,
Total W = 170 x 100 kJ/min
= - 17000kJ/min

Total work (W) = -17000kJ/min
δWa-b + δWb-c + δWc-d + δWd-a = -17000
Or, 2170 + 0 + 33900 + δWd-a = -17000
Or, δWd-a = -2170 – 33900 – 17000
$\therefore \delta$Wd-a = -53070kJ/min

Again
δQ = dE + δW
Or, δQ = 17170 + (-53070)
$\therefore \delta$Qd-a = 35900kJ/min

Chapter 4

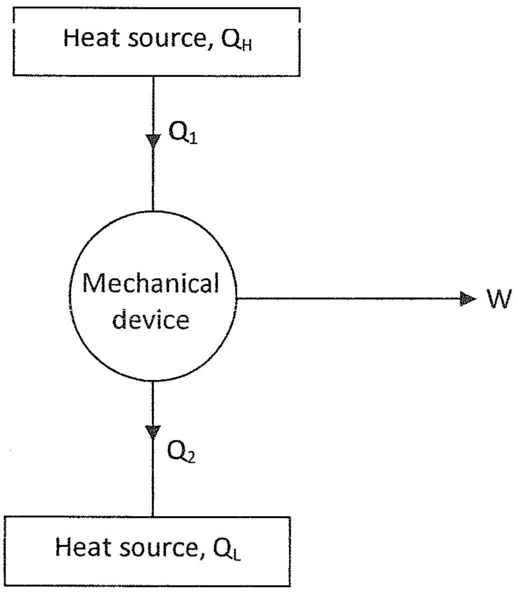
Second law of thermodynamic

Thermal reservoir:

- It is a hypothetical rigid body which is made of impermeable layer also consists of adiabatic boundary, from which and to which we can transfer or reject the enormous amount of heat.
- In thermodynamics there are two types of thermal reservoir.
  a) Heat source
  b) Heat sink

a) Heat Source
- It is defined as the higher temperature bodies from which we can transfer enormous amount of heat to any other system.
- Generally in thermodynamics it is represented by Q_H or Q_1
- Example: sun, boiler, combustion chamber, Nuclear reactor.

b) Heat sink
- It is defined as the lower temperature bodies to which we can supply enormous amount of heat in order to execute the cyclic process.
- Generally, in thermodynamics, it is represented by QL or Q2
- Example: River, ocean, Atmospheric air
- The schematic representation of thermal reservoir is given as



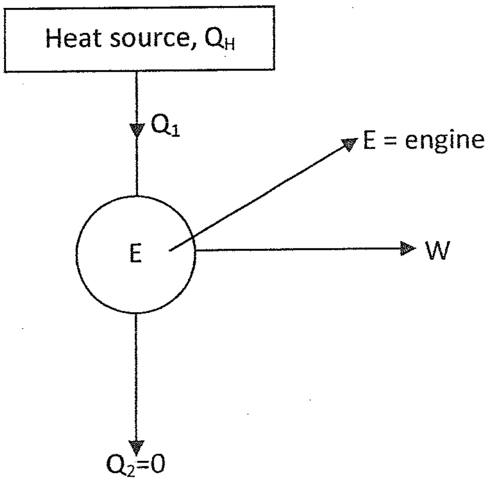
Statement of 2nd law of thermodynamics

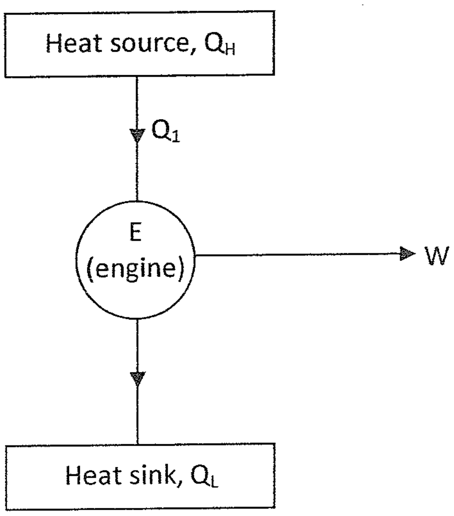
In thermodynamics 2nd is described by two different statements. They are

1. Kelvin – Plank statement
2. Clausius Statement

1) Kelvin – Plank statement

- Statement: It is impossible to construct a device, operating in a thermodynamic cycle which produce work having the heat interaction with single reservoir.
- So, there must be the heat interaction in between two different reservoir to produce work
- The schematic representation of this statement is given as.





- The performance of any power producing device is measured with the help of an efficiency (η) and it is given as

$$\eta = \frac{output}{input}$$
$$= \frac{work\ done(W)}{Heat\ supplied(Q_1)}$$

- If we made the impossible statement to the possible one then the performance of such devices is given as

$$\eta = \frac{output}{input}$$
$$= \frac{W}{Q_1}$$
$$= \frac{Q_1}{Q_1}$$
$$\therefore \eta = 100\%$$

- So, such devices does not exist in nature. So, they are also called perpetual machine of I[st] kind. (PPMI)

Clausius statement

- Statement: It is impossible to construct a device which is operating in a thermodynamic cycle, which transfer the heat from low temperature device to the high temperature device without the supply of external work on the particular system.
- So, the work done on the system is necessary condition to execute the thermodynamic cycle.
- The schematic representation of a clausius statement is given as:

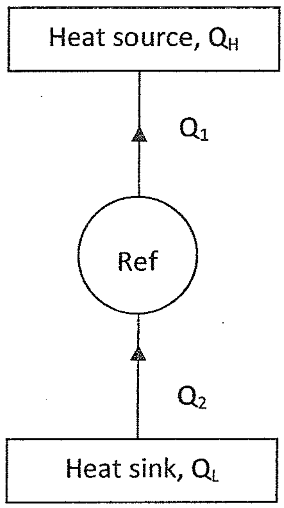


Fig. impossible

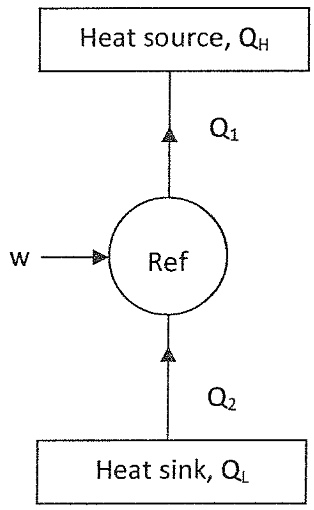


fig. possible

- The performance of such device is measured with the help co-efficient of performance (COP)
- The COP is given as,

$$COP = \frac{output}{input}$$

$$= \frac{cooling effect(Q_2)}{work\ done(W)}$$

- If we make the impossible statement to the possible one then the co-efficient of performance (COP) is given as

$$COP = \frac{output}{input}$$
$$= \frac{Q_2}{W}$$
$$= \frac{Q_2}{O}$$
$$\therefore COP = \infty$$

- So, such devices doesn't exist in nature. Such devices is also known as perpetual machine of IInd kind (PPM II)

Heat engine

- It is a power producing device which is operating in a thermodynamic cycle, in which the net amount of heat(δQ) is converted into the next work(δW)
- The components which is associated with a heat engine is given as:
  1. Boiler
  2. Turbine
  3. Condensor
  4. Pump
- The line diagram of a heat engine is given as:

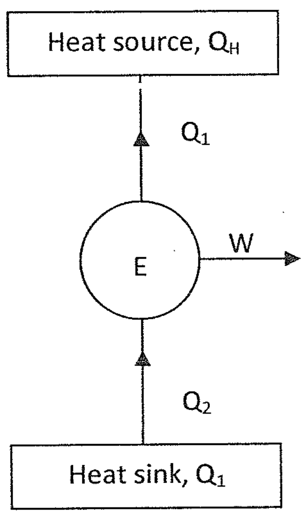
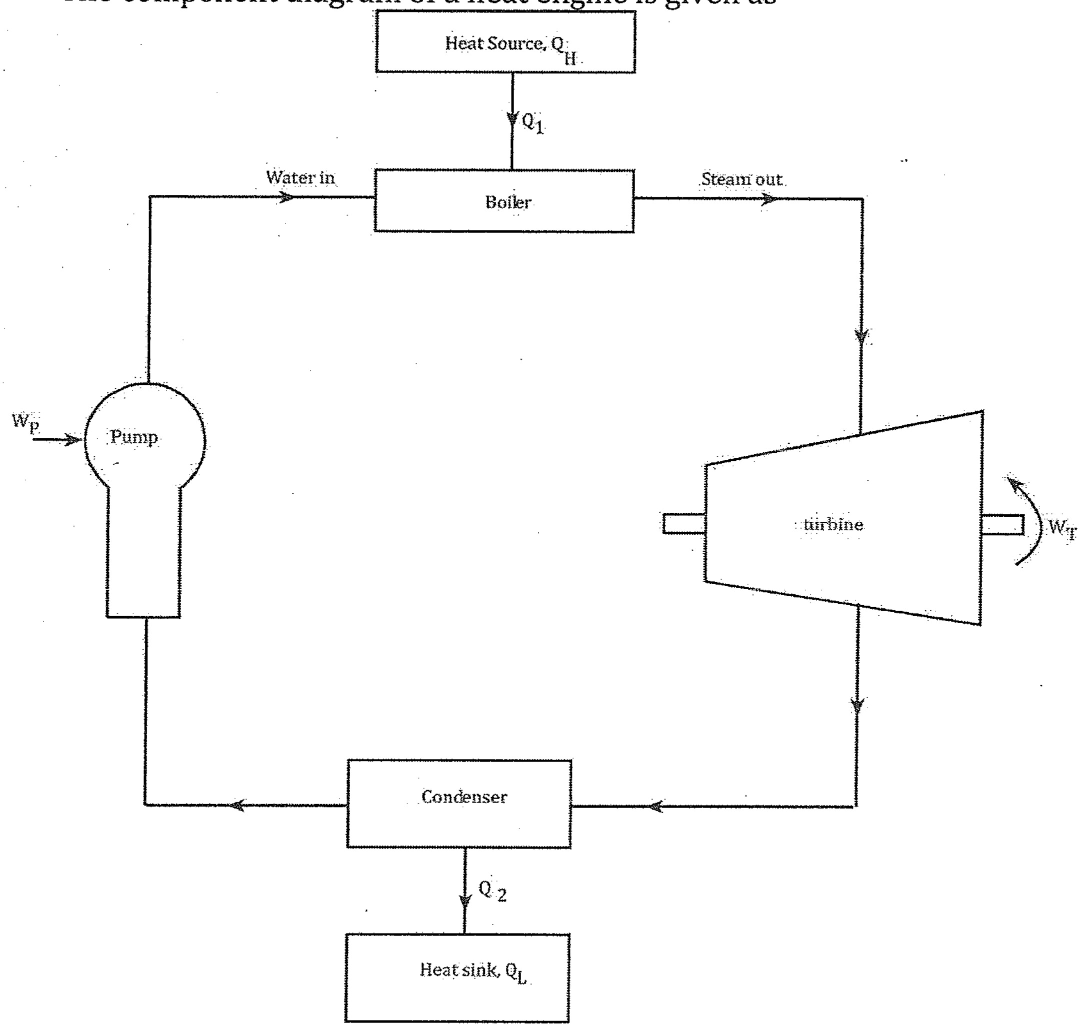


fig. line diagram

- The component diagram of a heat engine is given as



Working Principle

- In any heat engine boiler is considered as a heat source and it is represented by Q_1. The main function of boiler is to produce the steam with the help of fresh water.
- The steam which exits from the boiler is supplied to the turbine where δQ amount of heat is converted into δW amount of work. Generally, in the turbine heat energy is converted into the mechanical forms of energy.
- The low quality of steam which exits from the turbine is then supplied to the condenser where it rejects Q_2 amount of heat to the heat sink and forms in the liquid state i.e. water.
- Now, the water is supplied to the boiler thorough pump where it increases the pressure of the water and pumping process is also required to carry out the water from condenser to the boiler.

Performance:

- The performance of such devices is measured with the help of an efficiency and it is given as

$$\therefore \eta = \frac{output}{input}$$
$$= \frac{work\ done}{heat\ supplied}$$
$$= \frac{W}{Q_1}$$

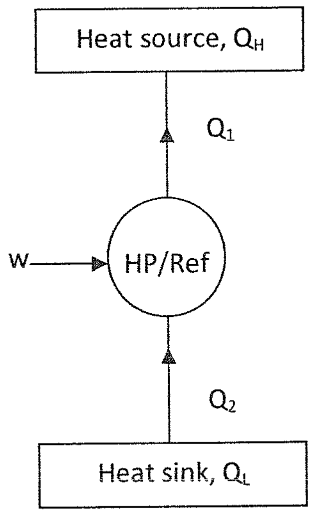
From the first law of thermodynamics

$$\oint \delta Q = \oint \delta W$$
$$Q_1 - Q_2 = W_T - W_P$$
$$\eta = \frac{Q_1 - Q_2}{Q_1}$$
$$\therefore \eta = 1 - \frac{Q_2}{Q_1}$$

So, the efficiency of any heat engine is always less than one.

Heat pump and refrigerator

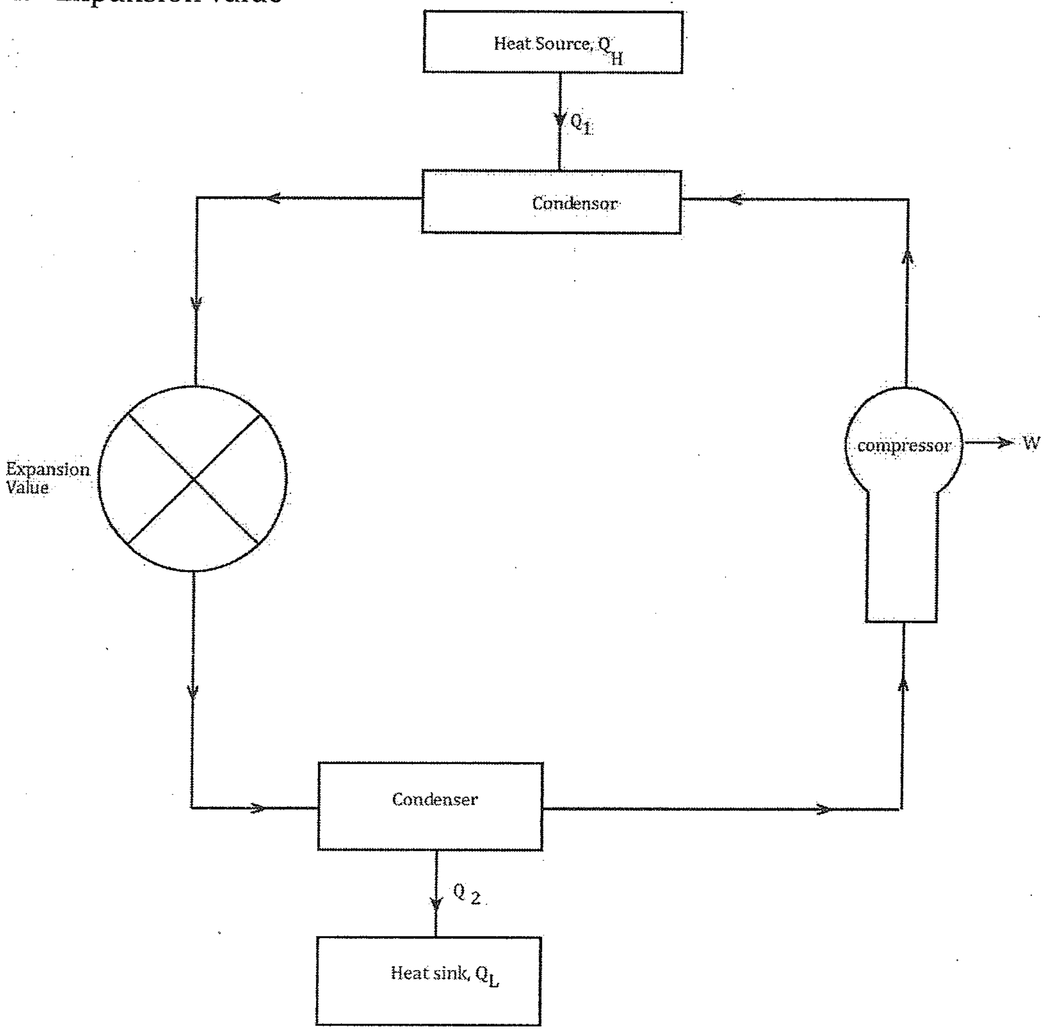
- Both are the power consuming devices, which gives their different forms of output by taking δW as a input
- Heat pump is a cyclic device which gives the heating effect as output and work as input
- Refrigerator is a cyclic device which gives the cooling effect as a output by consuming work as a input.
- The line diagram of both heat pump & refrigerator is given as



- Component diagram of a refrigerator is given as

Components

1. Evaporator
2. Compressor
3. Condenser
4. Expansion value



Working principle

- At first the liquid refrigerant is converted into the vapour refrigirant by absorbing Q_2 amount of heat from the heat sink. This process of phase change occurs in evaporator section
- The vapor refrigirant which is discharged by evaporator is supplied to the compressor where the vapor refrigerant gets compressed which results the increment in pressure & decrement of volume. This process is carried out in the compressor with the application of work.
- The high pressure vapor refrigerant is then supplied to the condenser where the high pressure vapor refrigerant is converted into the liquid state by rejecting Q_1 amount of heat to the heat source.
- The high pressure liquid which was discharged by the condenser is supplied to the expansion value where the high pressure liquid is converted into the low pressure liquid and further supplied to the evaporator.

Performance

Heat pump

The performance of heat pump is measured with the help of coefficient of performance (COP) & it is given as:

$$COP = \frac{output}{input}$$

$$= \frac{Heating\ effect}{Work\ supplied}$$

$$= \frac{Q_1}{W}$$

This expression can be further expressed as,

From the first law of thermodynamics,

$$\oint \delta Q = \oint \delta W$$

Or, $Q_1 - Q_2 = W$

So, the Cop is written as

$$COP_{HP} = \frac{Q_1}{Q_1 - Q_2} \rightarrow A$$

Refrigerator

The performance of refrigerator is measured with the help of co-efficient of performance which is given as

$$(COP)_{Ref} = \frac{output}{input}$$

$$= \frac{cooling\ effect}{work\ supplied}$$

- In refrigerator, cooling effect occurs in heat sink i.e. Q2

  So, $(COP)_{ref} = \frac{Q_2}{W}$ (W x actual work done)

From the 1st law of thermodynamics

$$\oint \delta Q = \oint \delta W$$

Or, $Q_1 - Q_2 = W$

So, $(COP)_{Ref} = \frac{Q_2}{Q_1 - Q_2} \rightarrow B$

From equation A & B

$(COP)_{HP} - (COP)_{Ref} = \frac{Q_2}{Q_1 - Q_2} - \frac{Q_2}{Q_1 - Q_2}$

$$= \frac{Q_1 - Q_2}{Q_1 - Q_2}$$

$$= 1$$

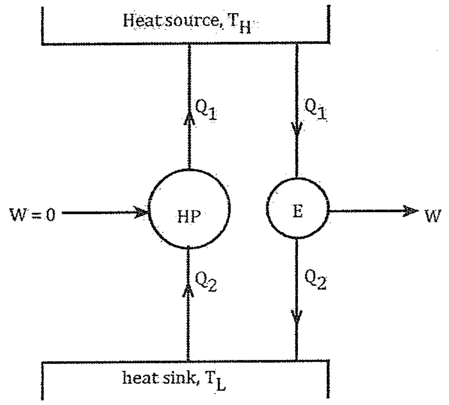
$\therefore (COP)_{HP} = (COP)_{ref} + 1$

From this expression we can conclude that the cop of heat pump is always greater than that of cop of refrigerator.

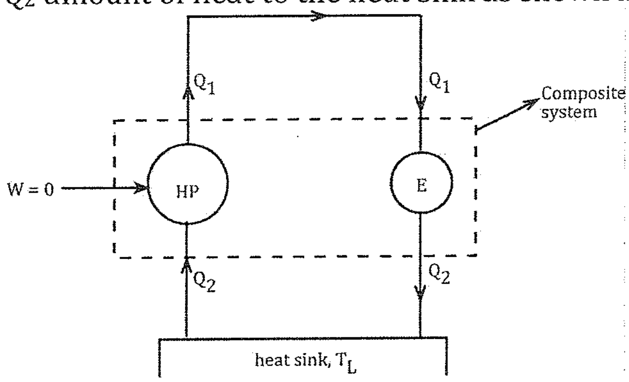
Equivalence of the two different statements of second law of thermodynamics.

- When we see the two different statements of second law for the first time, they became totally different and independent to each other.
- If we made the deep analysis of the two statements then they became dependent on each other and seem they are equivalent to each other. Their equivalence can be proved with the help of term "violation"

i) Violation of Celsius statement leads to the violation of Kelvin plank statement



- Let us consider a heat pump which is operating between T_H & T_L, which transfer the heat from low to high temperature i.e. Q_2 to Q_1, without any application of work, which completely violets clauses statement.
- Another heat engine which is operating in between the same two reservoir produce W amount of work by absorbing Q_1 amount of heat from the heat source and rejects Q_2 amount of heat to the heat sink as shown in fig. (i)



- The heat pump absorbs Q_2 amount of heat from the heat sink and supplied Q_1 amount of heat to the heat source.

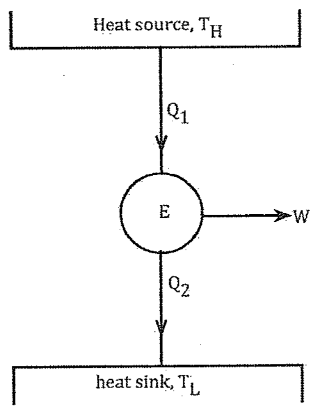
- The same amount of heat i.e. Q_1 is used to produce W amount of work by the engine. If this process continues then we can eliminate heat source, which execute a complete cycle and termed as composite system, which violets Kelvin plank statement of 2nd law.

Tutorial No. 3

1. A reversible so $n_{th} = n_{actual}$ heat engine which is operating between two different thermal reservoir operating between 800^0C and 30^0C. Determine the amount of heat supplied per kilowatt of the work done.

2. Derive an expression for the heat transfer during an isothermal process.

1. Solution



T1 = 8000C
T2 = 300C
Now

Efficiency (η_{Th}) = $\frac{T_1 - T_2}{T_1} \times 100\%$

η Heat engine = $\frac{output}{Input}$

$\frac{workdone}{Heat\ supplies}$

$\therefore \eta_{actual} = \frac{w}{Q_1}$

$\therefore \eta_{Th} = \frac{Q_1 - Q_2}{Q_1}$

$\therefore \eta_{Th} = \frac{T_1 - T_2}{T_1}$

$\eta_{Th} = \frac{800 - 30}{800} \times 100\%$
$= 96.25\%$

Again,

$\eta_{Actual} = \frac{w}{Q_1}$

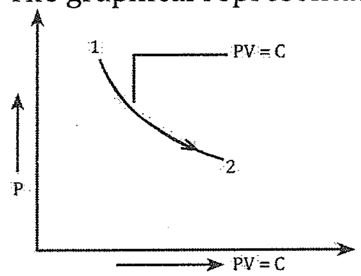
Or, $\frac{96.25}{100} = \frac{w}{Q_1}$

Or, $0.9625 = \frac{w}{Q_1}$

$\therefore Q_1 = 1.038KJ$

2. i) Isothermal process: The process in which temperature remains constant is called isothermal process.

ii) The graphical representation of isothermal process is



iii) The work done during an isothermal process is Given as:

$$\therefore \delta w = \int_{v_1}^{v_2} P.dv$$

As we know that,

$PV = C$

$P_1V_1 = P2V_2 = PV$

$\therefore$ The intermediate pressure P is given as

$$P = \frac{P_1V_1}{V}$$

Now

$$\delta w = \int_{V_1}^{V_2} \frac{P_1V_1}{V}.dv$$

$$= P_1V_1 \int_{V_1}^{V_2} \frac{dv}{V}$$

$$W = P_1V_1 in \left[\frac{V_2}{V_1}\right]$$

From the first law of thermodynamics we have

$\delta Q = dU + \delta w \rightarrow i$

For isothermal process

$dU = mc_v dt$ 'T' is constant so $dt = 0$)

From equation – i)

$\delta Q = 0 + \delta w$

$$\delta Q = P_1V_1 \ln \left[\frac{V_2}{V_1}\right]$$

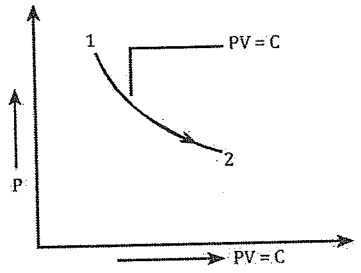
Hence the expression for heat transfer during as process goes on

$$\delta Q = P_1V_1 \ln \left[\frac{V_2}{V_1}\right]$$

3. Derive an expression for the work done during a polytrophic process, where the final equation is $\frac{0.287(T_2 - T_1)}{1-n}$, where m = 1

The work done during a polytrophic process is given as

$\delta w = \int_{v_1}^{v_2} P.dv$



As we know that
$PV^n = C$
$P_1V_1^n = P_2V_2^n$

$\therefore P = \frac{P_1V_1^n}{V^n}$

Now,

$$\therefore \delta w = \int_{v_1}^{v_2} P.dv$$

$$= \int_{v_1}^{v_2} \frac{P_1V_1^n}{v^n}.dv$$

$$= P_1V_1^n \int_{v_1}^{v_2} \frac{1}{v^n}.dv$$

$$= P_1V_1^n \int_{v_1}^{v_2} V^{-n}.dv$$

$$= P_1V_1^n \left[\frac{V^{-n+1}}{-n+1}\right]_{V_1}^{V_2}$$

$$= P_1V_1^n \left[\frac{V^{1-n}}{1-n}\right]_{V_1}^{V_2}$$

$$= \frac{P_1V_1^n}{1-n}[V_2^{1-n} - V_1^{1-n}]$$

$$= \frac{P_1V_1^n.V_2^{1-n} - P_1V_1^n.V_1^{1-n}}{1-n}$$

$$= \frac{P_1V_1^n.V_2^1.V_2^n - P_1V_1^n.V_1^{1-n}}{1-n}$$

$$= \frac{P_2V_2^n.V_2^1.V_2^n - P_1V_1^n.V_1^1.V_1^{-n}}{1-n}$$

$$\therefore W = \frac{P_2V_2 - P_1V_1}{1-n}$$

From ideal gas equation

PV = mRT

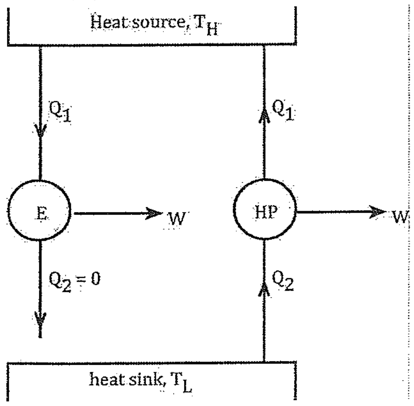
$$W = \frac{mRT_2 - mRT_1}{1-n}$$

$$= mR(T_2-T_1)/1-n$$

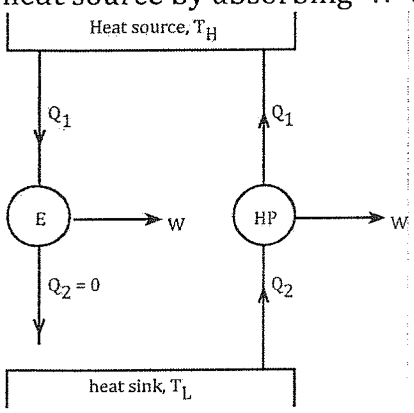
$$= \frac{1\times 0.287(T_2 - T_1)}{1-n}$$

$$= \frac{0.287(T_2 - T_1)}{1-n}$$

ii) Vidation of Kelvin Plank statement leads to the violation of clausius statement



- Let us consider a heat engine which produce 'W' amount of work by absorbing Q_1 amount of heat from the heat source and reject 0 (zero) amount of heat to the heat sink i.e. $Q_2 = 0$, which ultimately violets Kelvin Plank statement.
- In between the same two reservoirs we have another device which is heat pump, which transfer Q_2 amount of heat from the heat sink to the Q_1, amount of heat in the heat source by absorbing 'W' amount of external work.

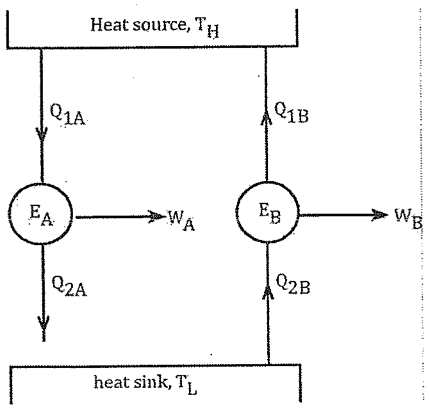


- If the heat engine produce w amount of work by the heat interaction with single reservoir then we must consider that particular work to be an internal work.
- The internal work which is produced by an engine is directly supplied to the heat pump which results the transfer of heat from low temperature to high temperature i.e. (TL to TH).
- In this mechanism there is the absence of external work, which violets clasusus statement of second low of thermodynamics.

Carnot theorem

Statement: For all the heat engines which is operating between a constant temperature heat source and a constant temperature heat sink none as a higher efficiency than the reversible heat engine.

Verification of the statement:



- Let us consider a heat engine which is operating between T_H & T_L. In the above figure (i) engine A (E_A) which produces W_A amount of work by absorbing Q_{1A} amount of heat and rejecting Q_{2A} amount of heat from the heat source and to the heat sink respectively.
- In between the two same reservoir another engine i.e. engine B(E_A) which produce W_B amount of work by absorbing Q_{1B} amount of heat from the heat source & rejecting Q_{2B} amount of heat to the heat sink.
- We have to prove that, η of a reversible heat engine >η of a normal heat engine
- Let us assume engine A(E_A) is normal heat engine and engine B(E_B) is reversible heat engine.

$\therefore \eta EB > \eta EA$

For a instant let us assume the statement is wrong.

$\therefore \eta EB > \eta EA$

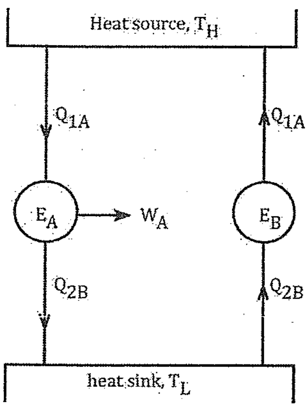
Or, $\frac{W_A}{Q_{1A}} > \frac{W_B}{Q_{1B}}$

Or, from the figure (i) $Q_{1A} = Q_{2B}$

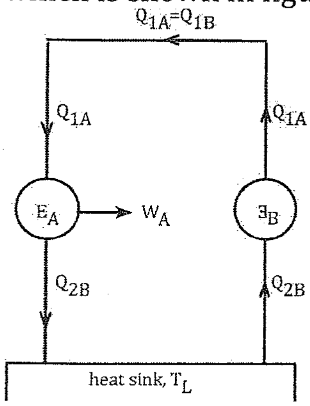
$\frac{W_A}{Q_{1A}} > \frac{W_B}{Q_{1A}}$

$\therefore WA = WB$

- As we know that the reversible heat engine also works like a heat pump. This is because any reversible process can change its direction which ultimately affects the working principle of that particular device which is shown in figure below.



- It has been proved that the work produced by the engine A is greater than that of the work produced by the engine B. If it exist the greater amount of work i.e. W_A is supplied to the reversible engine which helps to perform the reversed cycle. Better which is shown in figure below.



- From the figure the engine produce W_A amount of work by the interaction with Q_{1A} and Q_{2A}. The same amount of heat (i.e. Q_{2A}) is absorbed by the reversible engine and it reject Q_{1B} which is further directly supplied to the engine $A(E_A)$
- If this continues we don't further need of heat source which completely violets Kelvin plank statement of second law of thermodynamics so our assumption is wrong.
- Finally we can conclude that,
  $\eta E_B > \eta E_A$

Absolute thermodynamic temperature scale.

As we know that the efficiency of the heat engine which is operating between T_H & T_L is given by $\eta_{rev} = \frac{1-Q_2}{Q_1} \rightarrow$(i)

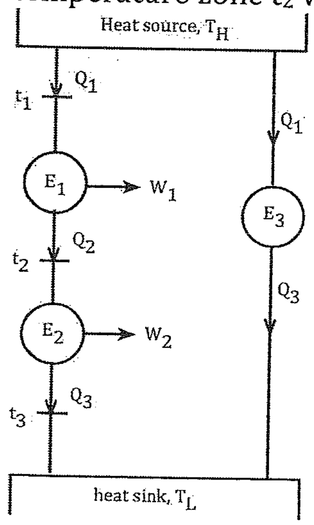
- From the second law of thermodynamics, it has been proved that there should be the needed of two different thermal reservoir in order to produce work 'W'. This indicates that work is particularly dependent upon the two different temperatures. So, the efficiency is also expressed in terms of a function
  $\therefore \eta rev = f(t_1, t_2) \rightarrow (2)$
  From equation (i) & (ii)

$$\frac{1-Q_2}{Q_1} = f(t_1, t_2) \rightarrow (3)$$

Or, $\frac{Q_2}{Q_1} = F(t_1, t_2) \rightarrow (4)$

Where F is a new function

- Let us consider a heat engine which absorbs Q_1 amount of heat from the heat source, particularly this is the temperature zone T_1 and the engine produce W amount of work by rejecting Q_2 amount of heat to the heat sink, particularly this is the temperature zone t_2 which is shown in figure.



From equation no. 4

$$\frac{Q_1}{Q_2} = F(t_1,t_2)$$

Or, $\frac{Q_2}{Q_3} = F(t_2,t_3)$

Or, $\frac{Q_1}{Q_3} = F(t_1,t_3)$

Now,

$$\frac{Q_1}{Q_2} = F(t_1,t_2) = \frac{Q_1/Q_3}{Q_2/Q_3}$$

Or, $\frac{Q_1}{Q_2} = F(t_1,t_2) = \frac{F(t_1,t_3)}{F(t_2,t_3)}$

Or, $\frac{Q_1}{Q_2} = F(t_1,t_2) = \frac{F(t_1)}{F(t_2)}$

Or, $\frac{Q_1}{Q_2} = F(t_1,t_2) = \frac{F(t_1)}{F(t_2)}$

- When the function of temperature is used to represent the quantity of heat then it becomes the definition of absolute temperature scale which is proposed by Kelvin. So, it is also called Kelvin scale.

- Let us assume,

$$\frac{F_1}{F_2}=\frac{\phi(t_1)}{\phi(t_2)}$$

Now, the final equation becomes,

$$\frac{Q_1}{Q_2}=\frac{\phi(T_1)}{\phi(T_2)}$$

Or, $\frac{Q_1}{Q_2}=\frac{T_1}{T_2}$

This is the required expression to represent the equation of absolute thermodynamic temperature scale.

Numerical

An inventor makes the following claims. Determine whether the claims are valid or invalid and explain why or why not?

Heat engine

$$\therefore \eta=\frac{output}{input}$$

$$\eta_{act}=\frac{Q_1}{W}$$

$$\therefore \eta=\frac{Q_1-Q_2}{Q_1}$$

$$\therefore \eta_{Th}=\frac{T_1-T_2}{T_1}$$

Heat pump

$$(COP)_{act}=\frac{Q_1}{W}$$

$$(COP)_{Th}=\frac{Q_1}{Q_1-Q_2}$$

$$(COP)_{Th}=\frac{T_1}{T_1-T_2}$$

Refrigerator $(COP)_{Th}=\frac{T_1}{T_1-T_2}$

a) A building receives a heat transfer of 50,000KJ/hr from a heat pump. The inside temperature is maintained at 21^0C & the surrounding is at -1^0C the inventor claims a

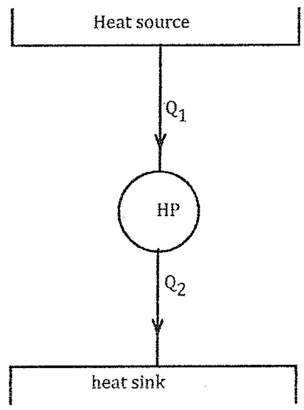
To be valid

$\therefore$ Theoretical performance > Actual performance

To be invalid

$\therefore$ Theoretical performance < Actual performance

Work input of 7,000KJ/hr is required.



Solution:

Heat transfer (Q_1) = 50,000KJ/hr

$T_1 = 21^0C = (21 + 273)k = 294k$

$T_2 = -1^0C = (-1 + 273)k = 272k$

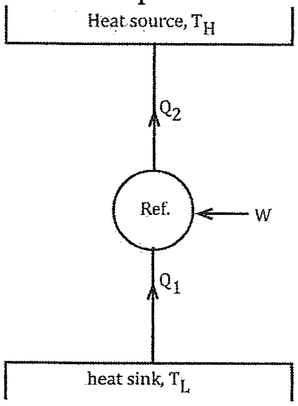
W = 7,000KJ/hr.

$$(COP)act = \frac{Q_1}{W} = \frac{50000}{7000} = 7.14$$

$$(COP)TH = \frac{T_1}{T_1 - T_2} = \frac{294}{294-273} = 13.36$$

The claim is valid because theoretical performance > Actual performance

A refrigeration unit maintains -10^0C in the refrigerator. This is kept in a room where the room temperature is 250C and has the COP of 8.5



Solution:

$T_2 = -10^0C = (-10 + 273)k = 263k$
$T_1 = 25^0C = (25 + 273)k = 298k$
Then,

$$(COP)_{Th} = \frac{T_2}{T_1 - T_2} = \frac{263}{298-263} = 7.5$$

A petrol engine operating between the temperatures 200^0C and 600^0C will produce 0.75 kw-hr consuming 0.12kg of petrol of 43,000kJ/kg calorific value.

$\therefore$ w = 0.75kw-hr

= 0.75 x 3600

W = 2700kw

0.12kg of petrol of calorific value 43,000kJ/kg

$Q_1 = 0.12 \times 43000$

$Q_1 = 5160kJ$

$$\therefore \eta_{Th} = \frac{Q_1 - Q_2}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1}$$

$$= 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{(600 + 273)}{(2000 + 273)}$$

$$\therefore \eta_{Th} = 0.61 = 61\%$$

$$\therefore \eta_{act} = \frac{w}{Q_1} = \frac{2700}{5160}$$

$$\therefore \eta act = 0.51 = 51\%$$

The claim is valid

Chapter 5

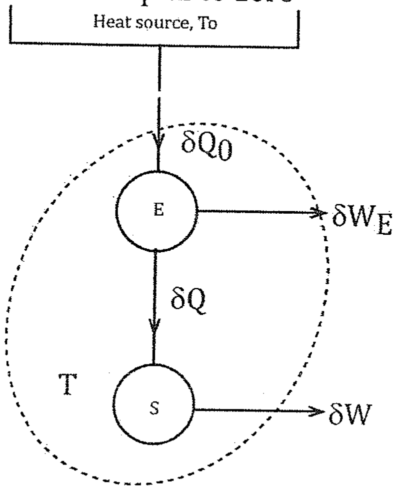
Entropy

Introduction

- When we apply 1st law of thermodynamic to a cyclic process then the amount of heat supply is converted into 'w' amount of work resulting a new property i.e. internal energy.
- So, basically 1st law of thermodynamic can be studied with the help of enthalpy which is represented by 'H'
- Likewise, when we apply the second law of thermodynamics to a cyclic process then it results a new kind of property which is entropy.
- Entropy is designed as the measurement of randomness of molecules present in any thermodynamic system due to the application of heat.
- Entropy always deals with the actual conversion of heat into studied which are energy, entropy and equilibrium.
- Entropy is an extensive property, if further the entropy is expressed per unit mass of a system then it becomes specific property which is specific entropy.

Clausius Inequality:

When the system undergoes a cyclic process then the cyclic integral of $\frac{\delta Q}{T}$ is always less than or equal to zero



- Let us consider a heat engine which absorbs δQ_0 amount of heat from the heat source and produce δW_E amount of work by rejecting δQ amount of heat.
- That rejected amount of heat is further absorbed by a thermodynamic system which produce δw amount of work.
- At first, by applying the first law of thermodynamics to a thermodynamic system, then it gives, $\oint \delta Q = \oint \delta W \rightarrow$(i)
- The system and engine constitute of single system which is shown by the doffed line in fig (1). These violets the Kelvin plank statement of second law so we can write.

$\oint \delta Q \le 0$

Single reservoir

Or, $\oint(\delta W_E + \delta W) \le 0 \rightarrow (2)$

Now,

By applying the first law of thermodynamics to a heat engine then,

$\delta Q0 - \delta Q = \delta W_E \rightarrow (3)$

From the definition of the absolute temperature scale we have,

$$\frac{\delta Q_0}{\delta Q} = \frac{T_0}{T}$$

Or, $\delta Q_0 = \delta Q(\frac{T_0}{T}) \rightarrow (4)$

From equation no. 3 & 4 we get,

Or, $\delta Q(\frac{T_0}{T}) - \delta Q = \delta W_E \rightarrow (5)$

Again,

By using the value of equation no. 1 & 5 in equation 2

We get

Or, $\oint\left(\delta Q\left(\frac{T_0}{T}\right) - \delta Q + \delta Q\right) \le 0$

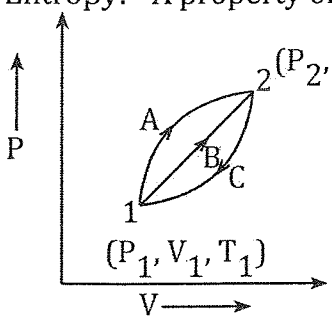
Or, $\oint\left(\delta Q\left(\frac{T_0}{T}\right) - \delta Q + \delta Q\right) \le 0$

Or, $\oint\left(\delta Q\left(\frac{T_0}{T}\right)\right)$

As we know that, T_0 is a heat source which cannot be negative or equal to zero. So, T_0 is eliminated.

$$\oint\left(\frac{\delta Q}{T}\right) \le 0$$

Entropy: - A property of a system



- Let us consider a cyclic process which consists of two different cycle formed through the three different paths usually A, B and C.
- As we know that path A & B is forward path & path C is backward path
- The two different cyclic process are
  1 – A – 2 – C – 1 & 1 – B – 2 – C – 1.

- This section we are going to verify the entropy as a property with the help of reversible process so, the equation is

  Or $\oint \frac{\delta Q}{T} = 0$

  At first let us take 1 – A - 2 - C – 1

  Or, $\int_1^2 \left(\frac{\delta Q}{T}\right)_A + \int_1^2 \left(\frac{\delta Q}{T}\right)_c = 0 \rightarrow (1)$

  Again, let us take another cyclic process
  i.e. 1 – B – 2 – C – 1

  or, $\int_1^2 \left(\frac{\delta Q}{T}\right)_B + \int_1^2 \left(\frac{\delta Q}{T}\right)_C = 0$

  Now, subtracting equation no. 2 from 1 we get,

  Or, $\int_1^2 \left(\frac{\delta Q}{T}\right)_A + \int_1^2 \left(\frac{\delta Q}{T}\right)_C - \int_1^2 \left(\frac{\delta Q}{T}\right)_B - \int_1^2 \left(\frac{\delta Q}{T}\right)_C = 0$

  Or, $\int_1^2 \left(\frac{\delta Q}{T}\right)_A + \int_1^2 \left(\frac{\delta Q}{T}\right)_C - \int_1^2 \left(\frac{\delta Q}{T}\right)_B - \int_1^2 \left(\frac{\delta Q}{T}\right)_C = 0$

  Or, $\int_1^2 \left(\frac{\delta Q}{T}\right)_A = \int_1^2 \left(\frac{\delta Q}{T}\right)_B \rightarrow (A)$

- From this equation we can conclude that the resultant property i.e. $\frac{\delta Q}{\pi}$ is independent on path so, it is termed as a thermodynamic property.
- $\frac{\delta Q}{T}$ is always dependent upon the initial & final states i.e. (i) & (ii) so, the resultant property is a point function. Finally, it generates a new property which is given as δQ = 0

  Or d S = 0

  ∴ S is constant

  So, the entropy of an isolated system is always constant.

Entropy generation during an irreversible process:

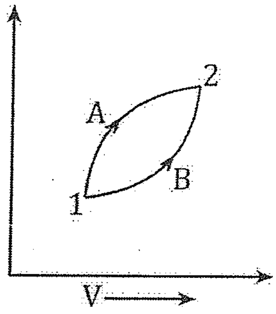
- Entropy is an extensive property which can be occurred in any types of process which exist in real nature.
- In thermodynamics, we have two types of process, i.e. reversible & irreversible process.
- According to the statement clausius inequality, the inequality sign present in irreversible process.
- In this section, we are going to find out the net entropy generation during an irreversible process.
- From the definition of clausius inequality,

  $\oint \frac{\delta Q}{T} \leq 0$

For an irreversible process

$$\oint \frac{\delta Q}{T} < 0 \rightarrow \text{(i)}$$

- Let us consider an irreversible cycle which is the combination of irreversible process and a reversible process. The state (1) changes its state up to (ii) by an irreversible process and the backward motion is directed by reversible process as shown in fig. (1).



From equation no. 1

$$\oint \frac{\delta Q}{T} < 0$$

Or, $\int_2^2 \left(\frac{\delta Q}{T}\right)_A + \int_2^1 \left(\frac{\delta Q}{T}\right)_B < 0$

As we know that
Path B indicates the reversible process

So, $\int_2^1 \left(\frac{\delta Q}{T}\right)_B = -\int_1^2 \left(\frac{\delta Q}{T}\right)_B$

Now,

Or, $\int_2^1 \left(\frac{\delta Q}{T}\right)_A - \int_1^2 \left(\frac{\delta Q}{T}\right)_B < 0$

Or, $\int_2^1 \left(\frac{\delta Q}{T}\right)_A < \int_1^2 \left(\frac{\delta Q}{T}\right)_B$

Or, $\int_2^1 \left(\frac{\delta Q}{T}\right)_B > \int_1^2 \left(\frac{\delta Q}{T}\right)_A$

Again because of the reversible process

$dS_B = \left(\frac{\delta Q}{T}\right)_B$

or, $\int_1^2 \delta S_B > \int_2^2 \left(\frac{\delta Q}{T}\right)_A$

Because of the entropy as a property, so, $dS_B = dS_A$

Or, $\int_1^2 dS_A > \int_2^2 \left(\frac{\delta Q}{T}\right)_A$

Or, $dS_A > \left(\frac{\delta Q}{T}\right)_A$

Or, $dS > \frac{\delta Q}{T}$, for an irreversible process.

* for a reversible process

$dS = \frac{\delta Q}{T}$

* any process,

$dS \geq \frac{\delta Q}{T}$

To balance the equation of irreversible process in which the entropy change is always greater than that of $\frac{\delta Q}{T}$. So, to balance the equation some amout of extra entropy should be generated which is given as,

$dS = \frac{\delta Q}{T} + \Delta S_{gen}$

where, ΔS_{gen} = entropy generation during an irreversible process.

Principle of increase of entropy:

- Statement: Entropy is an extensive property which must remain constant or increase of an isolated system.
- In this section, we are going to verify the statement and we have to find the amount of increasing entropy by using the statement.
- From the statement, let us take the example of an isolated system, i.e. the universe in universe, there are two types of process that exist which is reversible process and irreversible process.

For any process,

$dS\ dS = \frac{\delta Q}{T}$

For an isolated system

$\delta Q = 0$

Or, $\delta S_{iso} \geq 0$

For a reversible process

$dS_{iso} = 0$

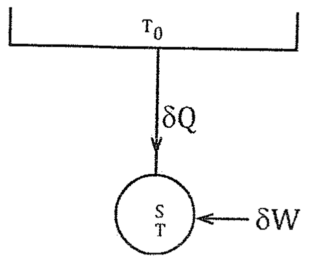
S = must be constant

For an irreversible process

Or $dS_{iso} > 0$

S must be increased

Let us verify this statement which is given as



- Let us consider a thermodynamic system which is operating between two different thermal reservoir i.3. T_0 & T

- The system absorbs 'δQ' amount of heat from the heat source and produce 'δW' amount of work.

As we know that,
Universe = system + surrounding
Or, dS = universe = dS system + dS surrounding

Or, dS system = $\frac{\delta Q}{T} \rightarrow (i)$

From equation no. (i) & (ii)

$dS_{univ} = \frac{\delta Q}{T} - \frac{\delta Q}{T_0}$

$dSuniv = \delta Q \left[\frac{1}{T} - \frac{1}{T_0}\right] \rightarrow (A)$

At first, T_o is a heat source & T is a heat sink. In this condition, $T_0 > T$

Or, $\left[\frac{1}{T} > \frac{1}{T_0}\right]$

Or, $\frac{1}{T} - \frac{1}{T_0} > 0 \rightarrow (i)$

Again if $T_0 = T$

$\frac{1}{T} - \frac{1}{T_0} = 0 \rightarrow (ii)$

From (i) & (ii)

Or, $\frac{1}{T} - \frac{1}{T_0} \geq 0 \rightarrow (B)$

From equation no. (A) & (B)
$dS_{univ} \geq 0$
From this equation we can conclude that the entropy must remain constant or increase of an isolated system.

Entropy change during an ideal gas

- In thermodynamics, every gas dynamics and their application can be studied with the help of ideal gas equation which is given as PV = mRT
- This approach gives a constant view about the matter of a thermodynamics system.
- In this section we are going to find out the net entropy change of an ideal gas.
- From the first law of thermodynamics,
  $\delta Q = dU + \delta W \rightarrow (i)$
  If the process is reversible, then
  $\delta q_{rev} = dU + \delta W_{rev}$
  Where, $\delta W_{rev} = P.\ dv$
  $dU = C_V dT$
  or, $\delta q_{rev} = C_V dT + P.dv$

Dividing the equation by T, we get,

Or, $\frac{\delta q_{rev}}{T} = C_V \frac{dT}{T} + \frac{P}{T} dv$

Or, dS = $C_V \frac{dT}{T} + \frac{P}{T} dv$

By applying the ideal gas equation we have

PV = mRT

P.V = RT

$\frac{P}{T} = \frac{R}{V}$

Or, dS = $C_V \frac{dT}{T} = \frac{R}{V}.dv$

- If the system changes its states from (i) to (ii) then the net entropy change during an ideal gas is given as,

$$\int_1^2 dS = C_V \int_1^2 \frac{dT}{T} + R \int_1^2 \frac{d\vartheta}{\vartheta}$$

Or, $[S]^2 = C_V (\ln T]^2{}_1 + R[\ln \upsilon]^2{}_1$

Or, $S_2 - S_1 = C_V (\ln T_2 - \ln T_1] + R[\ln \upsilon_2 - \ln \upsilon_1]$

$$S_2 - S_1 = C_V \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{\vartheta_2}{\vartheta_1}\right)$$

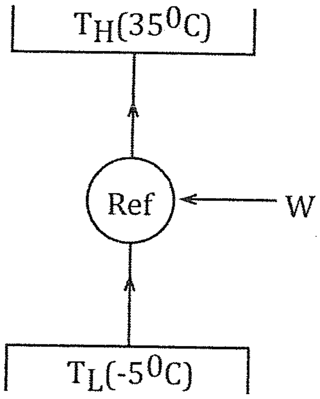
Tutorial:6

2. A cold storage is to be maintained at -5^{0}C while the surrounding temperature is 35^{0}C. The heat leakage from the cold storage to the surrounding is estimated to be 29kw. The actual COP of the refrigeration plant is 1/3rd of an ideal plant. Working between the same temperatures. Find the power required to drive the plant.

Given data:

T_H = 35^{0}C + 273 = 308k

T_L = 5^{0}C + 273 = 268k



$(cop)_{th} = \frac{Q2}{W}$

$= \frac{Q2}{Q1-Q2}$

$= \frac{T2}{T1-T2}$

$= \frac{268}{308-268}$

=6.7

$(COP)_{actual} = \frac{1}{3} x (cop)_{th}$

$= \frac{1}{3} x 6.7$

⍰ $(cop)_{actual}=2.233$

⍰ $(cop)_{actual} = \frac{Q2}{W}$

$2.33 = \frac{29}{W}$

⍰ W= 12.98 kw

3. A carnot engine which is operating between two reservoirs at temperature T_H & T_L. The work output of the engine is 0.6 times of the heat rejected. The difference in temperatures between the heat source & heat sink is 200^0c. Calculate the thermal efficiency, source temperature & sink temperature.

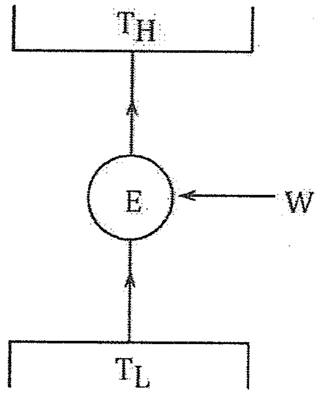
Given data:

$W = 0.6 \times Q_2$

⍰ $T_H-T_L = 200^0c$

To find

1. ⍰th
2. T_H ,T_L



Now

$⍰_{th} = \frac{W}{Q1}$

$= \frac{0.6*Q2}{W+Q2}$

$= \frac{0.6*Q2}{0.6Q2+Q2}$

$= \frac{0.6*Q2}{1.6Q2}$

=0.375%

In case of Carnot engine

$⍰_{th} = \frac{W}{Q1}$

$= \frac{Q1-Q2}{Q1}$

$⍰_{th}= \frac{T1-T2}{T1}$

Or,$0.375= \frac{200}{T1}$

⍰$T_1= 533.33^0c$

4) Air in a cylinder is compressed reversibly and isothermally from 95kpa, 25^0c to 290kpa. The initial volume is $0.162m^3$. Determine the work input , the heat transer & the entropy change of the air being compressed.

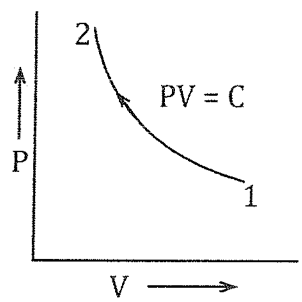
Process is reversibly isothermal process

Given data

P_1=95kpa

$=95*10^3$

$T_1= 25^0c$



= 25+273=298k

P_2= 290kpa = 290*10^3pa
V_1= 0.162m^3.

We have,

PV=C
$P_1V_1=P_2V_2$.
95*103*0.162 = 290*103*V2
V_2=0.053m^3

Work done during isothermal process is

$W= P_1V_1\ln[\frac{V2}{V1}]$
$= 95*103*0.162 \ln[\frac{0.053}{0.162}]$
= -17195.31W
= -17.195KW

Now, Q =?

⍰ SQ = du + SW (In isothermal process du=0)
⍰ SQ = SW
⍰ Q = -17.195KW

ds = ?

$ds = \frac{SQ}{T}$
$= \frac{-17.195}{298}$
= -0.0577KJ/K:

Q.5) Two blocks A and B which are initially at 95^0c and 540^0c respectively are brought together into contact and isolated from the surrounding. They are allowed to reach a final state of thermal equilibrium. Determine the entropy change of each block and of the isolated system.

Block A
Aluminium
T_A = (95+273) C_P = 0.9KJ/kgK
= 368 M_A = 0.045kg

Block B
copper
C_P = 0.385KJ/kgK T_B =(540+273)K
M_B = 0.09kg = 813K

At equilibrium condition

$Q_A = Q_B$
$(mcpdt)_A = (mcpdt)_B$
Or, $m_Acp(T-368) = m_Bcp(813-T)_B$
Or, 0.45 x 0.9(T-368) = 0.9 x 0.385(813-T)
Or, 0.425T -1490.4 = 281.7 – 0.346T
Or, 0.751T = 430.74

⍰ $T = 573.55K.$

$dS_A = m_A C_P \ln\left[\frac{T2}{T1}\right]$

$= m_A C_P \ln\left[\frac{T}{T1}\right]$

$= 0.45 \times 0.9 \ln\left[\frac{573.55}{368}\right]$

$= 0.179$ KJ/K

$dS_B = m_B C_P \ln\left[\frac{T2}{T}\right]$

$= 0.9 \times 0.385 \ln\left[\frac{813}{573.55}\right]$

$= 0.12$ KJ/K

$dS = dS_A + dS_B$

$= (0.179 + 0.12)$ J/K $= 0.299$ KJ/K

Q.6) 5 kg of water at 300c is mixed with 1kg of ice at 00c. The system is opened to atmosphere. Assuming the process of mixing is adiabatic, find the change of entropy where is equals to 4.2 KJ/kg K.

Soln:

At condition

⍰ Heat loss = Heat gain

Or, $(mC_p dT)_w = mL + (mC_p dT)_{ice}$

⍰$T=$

⍰$dS_{water} = mC_p \ln\left[\frac{T1}{T}\right]$

⍰$dS_{ice} = \frac{mL}{273} + mC_p \ln\left[\frac{T}{T2}\right]$

⍰ $dS = dS_W + dS_{ice}$

$=$

Entopy(contd....)

*Entropy change during the different reversible process of an ideal gas equation:

Isochoric process

-: As we know that, in isochoric process, the volume of the system remains constant
i.e. V=Constant

-: so the heat transfer during an ideal gas is given by,

$S_Q = mCvdT$ -------- (1)

Or, $\frac{SQ}{m} = CvdT$

Or, $SQ = CvdT$

If the process is reversible, then,

Or, $SQ_{rev} = CvdT$

Now, dividing the about equation by 'T', we get,

Or, $\frac{SQrev}{T} = \frac{CvdT}{T}$

Or, $dS = \frac{CvdT}{T}$

-: If the system changes its states from (1) to (2) then the net entropy change during an isochoric process is given by:

$$\int_1^2 dS = Cv \int_1^2 \frac{dT}{T}$$

Or, $[S]_1^2 = Cv[\ln T]_1^2$

Or, $S_2 - S_1 = Cv\,[\ln T_2 - \ln T_1]$

Or, $S_2 - S_1 = Cv\ln\left[\frac{T2}{T1}\right]$-----(A)

-: In isochoric process, the ideal gas is converted into,

$$Pv = mRT$$
$$p.v = RT$$
$$\frac{P}{T} = \frac{R}{u}$$
$$\frac{P}{T} = \text{constant}$$
$$\text{Or, } \frac{P1}{T1} = \frac{P2}{T2}$$
$$\text{Or, } \frac{P1}{P2} = \frac{T1}{T2}$$

Also,

$$S_2\text{-}S_1 = Cv\ln\left[\frac{P2}{P1}\right] \text{ ----- (B)}$$

Equation (A) & (B) represents the required equation of entropy change during the different reversible process of an ideal gas equation. (in isochoric process)

-: Isobaric process:

-: As we know that, in isobaric process, pressure of the system remains constant
i.e. P = constant
-: so, the heat transfer during an ideal gas is given by

$$SQ = mCpdT \text{----- (1)}$$
$$\text{Or, } \frac{SQ}{m} = CpdT$$
$$\text{Or, } SQ = CpdT$$

If the process is reversible, then,

$$\text{Or, } SQ_{rev} = CpdT$$

Now, dividing the above equation by 'T' we get,

$$\frac{SQrev}{T} = CpdT$$
$$\text{Or, } ds = \frac{CpdT}{T}$$

If the system changes its state from (1) to (2) then the net entropy change during an isobaric process is given as

$$\int_1^2 ds = Cp\int_1^2 \frac{dT}{T}$$
$$\text{Or, } [S]_1^2 = Cp[\ln T]_1^2$$
$$\text{Or, } [S_2 - S_1] = Cp[\ln T_2] - \ln T_1]$$
$$\text{Or, } S_2\text{-}S_1 = Cp\ln\left[\frac{T2}{T1}\right] \text{-----(A)}$$

In isobaric process, the ideal gas is converted into,

$$PV = mRT$$
$$\text{Or, } P.v = RT$$
$$\text{Or, } \frac{v}{T} = \frac{R}{P}$$
$$\text{Or, } \frac{v}{T} = c$$

So, $\frac{v1}{T1} = \frac{v2}{T2}$

Or, $\frac{v1}{v2} = \frac{T1}{T2}$

So,

S_2-S_1 =Cpln$[\frac{v2}{v1}]$ -----(B)

Equation (A) & (B) represents the basic equation of entropy change in an ideal gas during isobaric process.

-: Isothermal process:

As we know that, the temperature of the system remains constant. i.e.T= constant

-: From the 1st law of thermodynamics

SQ =du + SW

Representing the above equation in terms of mass 'm' we get,

SQ =du +SW

If the process is reversible

Or, SQ_{rev} =du + SW_{rev}

If the process is isothermal ,then

du = 0

so,SQ_{rev} = p.du

Dividing the equation by 'T' we get,

Or, $\frac{SQrev}{T} = \frac{P.du}{T}$

By using an ideal gas eqn, then

Pv =mRT

Or, p.u = RT

Or, $\frac{p}{T} = \frac{R}{u}$

Or, ds =$\frac{R}{u} du$

If the system changes its state from (1) to (2) then the eq^n becomes

$\int_1^2 ds = R \int_1^2 \frac{du}{u}$

Or, $[s]_1^2 = R[lnu]_1^2$

Or, s2-s1 = Rln$[\frac{u2}{u1}]$ ----- (A)

Again,

PV = mRT

P.v = RT

p.v = c

p1v1 = p2v2

$\frac{p1}{p2} = \frac{v2}{v1}$

So the eq^n (A) becomes,

S2-S1 = Rln$[\frac{p1}{p2}]$ ----- (B)

Eq^n (A) & (B) represents the basic equation that represents the enropy change of an ideal gas during isothermal process.

***Availability**

-:The maximum useful work that can be achieved from the system surrounding phenomenon, until the system comes to a dead state

-: Dead state is that particular state in which the system in which the system & surrounding comes to an equilibrium condition.

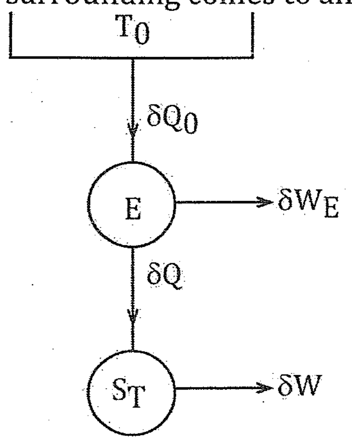


Fig:Available entropy from a system.

-: let us consider a heat engine which absorbs SQ amount of heat from the heat source. 'To' The engine produce SQ amount of heat to the system. The system absorbs SQ amount of heat and produce SW amount of water.

-: The maximum work that can be achieved from this particular system is given as,

$$SW_{max} = SW_E + SW \text{ ----- (i)}$$

By applying first law of thermodynamics to the heat engine, we have,

$$SQ_o - SQ = SW_E$$

By the definition of absolute temperature scale,

$$\frac{SQo}{SQ} = \frac{To}{T}$$

$$\text{Or, } SQo = SQ\left(\frac{To}{T}\right)$$

Now, eq^n becomes,

$$\text{Or, } SQ\left(\frac{To}{T}\right) - SQ = SW_E \text{ ----- (ii)}$$

-:Again, By Applying the 1st law of thermodynamics to the system, then,

$$\text{Or, } SQ = SE + SW \text{ ----- (iii)}$$

From eq^n (i)

$$SW_{max} = SQ\left(\frac{To}{T}\right) - SQ + SW$$

$$= SQ\left(\frac{To}{T}\right) + SWE - SQ$$

$$\text{Or, } SW_{max} = SQ\left(\frac{To}{T}\right) - SE$$

$$= To\left(\frac{SQ}{T}\right) - dE$$

$$\text{Or, } SW_{max} = To\, dS - dE$$

Now, the maximum work is given by,

Or, $SW_{max} = \int_0^1 TodS - dE$
$= To[S]_0^1 - [E]_0^1$
$= To[S1\text{-}S0] - [E1\text{-}E0]$
$W_{max} = ToS1 - ToS0 - E1 + E0$

Now, the work which is done by the surrounding to the system is given by,
$SW = P_oV_o - P_oV_1$
Now, finally the maximum useful work that can be achieved from any system is given by
SW_{max} (useful) $= W_{max} - SW$
$= [ToS1\text{-}ToSo\text{-} E1 + E0] - [PoVo\text{-}PoV1]$
$= [ToS1\text{-}ToSo\text{-} E1 + E0] - PoVo + PoV1]$
$= \frac{[Eo - ToSo - PoVo]}{A0} - \frac{[E1 - ToS1 - PoV1]}{A1}$
So, SW_{max} [useful] $= A_0A_1$
Where,
A_0 & A_1 are availability function.

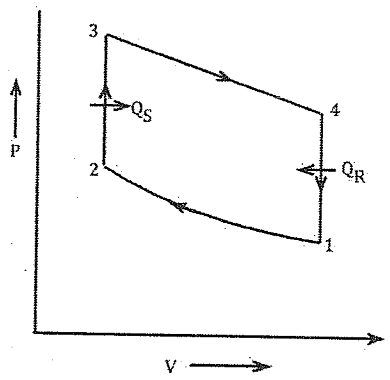
***Irreversibilty**
-: It is defines as the last work when we operate any actual system with compared to an ideal system.
-: Mathmatically, it is given as,
$I = SWmax - SW$
-: To find the expression of any irreversibility, we have,
SI = SWmax −SW
Or, SI =SWE +SW −SW
Or, SI =SWE
-: From the first law of thermodynamics, we have,
Or,SI =SQo −SQ
Or,SI = SQo $[1 - \frac{SQ}{SQo}]$
-: from the definition of absolute temperature scale,
SI =SQo$[1 - \frac{T}{To}]$
At first, To>T
$1 > \frac{T}{To}$
$1 - \frac{T}{To} > 0$ ----- (A)
Now,
To = T
Or, $1 - \frac{T}{To} = 0$ ----- (B)
From eqn (A) & (B)
SI ≥ 0

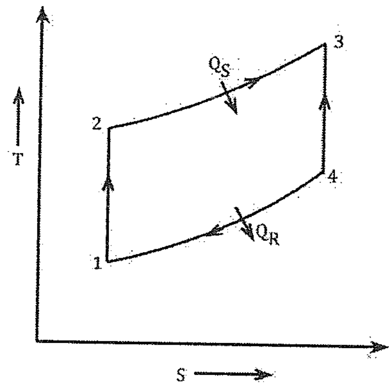
So, we can conclude that for any process there ie the presence of irreversibility factor.

Chapter: 6
Some power cycle

1.) Vapour power cycle $\Longrightarrow$ Rankine cycle
2.) Air standard cycle Carnot cycle
Brayton cycle
3.) Internal combination enginge Otto cycle
Diesel cycle

Internal Combination engine
1) Otto cycle
Introduction:
-: It was developed by DR.A.N.Otto
-: Now- a- days petrol engine are operating by using the principle of otto cycle.
-: The different process involved in the otto cycle are:-
i) Isentropic compression process $\longrightarrow$
ii) Constant volume heat Addition process
iii) Isentropic expansion process
iv) Constant volume heat rejection process
-: The P-V and T-S diagram of an otto cycle is





Working principle

-: (process 1-2) Isentropic Compression process:

In this process the mass of air and fuel gets compressed during compression stroke. During this process, the pressure increases and the volume decreases as shown in P-V diagram from the T-S diagram the entropy remains constant whereas temperature slightly increases.

-(process 2-3) Constant volume heat addition process:

In this process the heat is added on the system. During this process the pressure increases whereas the volume remains constant. From the T-S diagram both temperature and entropy increases.

So the heat addition is given as

$$QS = mCvdT$$
$$= m.Cv(T_3 - T_2)$$

-(process 3-4): Isentropic expansion process:

In this process expansion process occurs on the system. During this process the pressure decreases whereas the volume increases from the T-S diagram the entropy remains constant and temperature deceases.

-: (process 4-1): Constant volume heat rejection process:

This is the final process of the cycle in which heat is rejected by the system. During this process whereas the volume remains constant. From the T-S diagram, both temperature & entropy decreases.

So the heat rejection is given as

$$QR = mCvdT$$
$$= m.Cv(T_4 - T_1)$$

Performance:- The performance of an Otto cycle is measured with the help of an efficiency which is given as

$$⍰_{Thermal} = \frac{output}{input}$$
$$= \frac{Work\ done(W)}{Heat\ supplied(QS)}$$

-: By applying the first law of thermodynamics, we have,

$$QsW = QsQ$$
$$Or,\ W = Qs - QR$$
$$So,\ the\ ⍰_{Thermal} = \frac{Qs-QR}{Qs}$$

Now,

$$⍰_{Thermal} = \frac{m.Cv(T3-T2)- m.Cv(T4-T1)}{m.Cv(T3-T2)}$$
$$= 1 - \frac{m.Cv(T4-T1)}{m.Cv(T3-T2)}$$
$$⍰_{Thermal} = 1 - \frac{(T4-T1)}{(T3-T2)} \quad ----- (A)$$

-: Equation (A) represents the efficiency of an otto cycle in terms of temperature.

-: If the temperature is not given then the efficiency is also expressed in terms of compression ratio(r) which is given as

Total volume
$$r = \frac{V1}{V2} = \frac{V4}{V3}$$
clearance volume

process 1-2: Isentropic compression process

$$or,\ \frac{T2}{T1} = \left(\frac{V1}{V2}\right)^{r-1}$$
$$or,\ \frac{T2}{T1} = (r)^{r-1}$$
$$T_2 = T_1.\ (r)^{r-1} \quad ----- (1)$$

Process 3-4: Isentropic expansion process

$$or,\ \frac{T4}{T3} = \left(\frac{V4}{V3}\right)^{r-1}$$
$$or,\ \frac{T4}{T3} = (r)^{r-1}$$
$$T_3 = T_4.\ (r)^{r-1} \quad ----- (2)$$

By putting the value of eqn (1) & (2) in eqn (A) we get,

$$⍰_{\text{Thermal}} = 1 - \frac{(T4-T1)}{(T3-T2)}$$

$$= 1 - \frac{(T4-T1)}{\left(\mathrm{T4}.(r)^{r-1} - \mathrm{T1}.(r)^{r-1}\right)}$$

$$= 1 - \frac{(T4-T1)}{\left(\mathrm{T4}-\mathrm{T1}\right).(r)^{r-1}}$$

$$⍰⍰_{\text{Thermal}} = 1 - \frac{1}{\left(\,(r)^{r-1}\right)} \text{ ----- (B)}$$

Eqn (B) represents the efficiency of an otto cycle in terms of compression ratio.

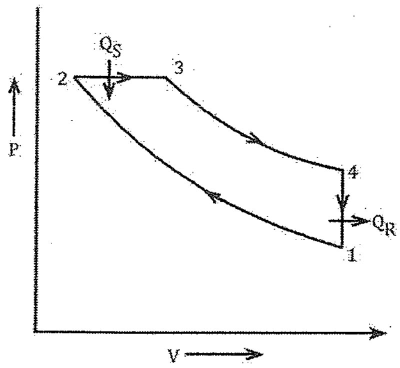
2.) Diesel cycle

Introduction

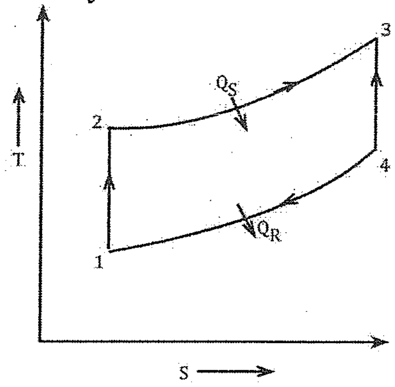
-: It was introduced by Rudalph Desiel
-: Now- a- days diesel engines are operating by using the principle a diesel cycle.
-: The different process involved in the diesel cycle is

1. Isentropic compression process
2. Constant pressure heat addition process
3. Isentropic expansion process
4. Constant volume heat rejection process.

-: The P-V and T-S diagram of a diesel cycle is



P - V diagram



T - S diagram

Working principle:

(Process 1-2) Isentropic Compression process
-: In this process mass of air is compressed which is termed as isentropic compression process. During this process the pressure increases whereas the volume decreases from the T-S diagram, the entropy remains constant & the temperature increases.
(Process 2-3) Constant pressure heat addition process
-: In this process, heat is added on the system. During this process the pressure remains constant whereas the volume increases slightly up to cut off volume (V_3). From the T-S diagram both temperature and entropy increases. Then heat addition is given by,

$$QS = m.CvdT$$
$$= m.Cv(T_3-T_2)$$

(Process 3-4) Isentropic expansion process
-: In this process, the expansion of piston takes place. During this process the pressure decreases whereas the volume increases. From the T-S diagram, the entropy remains constant & temperature slightly decreases.
(Process 4-1) Constant volume heat rejection process
-: The heat is rejected by the system by keeping the system volume constant, during the exhaust stroke of an engine. In this process the pressure decreases from the T-S diagram both temperature & entropy decreases. The heat rejection is given by

$$QR = mCvdT$$
$$= m.Cv(T_4-T_1)$$

Performance

The performance of a diesel cycle is measured with the help of efficiency and it is given as

$$\eta_{Thermal} = \frac{output}{input}$$
$$= \frac{Work\ done(W)}{Heat\ supplied(QS)}$$

from the first law of thermodynamics, we have,

QQ = QW

Or, Qs-QR = W

So, the $\eta_{Thermal} = \frac{Qs-QR}{Qs}$

Or, $\eta_{Thermal} = \frac{m.Cp(T3-T2)-m.Cv(T4-T1)}{m.Cp(T3-T2)}$

$= 1 - \frac{Cv(T4-T1)}{Cp(T3-T2)}$

$\eta_{Thermal} = 1 - \frac{(T4-T1)}{\psi(T3-T2)}$ ----- (A)

-: Equation (A) represents the efficiency of diesel cycle in terms of temperature.

-: If any of temperature is not given then the (r) efficiency is also expressed in terms of compression ratio(r) & cut-off ratio.

- Compression ratio (r) = $\frac{V1}{V2}$
- Cut-off ratio (s) = $\frac{V3(Cut-off\ volume)}{V2(Clearence\ volume)}$
- Expansion ratio = $\frac{V4}{V3} = \frac{V1}{V3} = \frac{V1}{V2} * \frac{V2}{V3}$
  $= \frac{r}{s}$
- Process 1-2:- Isentropic compression process
  or, $\frac{T2}{T1} = \left(\frac{V1}{V2}\right)^{r-1}$
  or, $T_2 = T_1.\ (r)^{r-1}$ ----- (1)
- Process 2-3:- Constant pressure heat addition process
  p = c
  $\frac{V}{T} = c$
  Or, $\frac{V2}{T2} = \frac{V3}{T3}$
  Or, $\frac{T3}{T2} = \frac{V3}{V2}$
  Or, $\frac{T3}{T2} = S$
  Or, T3 = T2.S
  T3 = T1.$(r)^{r-1}$.S ------ (2)

Process 3-4 :- Isentropic expansion process

or, $\frac{T3}{T4} = \left(\frac{V4}{V3}\right)^{r-1}$

or, $\frac{T3}{T4} = \left(\frac{r}{S}\right)^{r-1}$

or, T4 = $\frac{T3}{(r/S)^{r-1}}$

or, T4 = $\frac{T1.(r)^{r-1}.S}{(r/S)^{r-1}}$

or, T4 = $\frac{T1.(r)^{r-1}.(S)^{r-1}.S}{(r)^{r-1}}$

or, $T4 = T1.S.(S)^{r-1}$
or, $T4 = T1.S^{r}$ ----- (3)

From eqn no (A)

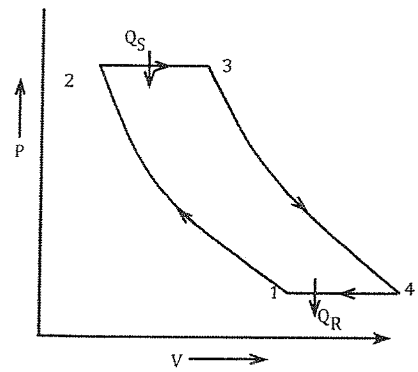
$$\eta_{Thermal} = 1 - \frac{(T4-T1)}{\psi(T3-T2)}$$
$$= 1 - \frac{1}{r}\left[\frac{T1(S)^{r}-T1}{T1.(r)^{r-1}.S-T1(r)^{r-1}}\right]$$
$$= 1 - \frac{1}{r}\left[\frac{T1.(S^{r}-1)}{T1.(r)^{r-1}.(S-1)}\right]$$
$$= 1 - \frac{1}{r}\left[\frac{(S^{r}-1)}{(r)^{r-1}.(S-1)}\right]$$
$$= 1 - \frac{1}{r(r)^{r-1}}\left[\frac{(S^{r}-1)}{(S-1)}\right] \text{ ----- (B)}$$

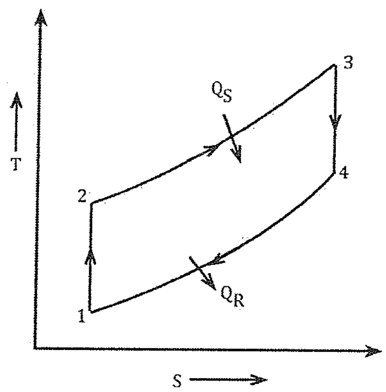
Eqn (B) represents the efficiency of diesel cycle in terms of cut-off ratio.

- Air Standard cycles
- **Brayton Cycle:** Introduction

-: It is a air standard cycle which use atmospheric air as working medium.
-: Now- a- day gas turbine power plant is operating by using the principle brayton cycle
-: The brayton cycle consists of combination chamber, turbine, condenser, compressor.
-: The brayton cycle consists of four different process.
1. Isentropic compression process.
2. Constant pressure heat addition process
3. Isentropic expansion process
4. Constant pressure heat reduction process.

-: The P-V & T-S diagram of a Brayton cycle is





Working principle:

1. Process (1-2) : Isentropic Compression process

In this process, the atmospheric air is compressed during the compression process. During this process the pressure increases whereas the volume decreases from the T-S diagram the entropy remains constant (S1 = S1) whereas the temperature slightly increases.

2. Process (2-3) : Constant pressure heat addition process

In this process, the heat is added on the system by keeping the system pressure constant. During this process, the volume increases from the T-S diagram both temperature and entropy increases.

The heat addition in this process is given by

$$QS = m.Cv.dT$$
$$= m.Cv.(T3-T2)$$

3. Process (3-4): Insentropic expansion process

In this process, the expansion of air takes places in the turbine. During this process the pressure decreases whereas volume further increases. From the T-S diagram the entropy remains constant (S3 = S4) and temperature decreases.

4. Process (4-1) Constant pressure heat rejection process

In this process, heat is rejected by the system by keeping the system pressure constant. During this process volume decreases from the T-S disgram both temperature and entropy decreases. The heat rejection in this process is given by

$$QR = m.Cp.dT$$
$$= m.Cp.(T4-T1)$$

Performance

The performance of a Brayton cycle is measured with the help of an efficiency and is given as

$$\text{?}_{Thermal} = \frac{output}{input}$$
$$= \frac{Work\ done(W)}{Heat\ supplied(QS)}$$

From the first law of thermodynamics, we get,,

$$QsQ = QsW$$
$$\text{Or, } Qs\text{-}QR = W$$
$$\text{So, the } \text{?}_{Thermal} = \frac{Qs-QR}{Qs}$$
$$\text{Or, } \text{?}_{Thermal} = \frac{m.Cp(T3-T2)-m.Cp(T4-T1)}{m.Cp(T3-T2)}$$
$$\text{?}_{Thermal} = 1 - \frac{(T4-T1)}{(T3-T2)} \quad \text{----- (A)}$$

-: Equation (A) represents the efficiency of a Brayton cycle in terms of temperature.

-: If any of temperature is unknown then the efficiency is also expressed in terms of pressure ratio which is given as

$$r_p = \frac{P2}{P1} = \frac{P3}{P4}$$

(Process 1-2): Isentropic compression process

$$\text{Or, } \frac{T2}{T1} = \left(\frac{P2}{P1}\right)^{\frac{r-1}{r}} \qquad \text{----- (1)}$$

(Process 3-4): Isentropic expansion process

$$\text{Or, } \frac{T3}{T4} = \left(\frac{P3}{P4}\right)^{\frac{r-1}{r}}$$

From the P-V diagram

P2 = P3

P1 = P4

$$\text{Or, } \frac{T3}{T4} = \left(\frac{P2}{P1}\right)^{\frac{r-1}{r}} \qquad \text{----- (2)}$$

From eqn (1) and (2)

$$\text{Or, } \frac{T2}{T1} = \frac{T3}{T4}$$

$$\text{Or, } \frac{T4}{T1} = \frac{T3}{T2}$$

By subtracting 1 on both sides, we get,

$$\text{Or, } \frac{T4}{T1} - 1 = \frac{T3}{T2} - 1$$

$$\text{Or, } \frac{T4-T1}{T1} = \frac{T3-T2}{T2}$$

$$\text{Or, } \frac{T4-T1}{T3-T2} = \frac{T1}{T2} = \left(\frac{P1}{P2}\right)^{\frac{r-1}{r}}$$

$$= \left(\frac{1}{rp}\right)^{\frac{r-1}{r}}$$

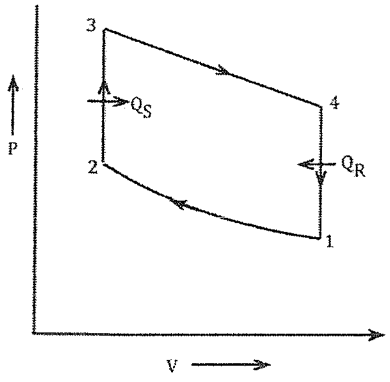
From eqn (A)

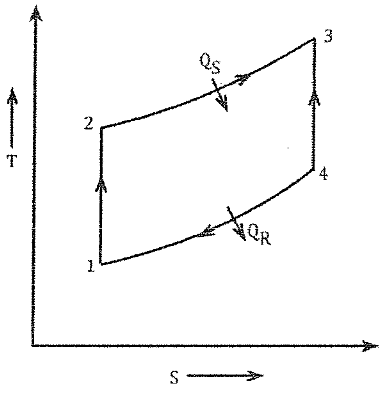
$$\eta_{th} = 1 - \frac{1}{(rp)^{\frac{r-1}{r}}} \qquad \text{----- (B)}$$

Eqn (B) represents the efficiency of an Brayton cycle in terms on pressure ratio.

Q) The compression ratio of an air standard otto cycle is 8. The pressure and temperature at the beginning of the compression stroke is 0.1 Mpa and 150c.
The heat supplied to the cycle is 1800 KJ/kg. Determine

1) The pressure and temperature at all the points of the cycle.
2) The thermal efficiency.





Soln:

$(r) = \frac{V1}{V2} = \frac{V4}{V3} = 8$

P1 = 0.1Mpa = $0.1*10^6$pa

T1 = 15 + 273 = 288 K

QS = 1800 KJ/kg

To find:

(1)P2 ,T2, P3, T3, P4, T4, η_{th} = ?

Process 1-2:

$\frac{P2}{P1} = \left(\frac{V1}{V2}\right)^r$

$P2 = P1\left(\frac{V1}{V2}\right)^r$

$P2 = 0.1*10^6*8^{1.4}$ r=1.4

= 1837917.368pa.

$\frac{T2}{T1} = \left(\frac{V1}{V2}\right)^{r-1}$

$T2 = T1*\left(\frac{V1}{V2}\right)^{r-1}$

$= 288*(8)^{1.4-1}$

$= 288*(8)^{0.4}$

= 814.587

(Process 2-3):

V=C

$\frac{P}{T} = C$

Or, $\frac{P2}{T2} = \frac{P3}{T3}$

Or, QS = mCv.(T3-T2)

Or, 1800 * 103 = 1* 0.718* 103(T3-T2)

Or, 1800 * 103 = 0.718* 103T3 - 0.718* 103T2

Or, T3 = 1800*103 + $\frac{0.718*103*8114.587}{0.718*1.3}$

Or, T3 = 1800 * 103 + $\frac{58487.466}{0.718*103}$

Or,T3 =

$\therefore$T3 = K

Or P3 = pa

(Process 3-4):

$\frac{P3}{P4} = \left(\frac{V4}{V3}\right)^r$

$P4 = P3\left(\frac{V3}{V4}\right)^r$

$\frac{T3}{T4} = \left(\frac{V4}{V3}\right)^{r-1}$

$T4 = T3 \cdot \left(\frac{V4}{V3}\right)^{r-1}$

Or $T4 =$ K

$\eta_{th} = 1 - \frac{1}{(r)^{\frac{r-1}{r}}}$

$= 1 - \frac{1}{(8)^{1.4-1}}$

$= 56\%$

Vapor power cycle

Ranking cycle

Introduction:

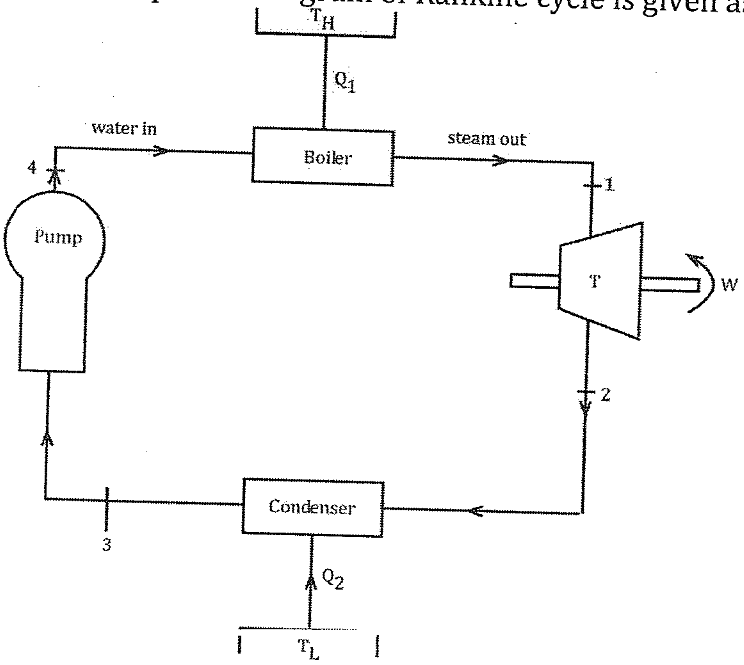
-: In this cycle vapor is used as a working medium.

-: Now a day's stream power plant, heat engine are operating by using these cycle.

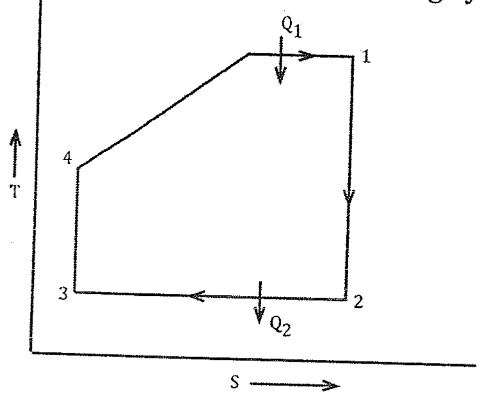
-: It consists of 4 different processes.

1) Isentropic expansion process
2) Constant pressure and temperature heat rejection process
3) Isentropic Compression process
4) Constant pressure and temperature heat addition process

-: The component diagram of Rankine cycle is given as



-: The T-S diagram of a Ranking cycle is given as



Working principle

Process 1-2: Isentropic expansion process.

The stream which exits from the boiler is subjected to the expansion process in turbine, where the heat 'Q' is converted into the work 'W' from the T-S diagram the entropy remains constant where temperature decreases slightly.

Process 2-3: Constant pressure and temperature heat rejection process

The stream (low quality stream) which exits from the turbine is supplied to the condenser where the heat rejection process takes place in the device know as condenser from the T-s diagram the temperature remains constant and entropy decreases.

Process 3-4: Isentropic Compression process

The water which exits from the condenser is supplied to the pump where the compression process takes place from the T-s diagram the entropy remains constant and temperature increases slightly under the working limit of the plant.

Process 4-1: Constant pressure and temperature heat addition process

The heat addition process takes place in the boiler where the water is converted into the stream. The high quality of stream which is produced by the boiler is supplied to the turbine from the T-S diagram the entropy increases while the temperature increases by linearly and constant temperature process.

Performance:

The performance of a Ranking cycle measured with the help of an efficiency which is given as

$$?_{\text{Thermal}} = \frac{output}{input}$$
$$= \frac{Work\ done(W)}{Heat\ supplied(QS)}$$
$$= \frac{WT - Wp}{Q1} \quad \text{------ (A)}$$

All of the devices which are used in a Ranking cycle follow the steady flow energy equation (SFEE)

$$\text{Or, } m(h1 + \frac{V1^2}{2} + gZ1) + Q = m(h2 + \frac{V2^2}{2} + gZ2) + W$$
$$\text{Or, } m(h1) + Q = m(h2) + W$$

Now applying the basc formula in the cycle, we have,

Process 1-2:

$$m(h1) + 0 = m(h2) + WT$$
$$WT = m(h1 - h2) \quad \text{----- (1)}$$

Process 2-3:

$$m(h1) + Q = m(h2) + W$$
$$\text{Or, } m(h2) + Q2 = m(h3) + 0$$
$$\text{Or } Q2 = m(h3 - h2)$$

Process 3-4:

$$m(h1) + Q = m(h2) + W$$

Or, m(h3) +0 = m(h4) – Wp
Or, Wp = m(h4-h3) ------ (3)

Process 4-1:

m(h1) + Q = m(h2) + W
Or, m(h4) + Q = m(h1) + 0
Or, Q1 = m(h1-h4) ------ (4)

From eqn no (A)

$$\eta_{Rankine} = \frac{WT-Wp}{Q1} = \frac{m(h1-h2)-m(h4-h3)}{m(h1-h4)}$$

Finally, h3 ≈ h4

$$\eta_{Rankine} = \frac{m(h1-h2)}{m(h1-h4)} = \frac{(h1-h2)}{(h1-h4)}$$

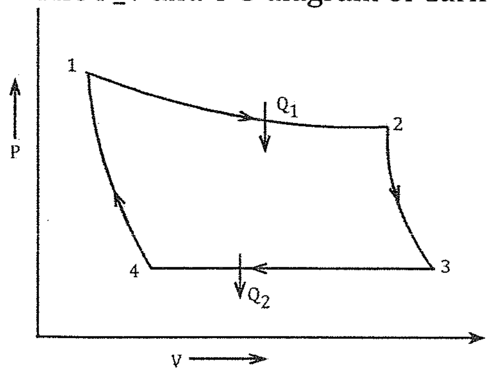
This is the required expression of a Rankine cycle.

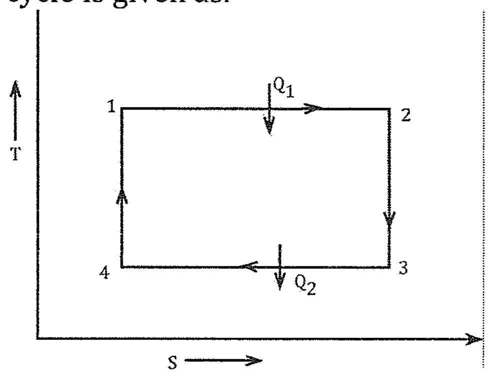
Carnot cycle

Introduction:
It is the first made cycle developed by Carnot.
In this cycle air is used as a working medium.
Now-a-days gas turbine power plants are operating by using this cycle.
It consists of four different processes.

1. Isothermal heat addition process.
2. Isentropic expansion process
3. Isothermal heat rejection process
4. Isentropic compression process

-: The P_V and T-S diagram of Carnot cycle is given as.





Process 1-2: Isothermal heat addition process

In this process, the heat addition takes place inside the combustion chamber. Atmospheric air is heated and supplied to the turbine. In those processes, pressure decreases whereas volume increases from the T-S diagram temperature is constant whereas entropy increases from s1 to s1.

Process 2-3: Isentropic expansion process

In this process, expansion is take place in the turbine where heat is converted into work. In this process, pressure further decreases whereas the volume further increases. From the T-S diagram entropy remains constant whereas temperature decreases.

Process 3-4: Isothermal heat rejection process

The low quality of air which is exit from the turbine is supplied to the condenser where heat is rejected during the constant temperature process. In this process volume decreases whereas the pressure slightly increases from the T-S diagram, temperature remain constant whereas entropy decreases.

Process 4-1 : Isentropic compression process

The compression process is carried out in pump. In this process pressure increases as the volume decreases from T-S diagram entropy remains constant whereas the temperature slightly increases.

Performance:

The performance of a Carnot cycle is measured with the help of an efficiency which is given as: ⍰$_{Thermal} = \frac{output}{input}$

$$= \frac{Work\ done(W)}{Heat\ supplied(QS)}$$

From the first law of thermodynamics,

⍰⍰$_{th} = \frac{Q1-Q2}{Q1}$ ----- (A)

The efficiency is also expressed in terms of temperature.

Process 1-2: Isothermal heat addition process

$W_{1-2} = P1V1\ln[\frac{V2}{V1}]$

From 1st law of thermodynamics

$SQ = du + SW$

$SQ_{1-2} = P_1V_1\ln[\frac{V2}{V1}]$

Process 3-4: Isothermal heat rejection process

$W_{3-4} = P_3V_3\ln[\frac{V4}{V3}]$

From 1st law of thermodynamics,

$SQ = du + SW$

$SQ_{3-4} = P_3V_3\ln[\frac{V4}{V3}]$

$= P_3V_3\ln[\frac{V3}{V4}]$

Again,

Process 2-3: Isentropic expansion process

$\frac{T3}{T2} = (\frac{V2}{V3})^{r-1}$

$\text{Or,} \frac{V2}{V3} = \left(\frac{T3}{T2}\right)^{\frac{1}{r-1}}$

Process 4-1: Isentropic compression process

$\text{Or,} \frac{V1}{V4} = \left(\frac{T4}{T1}\right)^{\frac{1}{r-1}}$

$\text{Or,} \frac{V1}{V4} = \left(\frac{T2}{T1}\right)^{\frac{1}{r-1}}$

$\text{So,} \frac{V2}{V3} = \frac{V1}{V4}$

$\text{Or,} \frac{V2}{V3} = \frac{V3}{V4}$

Now,

⍰$_{th} = \frac{Q1-Q2}{Q1}$

$$= \frac{mRT1\ln\left[\frac{V2}{V1}\right] - mRT3\left[\frac{V3}{V4}\right]}{mRT1\ln\left[\frac{V2}{V1}\right]}$$

Let m =1

$$= \frac{RT1\ln\left[\frac{V2}{V1}\right] - RT3\left[\frac{V3}{V4}\right]}{RT1\ln\left[\frac{V2}{V1}\right]}$$

$$?_{th} = \frac{RT1\ln\left[\frac{V2}{V1}\right] - RT2\left[\frac{V3}{V4}\right]}{RT1\ln\left[\frac{V2}{V1}\right]} \quad [? \; T3 = T2]$$

$$= \frac{RT1\ln\left[\frac{V2}{V1}\right] - RT2\left[\frac{V3}{V4}\right]}{RT1\ln\left[\frac{V2}{V1}\right]}$$

$$?_{th} = \frac{T1-T2}{T1}$$

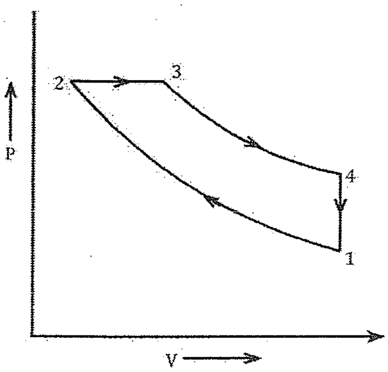
VVI) Numerical

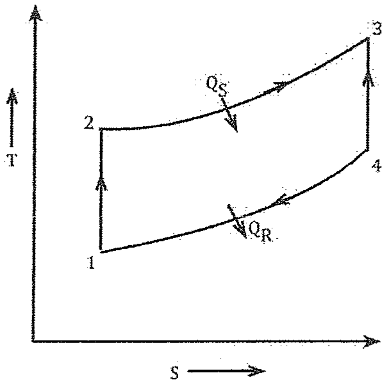
Q.2) The compression ratio of an air standard Otto cycle is 8. The pressure and atmosphere at the beginning of the compression stroke is 0.15 M pa and 18^0c respectively. The heat supplied per kg of the air is 1800 KJ/kg. Determine

I. The pressure and temperature at all the points of the cycle.

ii. The thermal efficiency of the cycle

iii. Mean efficiency pressure.





Sol^n

Given data:

$(r) = \frac{V1}{V2} = \frac{V4}{V3} = 8.$

$P_1 = 0.15$ M pa

$= 0.15 * 10^6$ pa

$T_1 = 18 + 273 = 291$ K

Qs = 1800 KJ/kg

$= 1800 * 10^3$ J/kg

To find

1. $P_2, T_2, P_3, T_3, P_4, T_4$
2. $?_{th}$

Process 1-2: ICP.

Or, $\frac{P2}{P1} = \left(\frac{V1}{V2}\right)^r$

Or, $P2 = P1 * \left(\frac{V1}{V2}\right)^{1.4}$

P2 = pa

Or, $\frac{T2}{T1} = (\frac{V1}{V2})^{r-1}$

Or, T2 = T1* $\left(\frac{V1}{V2}\right)^{r-1}$

Or , T2 = K

Process 2-3: CVHAP.

Qs = mCv(T3 – T2)

T3 = K

Again,

A+ V = C

$\frac{P}{T} = C$

$\frac{P2}{T2} = \frac{P3}{T3}$

Or,P3 = pa

Process 3-4 :ICP

$\frac{P2}{T2} = \left(\frac{V1}{V2}\right)^{r-1}$

Or, P4 = pa

Or, $\frac{T3}{P4} = \left(\frac{V1}{V2}\right)^{r-1}$

? T4 = K.

2.) ?th = 1- $\frac{1}{(r)^{r-1}}$

= 1- $\frac{1}{(8)^{1.4-1}}$

= 56%

4.) MEP = $\frac{W}{Stroke\ volume}$

= $\frac{W}{V1-V2}$

Or, ?th = $\frac{W}{Qs}$

?W =

Diesel cycle

Q) The compression ratio of an air standard diesel cycle is 18. The pressure and temperature at the beginning of the compression stroke is 0.1 Mpa and 250c respectively. The heat supplied per kg of the air is 2100 KJ/kg. Determine

1. The pressure and temperature at all the points of cycle
2. Thermal efficiency

Soln:

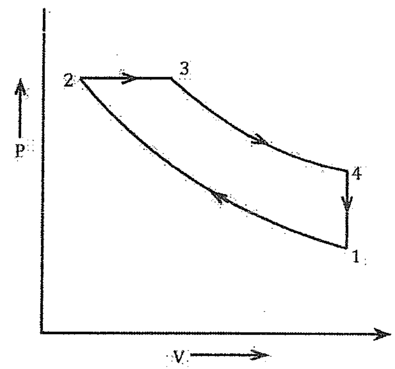
Given data

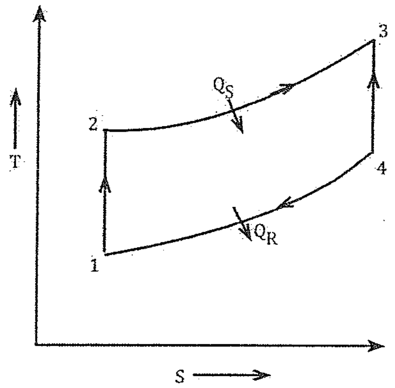
$(r) = 18 = \frac{V_1}{V_2}$

$Q_1 = 0.1MPa$

$T_1 = 25 + 273 = 398K$

$Q_2 = 2100 \times 10^3 J/kg$





Process 1 – 2: ICP

Or, $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)$

$P_2 = Pa$

Or, $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\vartheta-1}$

$T_2 = K$

Process 2 – 3 P = C

$QS = mcp\ (T_3 - T_2)$

$T_3 = k$

From PV diagram

$P_2 = P_3$

Process 3 – 4 IEP

Or, $\frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\vartheta}$

$V_4 = V1$

To find V_3

$P_3V_3 = mRT_3$

$V_3 = m3$

Or, P4 = Pa

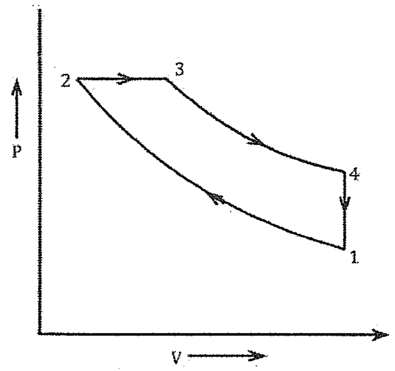
Or, $\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\vartheta-1}$

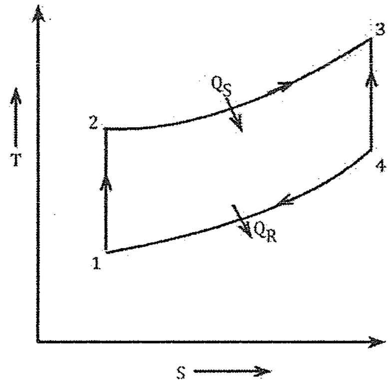
$\therefore T_4 = K$

2. $\eta_{Th} = \frac{Q_2 - Q_R}{Q_S}$

$Q_S = 2100 \times 10^3 J/kg$
$Q_R = mCv\ (T_4 - T_1)$
$Q_R =$
$\eta_{Th} = \%$

Q. The compression (r) ratio of an air standard otto cycle is 9. At the beginning of the compression stroke the pressure and temperature is given by 0.5 MPa, 15^0C respectively. The maximum temperature of the cycle is given as 1500^0C.
a) Determine the pressure and temperature at all the points of the cycle.
b) Thermal efficiency
c) Mean effective pressure.





Constant values
$C_P = 1.005kj/kgk$
$C_V = 0.718kJ/kgk$
$\gamma = 1.4$
$R = 0.287kJ/kgk$

Given data:
$(\gamma) = \frac{V_1}{V_2} = \frac{V_4}{V_3} = 9$
$P_1 = 0.5MPa = 0.5 \times 10^6Pa$
$T_1 = 15 + 273 = 288k$
$T3 = 1500 + 273 = 1773k$

To find:
1) $P_2, T_2, P_3, T_3, P_4, T_4$
2) η_{Th}
3) MEP

Process 1-2: ICP
$$\frac{P_3}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$
$$\text{Or, } P_2 = P_1 \times \left(\frac{V_1}{V_2}\right)^{\gamma}$$

Or, $P_2 = 0.5 \times 10^6 \times (9)^{1.4}$
$\therefore P_2 = 1 \times 10\text{-}5Pa$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

Or, $T_2 = T_1 \times \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

Or, $T_2 = 693.56k$

Process 2-3:
V = C

$$\frac{P}{T} = C$$

Or, $\frac{P_2}{T_2} = \frac{P_3}{P_3}$

$P_3 = \frac{P_2 \times P_3}{T_2}$

$$\frac{1\times10^{-5}\times1773}{693.56}$$

$= 2.5 \times 10^{-5}Pa$

Process 3-4: IEP

$$\frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\gamma}$$

Or, $P_4 = \frac{P_3}{\left(\frac{V_4}{V_3}\right)^{\gamma}}$

Or, $P_4 = \frac{2.5\times10^{-5}}{9^{1.4}}$

$\therefore P_4 = 1.15 \times 10^{-6}Pa$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1}$$

Or, $T_4 = \frac{T_4}{(V_4/V_3)^{\gamma-1}}$

Or, $T_4 = \frac{1773}{9^{0.4}}$

$\therefore T_4 = 736.22K$

3. MEP = $\frac{W}{V_1 - V_2}$

Or, $\eta = \frac{W}{Q_S}$

$Q_S = mC_v\ (T_3 - T_2)$
$= 1 \times 0.718 \times 10^3\ (T_3 - T_2)$
$= 1 \times 0.718 \times 10^3\ (1773 - 693.56)$
$= 775037.92$ J/Kg
$P_1V_1 = mRT_1$
$V_1 = \frac{mRT_1}{P_1}$
$= \frac{1 \times 0.287 \times 288}{0.5 \times 10^6}$
$= 1.653 \times 10^{-4}$

$P_1V_1 = mRT_1$
$V_1 = \frac{mRT_1}{P_1}$
$= \frac{1 \times 0.287 \times 288}{0.5 \times 10^6}$
$= 1.653 \times 10\text{-}4$

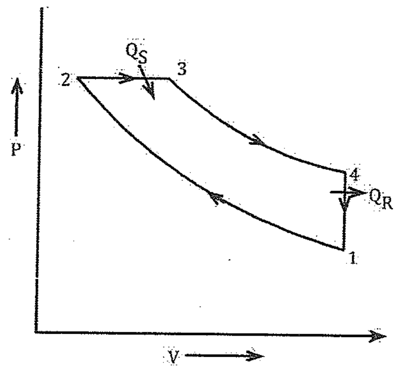
$P_2V_2 = mRT_2$
$V_2 = \frac{mRT_2}{P_2}$
$= \frac{1 \times 0.287 \times 693.56}{1 \times 10^{-5}}$
$= 19905172$

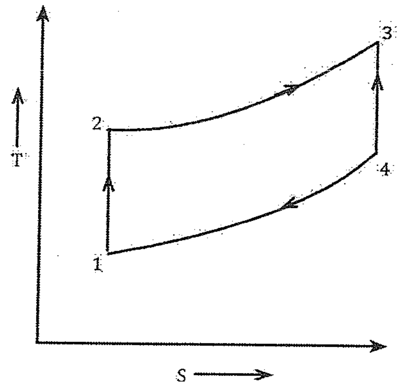
$$\text{MEP} = \frac{W}{V_1 - V_2} = \frac{45316467.18}{(1.653 \times 10^{-4} - 19905172)} = 58.47$$

$\eta = \frac{w}{Q_S} = \frac{w}{775037.92}$
$\therefore W = 45316464.18$ J

Q. At the beginning of the compression process of an air standard diesel cycle operating with the compression ratio of 18. The initial temperature is 300k and the pressure is 0.1 MPa. The out = off ratio of the cycle is 2. Determine
a) The temperature & the pressure at the end of process of the cycle

b) Thermal efficiency
c) Mean effective pressure





Solution
Given data

$(r) = 18 = \frac{V_1}{V_2} = \frac{V_4}{V_3}$

T1 = 300k

P1 = 0.1MPa = 0.1 x 106

$\delta = \frac{V_3}{V_2} = 2$

Process 1-2: ICP

Or, $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$

Or, $P_2 = P1\left(\frac{V_1}{V_2}\right)^{\gamma}$

Or, $P_2 = 0.1 \times 10^6 (18)^{1.4}$

$\therefore P_2 = 5719808.72Pa.$

$* \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$= 300(18)^{1.4-1}$

$= 953.3k$

Process 2-3: CPHAP

$P = C$

$\frac{V}{T} = C$

$\frac{V_2}{T_2} = \frac{V_3}{T_3}$

Or, $\frac{T_3}{T_2} = \frac{V_3}{V_2}$

$T_3 = T_2 \left(\frac{V_3}{V_2}\right)$

$= 953.3 \times (2)$

$= 1906.6k$

$P_3 = P_2$ from P – V diagram

Process 3-4: IEP

$* \ \frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\gamma}$

$P_3V_3 = mRT_3$

$V_3 = \frac{mRT_3}{P_3}$

$= \frac{1 \times 0.287 \times 10_3 \times 1906.6}{5719808.72}$

$= 0.09m3$

$P_4 = \frac{P_3}{(V_4 / V_3)^{\gamma}}$

$= \frac{5719808.72}{(18)^{1.4}}$

$= 2511886.422Pa$

$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\psi-1}$

$T4 = \frac{T_3}{(V_4 / V_3)^{\psi-1}}$

$= \frac{1906.6}{(18)^{1.4-1}}$

$= 599.99k$

2. $\eta_{diesel} = \frac{Q_S - Q_R}{Q_S}$

$= \frac{mCp(T_3 - T_2) - mCv(T_4 - T_1)}{mCp(T_3 - T_2)}$

$\eta_{diesel} = \frac{1-(T_4 - T_1)}{\psi(T_3 - T_2)}$ or, $\eta_{diesel} = 1 - \frac{1}{\psi(r)^{\psi-1}}\left[\frac{S^{\psi-1}}{S-1}\right]$

$= 1 - \frac{(599.99 - 300)}{1.4}$

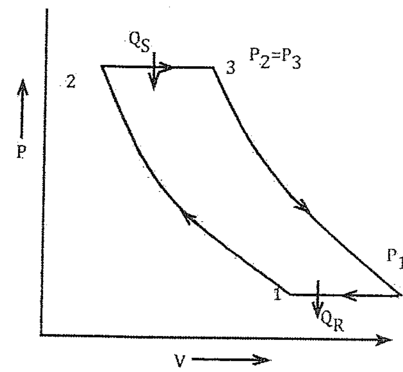
Q. In an air standard Bryaton cycle the air enters in the compressor at 0.1MPa, 15⁰C. The pressure having the compressor is 1MPa and the maximum temperature in the cycle is 1000⁰C. Determine

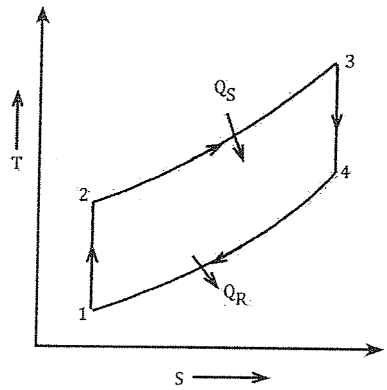
a) The pressure & temperature at each point of the cycle

b) The cycle efficiency.

Solution

Given data





$P_1 = 0.1MPa = 0.1 \times 10^6 Pa$

$T_1 = 150C + 278 = 288k$

$P_2 = 1MPa = 1 \times 10^6 Pa$

$T_3 = 100 + 273 = 1273$

Process 1-2: ICP

$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\psi-1}{\psi}}$

$T_2 = T_1 \times \left(\frac{P_2}{P_1}\right)^{\frac{\psi-1}{\psi}}$

2-3

$P_3 = P_2$

T_3 = Given = 1273k

3-4: IEP

Or, $\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\psi-1}{\psi}}$

Or, $T_4 = \frac{T3}{(P_3/P_4)^{\frac{\psi-1}{\psi}}}$

Or, T_4 = k

And $P_4 = P_1$ = Given

$\therefore \eta_{Bryaton} = \frac{1}{(rp)^{\frac{\psi-1}{\psi}}}$ pressure ration (P_2/P_1)

$= 1 - \frac{1}{(rp)^{\frac{\psi-1}{\psi}}}$

= %

$\therefore W_C = mCp(T_2 - T_1)$

$\therefore W_T = mCp\ (T_3 - T_4)$

Q1. In a gas turbine Bryaton system, the flows at rate of 5kg/sec. The inlet conditions of turbine is 90m/s with a specific volume of $0.85m^3/kg$. The exit condition of the turbine is specific volume of $1.45m^3/kg$. The exit area of the turbine is $0.04m^2$. In its passage through the turbine system, the specific enthalpy of the air is reduced by 250kJ/kg and there is heat transfer loss of 40kj/kg.

Determine.

a) the inlet area of the turbine

b) the exit velocity of the turbine

c) the power developed by the turbine

Q2. The compression ratio of a diesel cycle is 16. The initial temperature and pressure is 20∝, 0.2MPa. The maximum temperature of the cycle is 2500^0C. Determine

a) The pressure and temperature at the end of each process of the cycle.

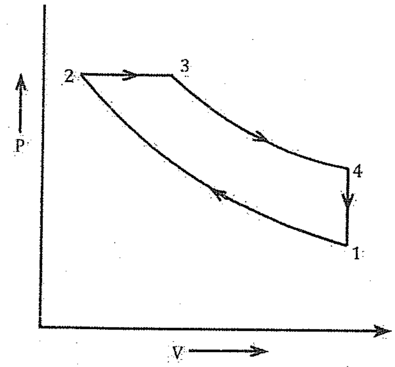
b) Thermal efficiency

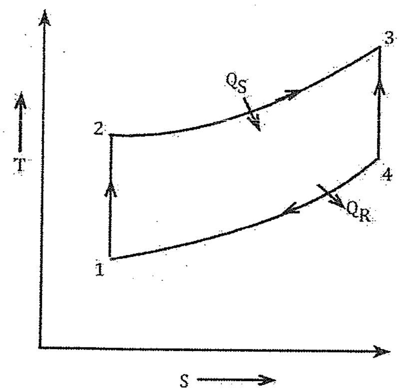
c) Mean effective pressure

Solution

Given data:

$(r) = 16 = \frac{V_1}{V_2}$





P1 = 0.2MPa = 0.2 x 106Pa
T1 = (20+273) = 293k
Maximum temperature (T3) = (2500 + 273)k = 2773k
Process 1-2: ICP

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\psi}$$

Or, $P_2 = P_1 \times \left(\frac{V_1}{V_2}\right)^{\psi}$

Or, $P_2 = 0.2 \times 106 \times (16)^{1.4}$

$\therefore P_2 = 9700586$Pa

$$* \ \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\psi-1}$$

Or, $T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\psi-1}$

Or, $T_2 = 293 \times (16)^{1.4-1}$

$\therefore$ T2 = 888.2

Process 2-3 CPHAP

$Q_S = mCpdT$

$= mCp(T_3 - T_2)$

$= 1 \times 1.005 \times 10^3 \times 1884.8$

= 1894224J/kg

P = C

$\frac{V}{T} = C$

$\frac{V_2}{T_2} = \frac{V_3}{T_3}$

V2 = $\frac{V_2}{T_2} \times T_2$ = 0.08 x $\frac{888.2}{2773}$ = 0.02m^3

Process 3-4: IEP

$\frac{P_3}{P_4} = \left(\frac{V_3}{V_3}\right)^{\psi}$

$P_1V_1 = mRT_1$

Or, 0.2 x 106 x V_1 = 1 x 0.287 x 10^3 x 297

Or, $V_1 = \frac{1 \times 0.287 \times 10^3 \times 293}{0.2 \times 10^6}$

$\therefore V_1 = V_4 = 0.42m^3$ (from the figure $V_1 = V_4$)

Now,

$P_3 = P_2$ (From the figure)

Or, 9700586 x V_3 = 1 x 0.287 x 10^3 x 2773

Or, $V_3 = \frac{1 \times 0.287 \times 10^3 \times 2773}{9700586}$

$\therefore$ V3 = 0.08m3

$P_4 = \frac{P_3}{(V_4/V_3)^{\psi}}$

$= \frac{9700586}{(0.42/0.08)^{1.4}}$

$= \frac{9700586}{10.19}$

$\therefore P_4 = 951971.14Pa$

$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\psi-1}$

Or, $T_4 = \frac{T_3}{\left(\frac{V_4}{V_3}\right)^{\psi-1}}$

Or, $T_4 = \frac{2773}{\left(\frac{o.42}{0.08}\right)^{1.4-1}}$

Or, $T_4 = \dfrac{2773}{1.94}$

$\therefore T_4 = 1429.38k$

$\eta_{Th} = \dfrac{Q_S - Q_R}{Q_S}$

$= \dfrac{1894224 - 815920.84}{1894224}$

$= 56.92\%$

$QR = mCV(T_4 - T_1)$

$= 1 \times 0.718 \times 103 \times 1136.38$

$= 815920.84$

$MEP = \dfrac{W}{V_1 - V_2}$

To find W

$\therefore \eta = \dfrac{W}{Q_S}$

Or, $\dfrac{56.92}{100} \times 1894224 = W$

$\therefore W = 1078192.3$

$MEP = \dfrac{W}{V_1 - V_2}$

$= \dfrac{1078192.3}{0.42 - 0.02}$

$= 2695480.75$

Chapter 7+8+5

Heat and mass transfer

Heat transfer:- The process of flow of energy in the forms of heat from one body to another body due to the difference in temperature is known as heat transfer.
- In thermodynamics, there are three types of heat transfer
1. Conduction
2. Convection
3. Radiation

1. Conduction: The process of heat transfer from higher temperature bodies to the lower temperature bodies due to the temperature difference in between two solid medium when they are physically contact to each other.
- There should be the requirement of any medium to make possible the conduction heat transfer.
- Conduction heat transfer is due to the kinetic energy present in the particular medium.
Eg. Heat transfer between iron rod with aluminium.

2. Convection: The process of heat transfer from higher temperature body to the lower temperature body in between solid medium and a fluid medium.
- There should be the requirement of a fluid medium in order to make possible the convection heat transfer Eg; water cooling system during an I.C engine.

3. Radiation: The process of heat transfer from higher temperature body to lower temperature body without any medium is known as radiation heat transfer.
- Radiation heat transfer is possible due to the electromagnetic, wave
- For eg: sunlight

* Fourier law of conduction Heat Transfer
Statement:
It states that the rate of heat transfer is directly proportional to the area measure normal to the direction of temperature gradient in that direction.
Mathematically,

$$Q \alpha - A\frac{dt}{dx}$$

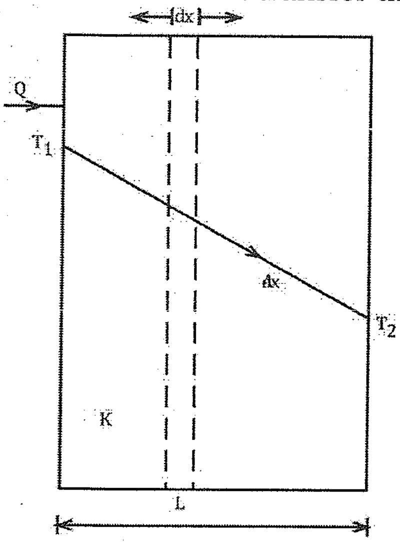
$$Q = -KA.\frac{dt}{dx}$$

Where,
K is a proportionality constant known as thermal conductivity (W/mk)
A is cross – sectional area (m2)
$\frac{dt}{dx}$ = Temperature gradient
Q = Rate of conduction heat transfer.

Thermal conductivity:

- The capacity of any material to conduct heat is known as thermal conductivity: is represented by K.
- The value of thermal conductivity is constant depending upon the material choice.

* Conduction heat transfer through a plane slab (or) plane wall



- Let us consider a plane wall having thickness 'L' as shown in figure. The wall is having cross sectional area of A and thermal conductivity K.
- Let dx is the small thickness of that particular wall. T_1 is the inner temperature which is generally a higher temperature and T_2 is the outer temperature which is a lower temperature.
- The rate of heat conduction through a plane wall is given as,
- From Fourier law of heat conduction. We have,

$$Q = -KA.\frac{dt}{dx}$$

Or, Q.dx = -K.A.dt

Now, integrating the equation from their respective limits, we have,

Or, $Q.\int_{O}^{L} dx = -K.A\int_{T_1}^{T_2} dt$

Or, $Q[L-O] = -K.A[t]_{T_1}^{T_2}$

Or, $Q[L-O] = -K.A[T_2 - T_1]$

Or, $Q.L = KA\,[T_1 - T_2]$

Or, $Q = \frac{KA[T_1 - T_2]}{L}$

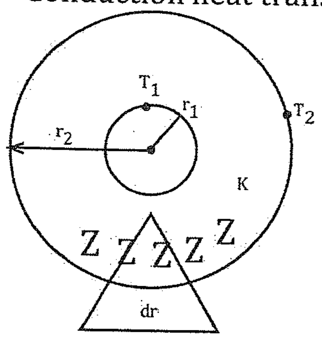
$\therefore Q = \frac{T_1 - T_2}{\frac{L}{KA}} \rightarrow$ (i)

Finally, $Q = \frac{\Delta T}{R}$

Where, $\Delta T = T_1 - T_2$

R = Thermal resistance equation that represents equation (i) represents the final conduction heat transfer through a plane slab.

* Conduction heat transfer through a hollow cylinder



fing. hollow cylinder

- Let us consider a hollow cylinder having inner radius r_1 & outer radius r_2
- Let dr is the small radius of that particular hollow cylinder.
- T_1 is the inner temperature & T_2 is the outer temperature.
- K is the thermal conductivity of that cylinder. The rate of conduction heat transfer through a cylinder is given by
- From Fourier law of heat conduction, we have,

$$Q = -KA.\frac{dt}{dx}$$

From the figure, dx is replaced by dr

Or, $Q = -KA.\frac{dt}{dx}$

Or, Q.dr = -KA.dt

Where A = cross – sectional area of cylinder but in this case, A = 2πrL

Or, Q.dr = -K. 2πrL.dt

Or, Q. $\frac{dr}{r}$ =-K.2πL.dt

Integrating the above equation from their respective limits.

Or, $Q.\int_{r_1}^{r_2}\frac{dr}{r} = -K.2\pi L\int_{T_1}^{T_2}dt$

Or, $Q.[Lnr]_{r_1}^{r_2} = -K.2\pi L[T]_{T_1}^{T_2}$

Or, $Q.\left[Ln[\frac{r_2}{r_1}]\right] = -K.2\pi L[T_2 - T_2]$

$Q.\left[Ln\left(\frac{r_2}{r_1}\right)\right] = -K.2\pi LK[T_2 - T_2]$

Or, $Q = \dfrac{T_1 - T_2}{\dfrac{1}{2\pi LK}\ln\left[\dfrac{r_2}{r_1}\right]}$

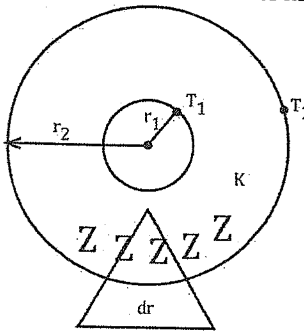
Or, $Q = \frac{\Delta T}{R}$

Where, $\Delta T = T1 - T2$

$$R = \frac{1}{2\pi LK}\left[\left(\ln\frac{r_2}{r_1}\right)\right] \rightarrow \text{(i)}$$

Equation (i) represents the final equation of conduction heat transfer through a hollow cylinder.

* Conduction heat transfer through a hollow sphere



fing. hollow sphere

- Let us consider a hollow sphere having inner radius r_1 and outer radius r_2.
- Let dr be the small radius of that particular hollow sphere
- T_1 is the inner temperature & T_2 is the outer Temperature
- K is the thermal conductivity of that cylinder. The rate of conduction heat transfer through a sphere is given by
- From Fourier law of heat conduction, we have

$$Q = -KA\frac{dt}{dx}$$

From the figure, dx is replace by dr

Or, $Q = -KA.\frac{dt}{dx}$

Or, Q.dr = -KA.dt

Where A = cross – sectional area of sphere,

= 4πr2

Or, Q.dr = -K.4πr^2.dt

Or, Q. $\frac{dr}{r^2}$ = -K.4π.dt

Integrating the above equation from their respective lengths,

Or, $Q\int_{r_1}^{r_2}\frac{dr}{r^2} = -K.4\pi[t]_{T_1}^{T_2}$

Or, $Q\left[\frac{-1}{r}\right]_{r_1}^{r_2} = -K.4\pi[T_2 - T_1]$

Or, $Q\left[\frac{-1}{r_2}+\frac{1}{r_1}\right] = -4\pi K[T_1 - T_2]$

Or, $Q\left[\frac{1}{r_2}-\frac{1}{r_1}\right] = -4\pi K[T_1 - T_2]$

Or, $Q = \dfrac{T_1 - T_2}{\dfrac{1}{4\pi K}\left[\dfrac{1}{r_1}-\dfrac{1}{r_2}\right]} \rightarrow$ (i)

Or, $Q = \dfrac{\Delta T}{R}$

Where, $R = \dfrac{1}{4\pi R}\left[\dfrac{1}{r_1} - \dfrac{1}{r_2}\right]$

Equation (i) represents the final equation of conduction of heat transfer through a hollow sphere.

Conduction Heat transfer through a multi-layer slab or wall

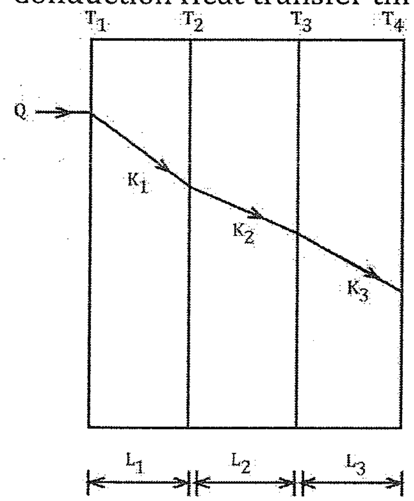


Fig. Multi-layer wall

- Let us consider a multi-layer slab having area A.
- The multilayer slab having their different thickness as represented by L_1, L_2 and L_3 as shown in figure
- Let K_1, K_2, and K_3 be the different thermal conductivities of the different material.
- When heat is transfer from the inside surface then T1 is the inner temperature and T4 is the outer temperature.
- Again, T_2 and T_3 are known as intermediate temperature.
- From Fourier law of heat conduction. The conduction heat transfer through a single slab is given as

$$Q = \frac{T_1 - T_2}{\frac{L}{KA}}$$

Or, $Q = \dfrac{KA(T_1 - T_2)}{L}$

Or, $QL = KA\,(T_1 - T_2)$

$\therefore T_1 - T_2 = \dfrac{Q.L_1}{K_1.A} \rightarrow$(i)

Again for 2nd slab

$$Q = \frac{T_2 - T_3}{\frac{L}{KA}}$$

Or, $Q = \dfrac{K_2 A (T_2 - T_3)}{L}$

$$\therefore T_2 - T_3 = \frac{Q.L_2}{K_2.A} \rightarrow \text{(ii)}$$

Also, for 3rd slab

$$Q = \frac{T_3 - T_4}{\dfrac{L_3}{K_3 A}}$$

Or, $Q.L_3 = K_3 A (T_3 - T_4)$

$$\therefore T_3 - T_4 = \frac{Q.L_3}{K_3.A} \rightarrow \text{(iii)}$$

Now equation adding (i), (ii) and (iii)

Or, $T_1 - T_2 + T_2 - T_3 + T_3 - T_4 = \dfrac{Q.L_1}{K_1 A} + \dfrac{Q.L_2}{K_2.A} + \dfrac{Q.L_3}{K_3.A}$

Or $T_1 - T_2 + T_2 - T_3 + T_3 - T_4 = \dfrac{Q.L_1}{K_1 A} + \dfrac{Q.L_2}{K_2.A} + \dfrac{Q.L_3}{K_3.A}$

Or, $T_1 - T_4 = \dfrac{Q}{A}\left[\dfrac{L_1}{K_1} + \dfrac{L_2}{K_2} + \dfrac{L_3}{K_3}\right]$

Or, $Q = \dfrac{T_1 - T_4}{\gamma A\left[\dfrac{L_1}{K_1} + \dfrac{L_2}{K_2} + \dfrac{L_3}{K_3}\right]} \rightarrow \text{(iv)}$

Equation represents the final condition of conduction heat transfer through a multi-layer slab or wall.

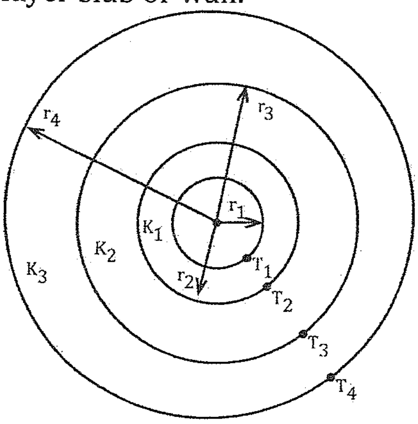


fig. Multi-layer cylinder

Conduction of heat transfer through a multi-layer cylinder.

- Conduction heat transfer through a hollow cylinder

$$\text{Or, } Q = \frac{T_1 - T_2}{\dfrac{1}{2\pi L}\left[\dfrac{\ln\left(\dfrac{r_2}{r_1}\right)}{K_1}\right]}$$

Or, $Q = \dfrac{T_1 - T_2}{\dfrac{1}{2\pi L}\left[\dfrac{\ln\left(\dfrac{r_2}{r_1}\right)}{K_1} + \dfrac{\ln\left(\dfrac{r_3}{r_2}\right)}{K_2} + \dfrac{\ln\left(\dfrac{r_4}{r_3}\right)}{K_3}\right]}$

Conduction heat transfer through a multi-layer sphere

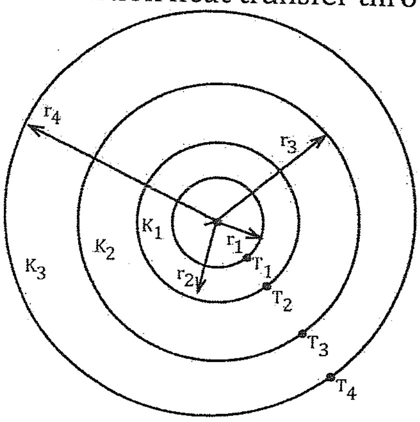


fig. Multi-layer cylinder

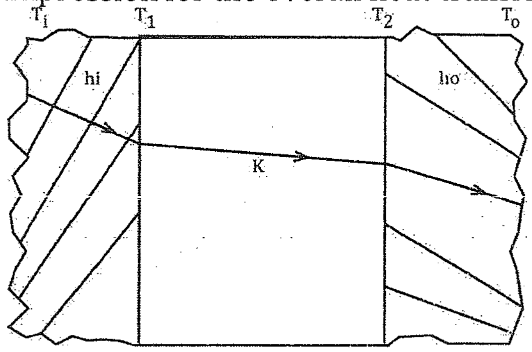
- For a single sphere $Q = \dfrac{T_1 - T_2}{\dfrac{1}{4\pi}\left[\dfrac{1}{K_1}\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right)\right]}$

- For a multilayer sphere

$$Q = \frac{T_1 - T_2}{\dfrac{1}{4\pi}\left[\dfrac{1}{K_1}\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right) + \dfrac{1}{K_2}\left(\dfrac{1}{r_2} - \dfrac{1}{r_3}\right) + \dfrac{1}{K_3}\left(\dfrac{1}{r_3} - \dfrac{1}{r_4}\right)\right]}$$

Convection:
Newton's law of convective heat transfer
* The convective heat transfer is given by
Q = hA (Ts – Tf)
Where, h = local heat transfer co-efficient (W/m^2k)
A = Area (m^2)
Ts = Temperature of solid
Tf = Temperature of fluid

Expression for the plane slab separating two fluid median
Or
Expression for the overall heat transfer coefficient



- Let us consider a plane slab having thickness L & thermal conductivity K.
- T_1 is the initial temperature of slab and T_2 is the outer temperature of slab.
- From the figure, the plane slab separates two different fluid media having hi the local heat transfer co-efficient of inlet fluid and ho is the local heat transfer co-efficient of outlet fluid.
- When heat is transferred the first resulting temperature is Ti & the final resulting temperature is To
- At first heat is transferred from fluid to solid which results the convective heat transfer and it is given by

$Q = hiA\,(Ti - T_1)$

Or $Ti - T_1 = \frac{Q}{hiA} \rightarrow$ (i)

- Again, heat is transferred from solid to solid which results the conduction heat transfer and it is given by

$$Q = \frac{T_1 - T_2}{\frac{L}{KA}}$$

Or, $Q = \frac{KA(T_1 - T_2)}{L}$

Or, $T_1 - T_2 = \frac{Q.L}{KA} \rightarrow$(ii)

Also, Heat transfer from solid to fluid so convective heat transfer is given as,

$Q = hoA(T2 - To)$

Or, $T_2 - To = \frac{Q}{hoA} \rightarrow$(iii)

Now adding equation no. i, ii and iii

Or, $Ti - T_1 + T_1 - T_2 + T_2 - To = \frac{Q}{hiA} + \frac{Q}{KA} + \frac{Q}{hoA}$

Or, $Ti - To = \frac{Q}{A}\left[\frac{1}{hi} + \frac{L}{K} + \frac{1}{ho}\right]$

Or, $Q = \dfrac{Ti - To}{\dfrac{1}{A}\left[\dfrac{1}{hi} + \dfrac{L}{K} + \dfrac{1}{ho}\right]} \rightarrow A$

Equation (A) represents the expression for the plane slab separating two fluid media.
Again
The overall heat transfer is given as,
Q = UA (Ti – To)
Where,
U = overall heat transfer co-efficient
Also,

$\dfrac{Q}{UA} = Ti - To \rightarrow B$

From equation no A and B

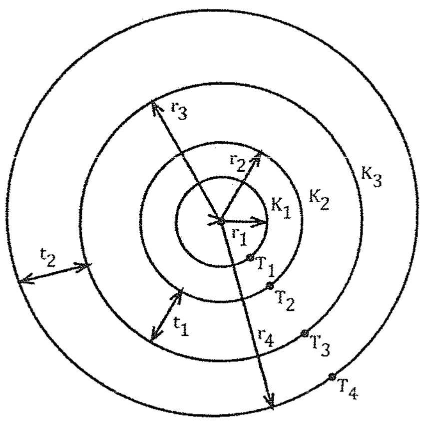
$$\frac{1}{UA} = \frac{1}{A}\left[\frac{1}{hi} + \frac{L}{K} + \frac{1}{ho}\right]$$

Or, $\dfrac{1}{A} = \dfrac{1}{hi} + \dfrac{L}{K} + \dfrac{1}{ho} \rightarrow (B)$

Equation (B) is the required expression that represents the equation of overall heat transfer co-efficient.

Q.1 A steel pipe having 10mm inner diameter and 25mm outer diameter which is carrying steam. The steel pipe consists of the insulating material composed of two layers having thickness of 5mm and 2mm respectively. The thermal conductivity of the steel is 50w/mk and the insulating material is 5w/mk and 2w/mk respectively. Determine the loss of heat from the pipe per meter of length. Also find the interface temperature if the initial temperature of the pipe is 500^0C and the final temperature is 30^0C.

Solution
Given data:
$d_1 = 10mm$
$r_1 = 5mm = 5 \times 10^{-3}m$
or $d_2 = 25mm$
or $r_2 = 12.5 \times 10^{-3}m$
or $r_3 = r_2 + t_1$
or $r_3 = 12.5 \times 10^{-3} + 5 \times 10^{-3}$
$\therefore r_3 = 17.5 \times 10^{-3}m$



$r_4 = r_3 + t_2$
$= 17.5 \times 10^{-3} + 2 \times 10^{-3}$
$= 19.5 \times 10^{-3}$

$K_1 = 50w/mk$
$K_2 = 5w/mk$
$K_3 = 2w/mk$

$T_1 = 500 + 273$
$= 773k$
$T_4 = 30 + 273 = 303$
To find
$\frac{Q}{L}$

Or, $Q = \dfrac{T_1 - T_4}{\dfrac{1}{2\pi L}\left[\dfrac{\ln(r_2/r_1)}{K_1} + \dfrac{\ln(r_3/r_2)}{K_2} + \dfrac{\ln(r_4/r_3)}{K_3}\right]}$

Or, $\dfrac{Q}{L} = \dfrac{T_1 - T_4}{\dfrac{1}{2\pi}\left[\dfrac{\ln(r_2/r_1)}{K_1} + \dfrac{\ln(r_3/r_2)}{K_2} + \dfrac{\ln(r_4/r_3)}{K_3}\right]}$

Or, $\dfrac{Q}{L} =$

Again,
To find T_2 and T_3

$Q = \dfrac{T_1 - T_2}{\dfrac{1}{2\pi L}\left[\ln\left(\dfrac{\frac{r_2}{r_1}}{K_1}\right)\right]}$

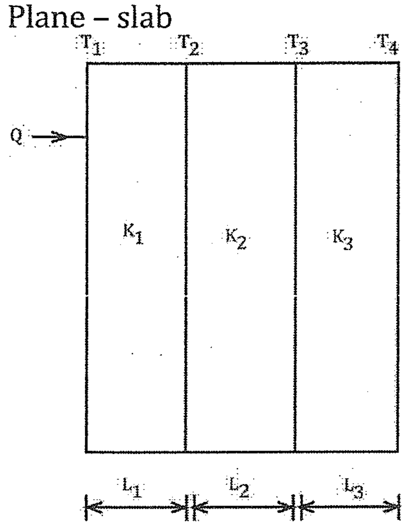
Or, $\dfrac{Q}{L} = \dfrac{T_1 - T_2}{\dfrac{1}{2\pi}\left[\ln\dfrac{\frac{r_2}{r_1}}{K_1}\right]}$

Or, T_2=K

Or, $Q = \dfrac{T_2 - T_3}{\dfrac{1}{2\pi L}\left[\ln\dfrac{\left(\frac{r_2}{r_3}\right)}{K_2}\right]}$

Or, $\frac{Q}{L} = \frac{T_2 - T_3}{\frac{1}{2\pi}\left[\ln \frac{\left(\frac{r_2}{r_3}\right)}{K_2}\right]}$

Plane – slab



1st case:

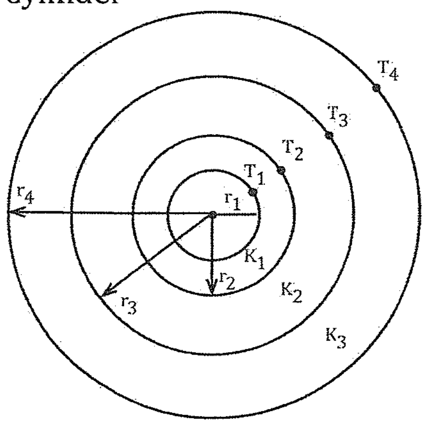
$$Q = \frac{T_1 - T_4}{\frac{1}{A}\left[\frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3}\right]}$$

2nd case:

If local heat transfer co-efficient (h) is given hi and ho is given

$$Q = \frac{Ti - To}{\frac{1}{A}\left[\frac{1}{hi} + \frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3} + \frac{1}{ho}\right]}$$

Cylinder



$$Q = \frac{T_1 - T_2}{\frac{1}{2\pi L}\left[\frac{\ln(r_2 / r_1)}{K_1} + \frac{\ln(r_3 / r_2)}{K_2} + \frac{\ln(r_4 / r_3)}{K_3}\right]}$$

If hi and h0 is given

$$Q=\frac{Ti-To}{\frac{1}{2\pi L}\left[\frac{1}{hir_1}+\frac{\ln(r_2/r_1)}{K_1}+\frac{\ln(r_3/r_2)}{K_2}+\frac{1}{hor_2}\right]}$$

Sphere:

$$Q=\frac{T_1-T_2}{\frac{1}{4\pi}\left[\frac{1}{K_1}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)+\frac{1}{K_2}\left(\frac{1}{r_2}-\frac{1}{r_3}\right)+\frac{1}{K_3}\left(\frac{1}{r_3}-\frac{1}{r_r}\right)\right]}$$

To find T_1

$$Q=\frac{Ti-To}{\frac{1}{4\pi}\left[\frac{1}{hir_1^2}\right]}$$

For T_2

$$Q=\frac{T_1-T_2}{\frac{1}{4\pi}\left[\frac{1}{\frac{1}{K_1}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)}\right]}$$

For T_3

$$Q=\frac{T_2-T_3}{\frac{1}{4\pi}\left[\frac{1}{\frac{1}{K_2}\left(\frac{1}{r_3}-\frac{1}{r}\right)}\right]}$$

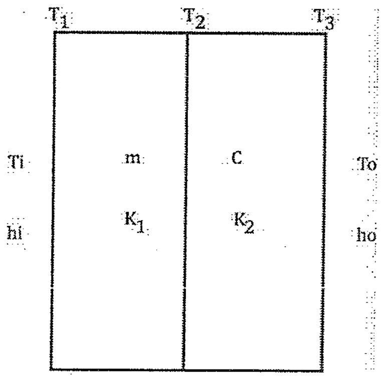
$$Q=\frac{T_3-T_4}{\frac{1}{4\pi}\left[\frac{1}{K_3}\left(\frac{1}{r_3}-\frac{1}{r_4}\right)\right]}$$

$$Q=\frac{Ti-To}{\frac{1}{4\pi}\left[\frac{1}{hir_1^2}+\dots\dots\dots+\frac{1}{hor_4^2}\right]}$$

Q.2 A finance wall is made of 20cm of magnetite brick & 20cm of common brick. The magnetite brick is exposed to not gases at 1355^0C and common brick outer surface is exposed to 45^0C of room air. The heat transfer co-efficient of the inside wall is 34 w/m^2k and that of the outside wall is 20 watt/m^2k respectively. Thermal conductivities of magnesite and common brick are 3.8 and 0.66w/mk respectively. Determine

i) heat loss per meter square of the furnaces

ii) maximum temperature to which common brick is subjected



Given data:
$L_1 = 20cm = 0.2m$
$L_2 = 20cm = 0.2m$
$Ti = 1355^0C = 1355 + 273 = 1628k$
$T0 = 45^0C = 45 + 273 = 318k$
$K_1 = 3.8w/mk$
$K_2 = 0.66w/k$
$hi = 34w/m^2k$
$ho = 20w/m^2k$
To find
Q = ?
We have

$$Q = \frac{Ti - To}{\frac{1}{A}\left[\frac{1}{hi} + \frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{1}{ho}\right]}$$

$$Q = \frac{1628 - 318}{\frac{1}{A}\left[\frac{1}{34} + \frac{0.2}{3.8} + \frac{0.2}{0.66} + \frac{1}{20}\right]}$$

$$\frac{Q}{A} = \frac{1310}{0.4350} = 3011.49 w/m^2$$

$$\therefore \frac{Q}{A}$$

To find temp:

$$\frac{Q}{A} = \frac{Ti - T_1}{\left[\frac{1}{hi} + \frac{L_1}{K_1}\right]} \text{ and } \frac{Q}{A} = \frac{T_1 - T_2}{\left[\frac{L_1}{K_1}\right]}$$

$$\text{Or, } 3011.49 = \frac{1628 - T_1}{\left[\frac{1}{34} + \frac{0.2}{3.8}\right]}$$

$$\text{Or, } 3011.49 = \frac{1628 - T_1}{0.082}$$

Or, $246.94 = 1628 - T_1$
$\therefore T_1 = 1381.06k$

Again

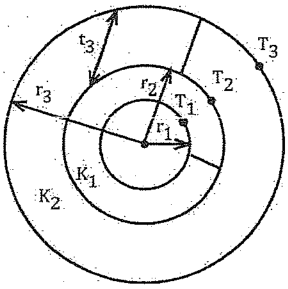
$$\frac{Q}{A} = \frac{T_1 - T_2}{\left[\frac{L_1}{K_1}\right]}$$

$$\text{Or, } 3011.49 = \frac{1381.06 - T_2}{\left[\dfrac{0.2}{3.8}\right]}$$

$$\text{Or, } 3011.49 = \frac{1381.06 - T_2}{0.052}$$

$\therefore T_2 = 1224.47\text{k}$

Q3. A pipe of stainless steel having thermal conductivity k = 1gw/m^{0}C with 2cm inside diameter & 4cm outer diameter is covered with a 3cm layer of asbestos insulation having thermal conductivity 0.2w/m^{0}C. If the inside wall temperature of the pipe is maintained at 600^{0}C. calculate the heat loss per meter of length. Also calculate the tube insulation interface temperature. Take outside temperature is 100^{0}C.



d_1 = 2cm so r_1 = 1cm = 0.01m
d_2 = 4cm so, r_2 = 2cm = 0.02m
t_3 = 3cm = 0.03m
$r_3 = r_2 + t_3$ = (0.02 + 0.03)m = 0.05m
K_1 = 19w/m0C
K_2 = 0.2w/m0C
T_1 = 600^{0}C
T_3 = 100^{0}C
To fine
$\frac{Q}{L}$, T2
Now,

$$Q = \frac{T_1 - T_3}{\frac{1}{2\pi L}\left[\frac{\ln(r_2/r_1)}{K_1} + \frac{\ln(r_3/r_2)}{K_2}\right]}$$

Or, $$\frac{Q}{L} = \frac{600-100}{\frac{1}{2\pi}\left[\frac{\ln(0.02/0.01)}{19} + \frac{\ln(0.05/0.02)}{0.2}\right]}$$

$$\frac{Q}{L} = \frac{500}{0.159[0.036+4.58}$$

$$\frac{Q}{L} = \frac{500}{0.733}$$

= 681.25W/m

Q4. A pipe of steel which is carrying steam having inside diameter 20cm and outside diameter 35cm. The pipe is covered with insulation with 5cm layer of 1st insulation & 2cm layer of 2nd insulation. The inner temperature of the pipe is 800^{0}C & the outer temperature is 30^{0}C. The thermal conductivity of the steel pipe is 50w/mk & the insulation is 5w/mk & 3w/mk respectively. Determine
i) the heat loss per meter length
ii) the % decrease or increase in the heat transfer with single layer of insulation compare to double layer of insulation.

$$Q=\frac{T_1-T_4}{\frac{1}{2\pi L}\left[\frac{\ln(r_1/r_2)}{K_1}+\frac{\ln(r_2/r_3)}{K_2}+\frac{\ln(r_2/r_3)}{K_2}\frac{\ln(r_4/r_3)}{K_3}\right]}$$

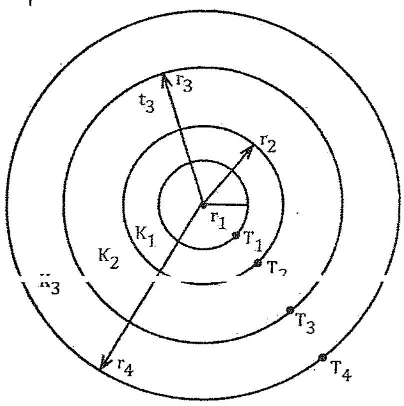
$$Q=\frac{T_1-T_3}{\frac{1}{2\pi L}\left[\frac{\ln(r_2/r_1)}{K_1}+\frac{\ln(r_2/r_2)}{K_2}\right]}$$

Q. A spherical pipe having 25mm inner diameter & 40mm outer diameter is covered with two layers of insulation with 2mm and 4mm thickness respectively. The inner temperature of the pipe is 540^{0}C & the ambient (atmospheric) temperature is 25^{0}C. The thermal conductivity of the pipe is 55w/mk and for the insulation the value of conductivity is 2w/mk & 5w/mk respectively. The local heat transfer coefficient (hi = h0) is 10w/m^{2}k. Determine.

i) the heat lost from the pipe.

ii) the inter phase temperature

$$Q=\frac{Ti-T0}{\frac{1}{2\pi}\left[\frac{1}{hir_1}+\frac{1}{K_1}+\left[\frac{1}{r_2}-\frac{1}{r}\right]+\frac{1}{K_2}\left[\frac{1}{r_2}-\frac{1}{r_3}\right]+\frac{1}{K_3}\left[\frac{1}{r_3}-\frac{1}{r_4}\right]+\frac{1}{h0r_4^{\,2}}\right]}$$



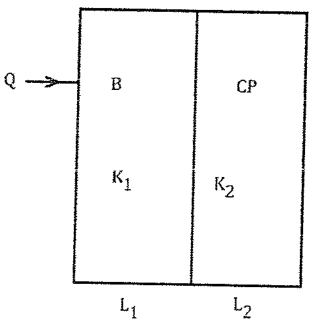
$$Q=\frac{Ti-T0}{\frac{1}{2\pi}\left[\frac{1}{hir_1^{\,2}}\right]}$$

$$Q=\frac{T_1-T_2}{\frac{1}{2\pi}\left[\frac{1}{K_1}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)\right]}$$

$$Q=\frac{T_2-T_3}{\frac{1}{2\pi}\left[\frac{1}{K_2}\left(\frac{1}{r_2}-\frac{1}{r_3}\right)\right]}$$

$$Q=\frac{T_3-T_4}{\frac{1}{2\pi}\left[\frac{1}{K_3}\left(\frac{1}{r_3}-\frac{1}{r_4}\right)\right]}$$

Q6. An exterior wall of house is made by a 10cm layer of common brick (k = 0.7 w/m^{o}c) which is followed by a layer of 3.8cm of cement plaster (k = 0.48w/m^{o}c). What thickness of another insulation (k = 0.065w/m0c) should be added to reduce the heat loss through the wall by 80%.



Solution:

Given data:

L_1 = 10cm = 10 x 10^{-2}m

L_2 = 3.8cm = 3.8 x 10^{-2}m

K_1 = 0.7 w/m^{0}c

$K_2 = 0.48 w/m^0c$

$$Q = \frac{\Delta T}{\frac{1}{A}\left[\frac{L_1}{K_1} + \frac{L_2}{K_2}\right]}$$

$A = 1m^2$

$$Q = \frac{\Delta T}{\left[\frac{L_1}{K_1} + \frac{L_2}{K_2}\right]}$$

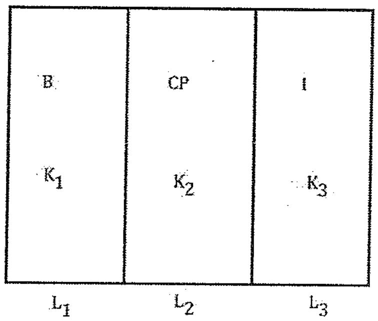
$$100 = \frac{\Delta T}{\left[\frac{10\times10^{-2}}{0.7} + \frac{3.8\times10^{-2}}{0.48}\right]}$$

$\Delta T = 22.2^0C$

2^{nd} case

When another layer is added the heat loss is reduced by 80% so,

$$Q = \frac{\Delta T}{\frac{1}{A}\left[\frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3}\right]}$$



A = 1m2

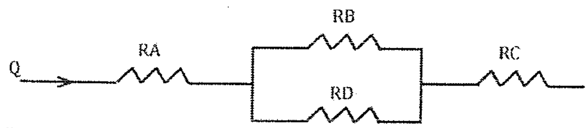
So,

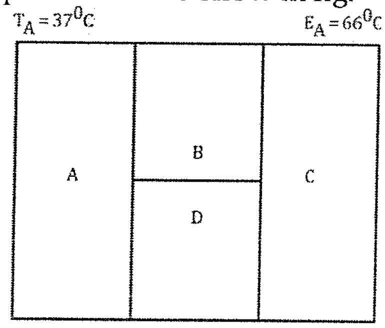
$$Q = \frac{\Delta T}{\left[\frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3}\right]}$$

$$\text{Or, } 20 = \frac{22.2}{\left[\frac{10\times10^{-2}}{0.7} + \frac{3.8\times10^{-2}}{0.48} + \frac{L_3}{0.65}\right]}$$

$\therefore L_3 = 5.7cm$

Q7. Find the heat transfer through the composite wall as show in fig.





$KA = 150 m/m^0C$, $L_A = 2.5cm$

$KB = 30 w/m^0C$, $L_B = 7.5cm$

$KC = 50 w/m^0C$, $L_C = 5.0cm$

$KD = 70 w/m^0C$, $L_D = 7.5$ com

$A_A = A_C = 2A_B + 2A_D = 0.1m^2$

$$Q = \frac{TA - TC}{R_A + R_B + R_D + | R_C}$$

$$R_A = \frac{L_A}{K_A.A_A}$$

$$= \frac{2.5\times10^{-2}}{150\times0.1}$$

$R_A = 1.66 \times 10^{-3}$

$$R_B = \frac{L_B}{K_B \times A_B}$$

$$\frac{7.5\times10^{-2}}{30\times0.05}$$

$= 5 \times 10^{-2}$

$$R_C = \frac{L_c}{K_c \times A_c}$$

$$\frac{5.0\times10^{-2}}{50\times0.1}$$

$= 1 \times 10^{-2}$

$$R_D = \frac{L_D}{K_D \times A_D}$$

$$\frac{7.5\times10^{-4}}{70\times0.05}$$

$= 2.14 \times 10^{-2}$

Now,

$$R\ net = Ra + \left(\frac{1}{R_B} + \frac{1}{R_D}\right) R_C$$

$$= R_A + \left(\frac{R_B}{R_B} + \frac{R_D}{R_D}\right) + R_C$$

$$\text{Or, } R\ Net = R_A \left(\frac{R_B.R_D}{R_B.R_D}\right)$$

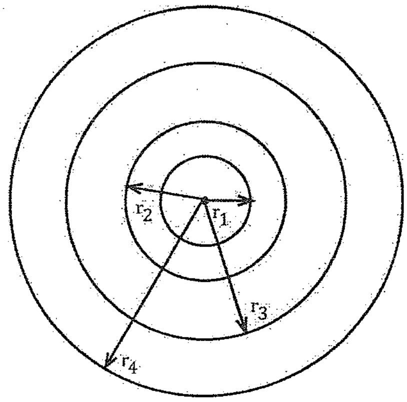
$$= 1.66 \times 10^{-3} + \left(\frac{5\times10^{-2}\times2.14\times10^{-2}}{5\times10^{-2}\times2.14\times10^{-2}}\right) + 1 \times 10^{-2}$$

$= 0.0266$

$$Q = \frac{\Delta Y}{Rnet} = \frac{T_A - T_C}{Rnet} = \frac{370-66}{0.0266}$$

= 11428.57w
= 11.4kw.

Q8. A steel pipe is made with 50cm inside diameter and 60cm outer diameter having thermal conductivity 45w/mk. Pipe is covered with 2 layers of insulation in which 4cm & 5 cm thickness respectively. The thermal conductive of the insulators are 4w/mk & 5w/mk respectively. Determine
i) The percentage of heat loss with the single layer of insulation compared to double layer of insulation.
Solution



$$Q = \frac{T_1 - T_4}{\frac{1}{2\pi L}\left[\frac{\ln(r_2/r_1)}{K_1} + ..\right]}$$

$$Q = \frac{\Delta T}{R}$$

$Q = \Delta T$

$$Q^1 = \frac{\Delta T}{2\pi\left[\ln\frac{(r_2/r_1)}{K_1} + \ln\frac{(r_3/r_2)}{K_2}\right]}$$

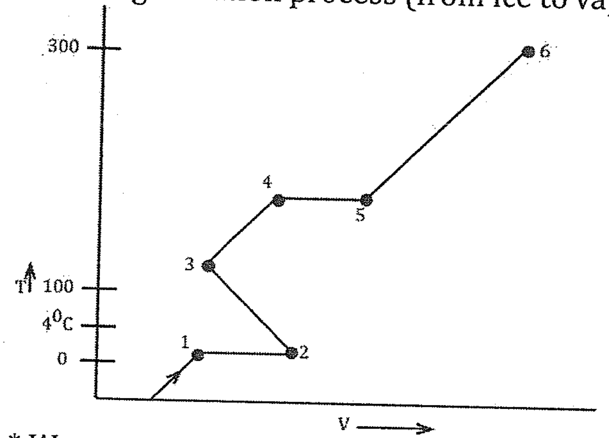
$Q^1 = - \Delta T$

$$Q^1 = \frac{Q^1 - Q}{Q^1} \times 100$$

Chapter 8
Pure Substances

1. Homogenous in chemical composition
2. Homogenous in chemical aggregation
3. Invariable in chemical aggregation

* Steam generation process (from ice to vapour)



* Wet steam
Steam which contains moisture
Two phase mixuter
i) liquid
ii) vapour

quality (x)

$$x = \frac{mass\ of\ gas}{Total\ mass} = \frac{m_g}{m}$$

where total mass = mass of liquid (mL) + mass of gas (mg)

$$\therefore x = \frac{mg}{ml + mg}$$

$\therefore$ x is always expressed in %

Sometimes, quality is expressed in terms of whole number. In this case, quality is also called as dryness fraction.
y = (1 – x)

* Determination of properties of a wet steam.
Specific volume
- Let us consider one kg of mass of a mixture is kept in a vessel. 'V' is the total volume of that mixture.
- The mixture consists of both liquid & gas so V_L represents the volume of liquid and Vg represents the volume of gas.

Mathematically,
$V = Vl + Vg \rightarrow$ (i)
Dividing both sides by mass 'm' we get

Or, $\frac{V}{m} = \frac{Vl}{m} + \frac{Vg}{m}$

Or, $\vartheta = \frac{Vl}{m} \times \frac{Vl}{m} + \frac{Vg}{mg} \times \frac{mg}{m}$

Or, $\vartheta = \vartheta l \times \left(\frac{Vl}{m}\right) + \vartheta g \times (x)$

Or, $\vartheta = \vartheta l \times \left(\frac{Vl}{m}\right) + \vartheta g.x$

Or, $\vartheta = \vartheta L \times \left(\frac{m - mg}{m}\right) + \vartheta g.x$

Or, $\vartheta = \vartheta_L \times (1-x) + \vartheta g.x$
Or, $\vartheta = \vartheta l - x\vartheta l + x\vartheta g$
Or, $\vartheta = \vartheta l + x\vartheta g - x\vartheta$
Or, $\vartheta = \vartheta l + x\vartheta lg$
Where $\vartheta lg = \vartheta g - \vartheta l$

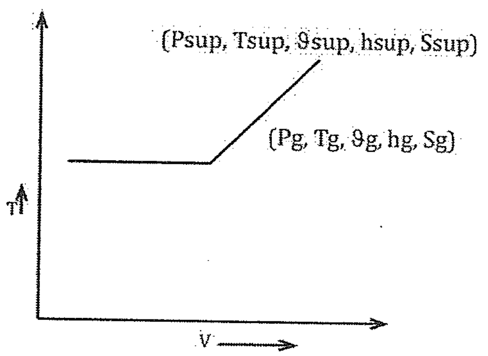
* Specific enthalpy (h)
$\therefore$ h = hl + xhlg

* Specific entropy (s)
S = Sl + xSlg
1 litre = 1 x $10^{-3}m^3$
* Specific internal energy (U)
Or, U = Ul + xulg

Dry steam super heated steam
- The steam in which percentage of moisture is negligible is known as dry steam
- Generally, the steam which is beyond the critical temperature is considered as dry steam
- It has wider applications in power production device such as thermal power plant.

* Critical point:
- The point in which both of the phase's i.e. liquid phase & vapour phase of water can co-exist is known as critical point.
- In this point the value of temperature, pressure and volume is given as
TC = 374.17^0C
Critical pressure (PC) = 222.12 bar.
(ϑ) critical volume = $0.00713m^3/kg$

* Determination of properties of dry steam
1) Specific volume



- Let us consider water is heated from solid to vapour.
- From fig (i) (Pg, Tg, ϑg, hg, Sg) represents pressure, temperature, specific volume, specific enthalpy, specific entropy at vapour condition and (Psup, Tsup, ϑsup, hsup, Ssup) represents the following properties at superheated condition.
- In this section we are going to find out the specific volume of dry steam.
- As we know that in vapour condition, the state behaves like a perfect gas. From the gas equation,

PV = mRT

$P.\frac{V}{m} = RT$

Or, $P.\vartheta = RT$

* From the graph, at constant pressure,

Or, $\frac{\vartheta}{T} = C$

Or, $\frac{\vartheta \text{sup}}{T \text{sup}} = C$

Or, $\frac{\vartheta \text{sup}}{T \text{sup}} = \frac{\vartheta g}{Tg}$

Or, $\vartheta g = \frac{\vartheta \text{sup}}{T \text{sup}} \times Tg$

$\therefore \vartheta \text{sup} = \frac{\vartheta g}{Tg} \times T \text{sup} \rightarrow (A)$

Equation (A) represents the specific volume of superheated steam.

2) Specific enthalpy
* At constant pressure

dh = CpdT

$\left[Cp = \frac{dh}{dt} \right]$

Now,
From the initial and final state

Or, (hsup – hg) = Cp.(Tsup – Tg)

Or, (hsup = hg + Cp(Tsup – Tg) → (B)

Equation (B) represents the specific enthalpy of dry steam

3) Specific entropy
As we know that,

ds = $\frac{sq}{T}$

where, sq = m.Cp dT

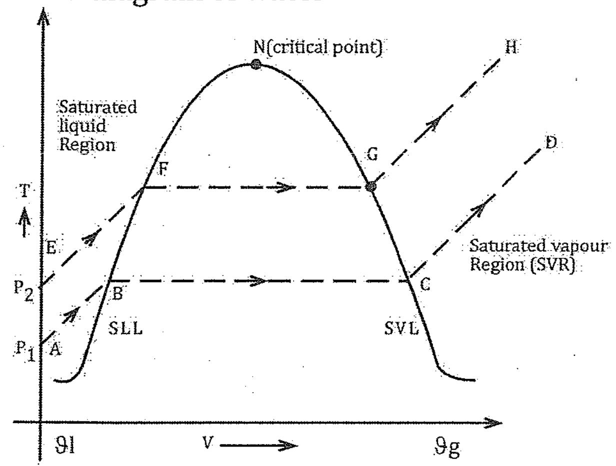
$$ds = \frac{Cp.dT}{T}$$

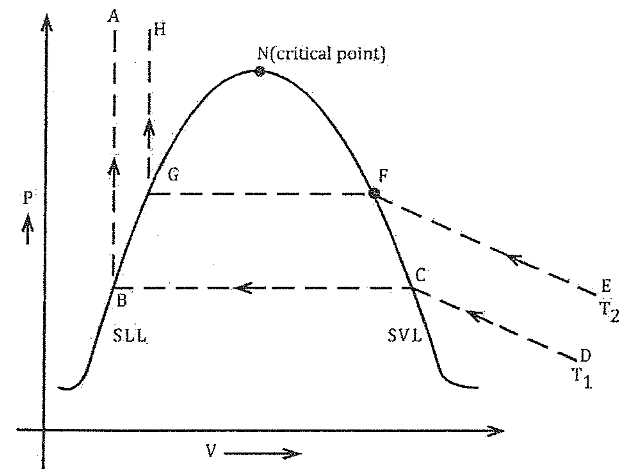
Integrating the above equation from Tg to Tsup

Or, $[S]_{Sg}^{Sup} = Cp\left[\ln\frac{T\,sup}{Tg}\right] \rightarrow (c)$

Equation (c) represents the specific entropy of dry steam.

* T-V diagram of water





Q. A steam is supplied through a steam turbine having the initial condition is 2mega pascal, 400^{0}C, 10m/sec. After expansion in the turbine, the steam is supplied to the condenser where the pressure is 0.75 mega pascal, 2m/s with 96% quality. The heat loss from the turbine to the surrounding is 40kJ/kg. Determine the work done by the turbine if the mass flow rate is 0.5 kg/sec.

Solution:

Given data:

Inlet	Outlet
P_1 = 2MPa	P_2 = 0.75MPa
T_1 = 400^{0}C	V_2 = 2m/s

$V_1 = 10m/s$ $x_2 = 0.96$

$$m\left(h_1 + \frac{V_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{V_2^2}{2} + gz_2\right) + w$$

from $P_1 = 2MPa$

$P2 = 0.75$

hL, hlg, hg

h = hL + xhlg

h = h2

Numerical

\# $\vartheta = VL + x\vartheta lg$

\# $x = \frac{mg}{m}$

Or, m = mL + mg

Or, $\vartheta = \frac{V}{m}$

Q1. A vessel of volume $0.4m^3$ contains 2kg of liquid water and water vapour mixture in equilibrium at a pressure of 0.6 mpa calculate:

a) The mass of liquid and vapour

b) The volume of liquid and vapour

Solution:

$V = 0.4m^3$

m = 2kg

p = 0.6mpa

$= 0.6 \times 10^6 pa$

= 600Kpa

To find

i) Vl, mg

ii) Vl, Vg

$$\vartheta = \frac{V}{m} = \frac{0.4}{2} = 0.2m^3kg$$

From the question.

Atp = 600kpa

$Vl = 0.001101m^3/kg$

$VLg = 0.3145m^3/kg$

$Vg = 0.3156m^3/kg$

$\vartheta = \vartheta l + x\vartheta Lg$

$\rightarrow 0.2 = 0.001101 + x \times 0.3145$

$\rightarrow x = 0.63$

Now $x = \frac{mg}{m}$

mg = 0.63 x 2

mg = 1.26kg

Again,

m = mL + mg

$2 = mL + 1.26$
$mL = 0.74kg$

Here,
$\vartheta = \frac{V}{m}$
$\vartheta L = \frac{Vl}{ml}$
Or, $0.001101 = \frac{VL}{0.74}$
$VL = 8.14 \times 10\text{-}4m^3$
$\vartheta g = \frac{Vg}{mg}$
Or, $0.3156 = \frac{Vg}{1.26}$
$\therefore Vg = 0.3976m^3$

Q2. A vessel of volume $0.04m^3$ contains a mixture of liquid water vapour at the temperature of 250^0C. The mass of the mixture is 9kg. Find the pressure mass of liquid and gas, the specific volume, the enthalpy and the internal energy.
Solution:
Given
$V = 0.04m^3$
$m = 9kg$
$T = 250^0C$
To find:
1) P, ml, mg, ϑ, h, U
$\vartheta l = 0.001251m^3/kg$
$\vartheta lg = 0.001251m^3/kg$
$\vartheta g = 0.05011m^3/kg$
UL = 1080.3kJ/kg
Ulg = 1521.3 kJ/kg
Ug = 2601.6kJ/kg
hL = 1085.3kJ/kg
hLg = 1715.4kJ/kg
hg = 2800.7kJ/kg
P = 3973.6kPa
P = 3979.6kPa

As we know that
$\vartheta = \vartheta l + x\vartheta lg$
Where, $\vartheta = \frac{V}{m}$
$\frac{0.04}{9}$

$= 4.4 \times 10^{-3} m^3/kg$

Again,

$\vartheta = \vartheta l + x\vartheta Lg$

Or, $4.4 \times 10\text{-}3 = 0.001251 + x \times 0.04886$

$\therefore x = 0.065$

Again,

$x = \frac{mg}{m}$

$0.065 = \frac{mg}{9}$

$\therefore mg = 0.585kg$

Again,

$m = mL + mg$

$mL = 8.415kg$

$\therefore h = hL = xhlg$

$\therefore h = 1085.3 + 0.065 \times 1715.4KJ/kg$

$= 1196.801kJ/kg$

Or, $U = Ul + xUlg$

$= 1080.3 + (0.065 \times 1521.3)kJ/kg$

$= 1179.1845kJ/kg$

Q3. Steam enters into a steam turbine which is operating at a steady state with a mass flow rate of 4.6700kg/hr. The turbine develops a power output of 1000kw. At the inlet the pressure is 60 bar & the temperature is 400°C with the velocity of 10m/s. At the exit the pressure is 0.1 bar, the quality is 0.9 and the velocity is 50m/s. Calculate the rate of heat transfer between the system and surrounding in kw.

Inlet

$m = 4600kg/h$

$= \frac{4600}{3600}$

$= 1.27kg/s$

$P1 = 60bar$

$= 60 \times 105Pa$

$= 6000kPa$

$T_1 = 400^0C$

$V_1 = 10m/s$

Outlet

$m = 1.27kg/s$

$P2 = 0.1$ bar

$= 0.1 \times 10^5Pa$

$= 10kPa$

$x_2 = 0.9$

$V2 = 50m/s$

At inlet

To find enthalpy (h_1)

From steam table

At P = 6000kPa

h1= hg = 2783.9kJ/kg

At outlet

h2 = ?

At P = 10kPa

Hl, hlg, hg

$\therefore$ h = hL + xhlg hL = 191.83, hlg = 2392.0

h = h2 = 191.83 + 0.9 x 2392.0 hg = 2583.8

= 2344.63kJ/kg

From steady flow energy equation

$$m\left(h_1 + \frac{V_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{V_2^2}{2} + gz_2\right) + w$$

$$\text{or, } 1.27\left(2783.9\times10^3 + \frac{10^2}{2} + 0\right) + Q = 1.27\left(2271.19\times10^3 + \frac{50^2}{2} + 0\right) + 1000 \times 10^3$$

or, 3535616.5 + Q = 3885998.8

$\therefore$ Q = 350382.3J/kg

= 350382.3w

= 350.3823kw

Q.1) A vessel of volume $0.04m^3$ contains a mixture of water and steam at the temperature of 250^0C. The mass of liquid present is 9kg. find the pressure, the mass, the specific volume the enthalpy & the internal energy.

Solution:

Given data:

$V = 0.04m^3$

$T = 2500C$

$mL = 9kg$

To find

P, m, ϑ, h, u

From the steam table

At temp 25^0

$\vartheta L = 0.00100m^3/kg$

$\vartheta Lg = 43.356m^3/kg$

$\vartheta g = 43.357m^3/kg$

$hL = 104.75kJ/kg$

$hLg = 2441.6\ kJ/kg$

$hg = 2546.3kJ/kg$

$UL = 104.75kJ/kg$

$ULg = 2304.1kJ/kg$

$Ug = 2408.9kJ/kg$

To find the volume of liquid

$\therefore \vartheta l = \frac{Vl}{mL}$

$0.001003 = \frac{Vl}{9}$

Or, $VL = 0.001003 \times 9$

$= 9.027 \times 10^{-3}$

$VL = m^3$

$\therefore\ Vg = m^3$

Again

$\vartheta g = \frac{Vg}{mg}$

$\therefore mg = -\ kg$

Now, $m = mL = mg$

$= -\ kg$

$\vartheta = \frac{V}{m}$

$\therefore \vartheta = m^3/kg$

$\therefore \vartheta = \vartheta L + x\vartheta Lg$

$\therefore h = hl + xhlg$

$h =$

$\vartheta = \vartheta L + xULg$

Q. A steam turbine is designed to operate at a mass flow rate of 1.5kg/sec. the inlet conditions are P_1 = 2MPa, T1 = 400^{0}C & V_1 = 60m/s and the outlet conditions are P_2 = 0.1MPa x2 = 0.98 & V_2 = 150m/s. The change in elevation from inlet to outlet is a drop of 1m & the heat loss is 50w. Determine

a) The power output of the turbine
b) Evaluate the power output if the changes in KE & PE are neglected.
c) What are the diameters of inlet & outlet pipe.

Solution:

$$m\left(h_1 + \frac{V_1^2}{2} + gz_1\right) + Q = m\left(h_2 + \frac{V_2^2}{2} + gz_2\right) + w$$

To find h,
From steam table
A + P = 2MPa
= 2000kPa
$h_1 = hg = 2798.7$kJ/kg

to find h_2
from S.T
$P_2 = 0.1$MPa
= 1000kPa
hL, hLg, hg
h2 = hL + xhLg
hL = 417.51kJ/kg
hlg = 2257.6kJ/kg
$h_2 = 417.51 + 0.98 \times 2257.6$ kJ/kg
= 2629.958kJ/kg

Inlet
$V_1 = Vg$

outlet
$V_2 = VL + xVLg$

$$1.5\left[2789.7\times10^3 + \frac{60^2}{2} - 2629.958\times10^3\,\frac{150^2}{2} + 1\right]$$

i) W = ..
ii) $mh_1 + Q = mh_2 + w$
ii) $\vartheta_1 = Vg$
$= \vartheta_2 = \vartheta L + x_2\vartheta Lg$

$$m = \frac{V_1 A_1}{\vartheta_1} \qquad m = \frac{V_2 A_2}{\vartheta_2}$$

$$A_1 = \frac{m\vartheta_1}{V_1} \qquad A_2 = \frac{m\vartheta_2}{V_2}$$

$$A_1 = \frac{\pi d_1^2}{4} \qquad A_2 = \frac{\pi d_2^2}{4}$$